Supply or Demand?*

Auditing Algorithmic Amplification of Antisemitic Content

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Abstract

Whether recommender systems amplify antisemitic content beyond what users seek, the question of supply or demand, is central to platform accountability, including the systemic-risk duties of the EU Digital Services Act, yet "amplification" is rarely defined. This methods paper defines it as a causal contrast: expected exposure to antisemitic content under the algorithmic policy relative to a named reference policy, at fixed demand. For real users, observed exposure splits into demand, supply, demand shaping and a feedback interaction; sock-puppet audits of fixed demand identify only supply. A Markov model of sessions gives long-run exposure, rabbit holes and a perturbation bound: small nudges can have large effects. We state when randomised audits identify amplification, bound the bias when puppets interfere through the system they audit, show that arm-dependent classifier error can put a naive estimate on the wrong side of one, correct it by arm, and give an approximately anytime-valid confidence sequence for stopping as soon as the evidence is clear. In a synthetic recommender, the registered analysis kept its error rates despite heavy-tailed puppet outcomes (time-uniform coverage 98.4–99.7% at nominal 95–97.5%), and the pilot check failed in 89.5–100% of pilots whenever the sequence for the ratio did not hold. Under plausible inputs a materiality audit of four personas needs thousands of accounts; the pre-registered protocol therefore commits only to a pilot and a registered feasibility gate, with ethical safeguards. We have audited no real platform and report no finding about any service.

Keywords algorithm audits; recommender systems; amplification; sock-puppet audits; potential outcomes; Markov chains; interference; misclassification; anytime-valid inference; Digital Services Act; antisemitism.

1 Introduction

Recommender systems decide much of what the users of large online services see. The European Union's Digital Services Act (DSA) puts it in its recitals: such systems "play an important role in the amplification of certain messages, the viral dissemination of information and the stimulation of online behaviour" (European Parliament and Council of the European Union, 2022, Recital 70). For the largest services, those with at least 45 million average monthly active recipients in the Union, the Act requires providers to assess the systemic

*Status. Methods paper; simulation study on a synthetic recommender; no audit of any real platform was conducted. Every number in Section 7 is computed from a synthetic recommender with known ground truth; the audit protocol of Section 8 is pre-registered and has not been carried out. Published audits are quoted only as their authors report them, and nothing here is a finding about any real service. The companion papers (Kornstein Lab for Antisemitism Research, 2026a,b,c) study recorded incidents and attitudes; this paper studies what recommender systems supply.

risks that stem from their services, among them negative effects on human dignity and non-discrimination, taking the design of their recommender systems into account (Articles 33 and 34). For Jewish users the question is not abstract. In the third survey of Jewish people by the EU Agency for Fundamental Rights, conducted online in 2023, 90% of almost 8,000 respondents in 13 Member States said they had encountered antisemitism online in the previous twelve months (European Union Agency for Fundamental Rights, 2024). That figure says that antisemitic content is widely seen. It does not say why.

There are two broad answers, and they call for different remedies. Users may see antisemitic content because they, or the accounts they follow, seek it out: the exposure is driven by *demand*. Or a recommender may put such content in front of them more often than an alternative way of choosing content would: the exposure is driven by *supply*, and the remedy lies in the design of the recommender. Public argument about whether algorithms "amplify" hateful content rarely says what amplification is compared with, or for whom, and published audits differ in what they compare, what they measure, the content they study and the period they cover (Section 2). A claim that an algorithm amplifies antisemitic content, or that it does not, often cannot be checked against another, because the two claims concern different estimands.

This paper supplies the estimands and the methods to estimate them. *Amplification* is a causal contrast: the expected exposure to antisemitic content under the algorithmic policy, divided by the expected exposure under a named reference policy, for a fixed demand profile. An audit with sock puppets (synthetic accounts whose behaviour the researcher scripts) then becomes a randomised experiment whose identifying assumptions can be stated and, in part, checked. Its design problems are those of any experiment on a live system that learns: puppets can change the model they audit, the model changes while it is audited, the outcome is measured by an imperfect classifier, and every session adds to the footprint of the audit on a real service. Figure 1 shows the audit design that results.

What this paper does, and does not, do. This is a methods paper. We define the estimands, prove what a sock-puppet audit can and cannot identify, derive estimators that correct for misclassification and a sequential procedure that stops when the evidence is clear, evaluate them in a simulation study on a synthetic recommender with known ground truth, and pre-register a protocol for an audit, with its ethical and legal conditions. We have run *no* audit of any real platform, and nothing here should be read as a finding that any service amplifies antisemitic content, or that it does not. Published audits are cited only as their authors report them. The protocol is platform-neutral: it is written for any service with a recommendation surface and a reference policy. Notation is collected in Table 1.

Contributions. Classical results are attributed where used; we flag what is, to our knowledge, new in the form given.

C1 Estimands. The amplification ratio $\mathrm{AR}_\rho(d)$ and additive amplification $\mathrm{AA}_\rho(d)$ compare mean exposure under the algorithm and under a reference policy ρ at fixed demand d (Definition 3.1; Imbens and Rubin, 2015; Rubin, 1974); the reference is part of the estimand. For real users, observed exposure splits into demand, supply, demand shaping and a feedback interaction, bounded by the heterogeneity of supply times the shift in demand (Theorem 3.3), and fixed-demand audits identify only supply (Proposition 3.4). The bilinear form, the interaction bounds and Proposition 3.4 are new in this form.

C2 A Markov model of exposure. Long-run and finite-horizon exposure, hitting times and *rabbit holes* follow from the fundamental matrix (Proposition 4.1; Kemeny and Snell, 1960; Meyer, 1975). Redirecting a share ε of recommendations to an antisemitic state raises long-run exposure, to first order, by the relative amount ε times the mean number of items needed to reach it from equilibrium,

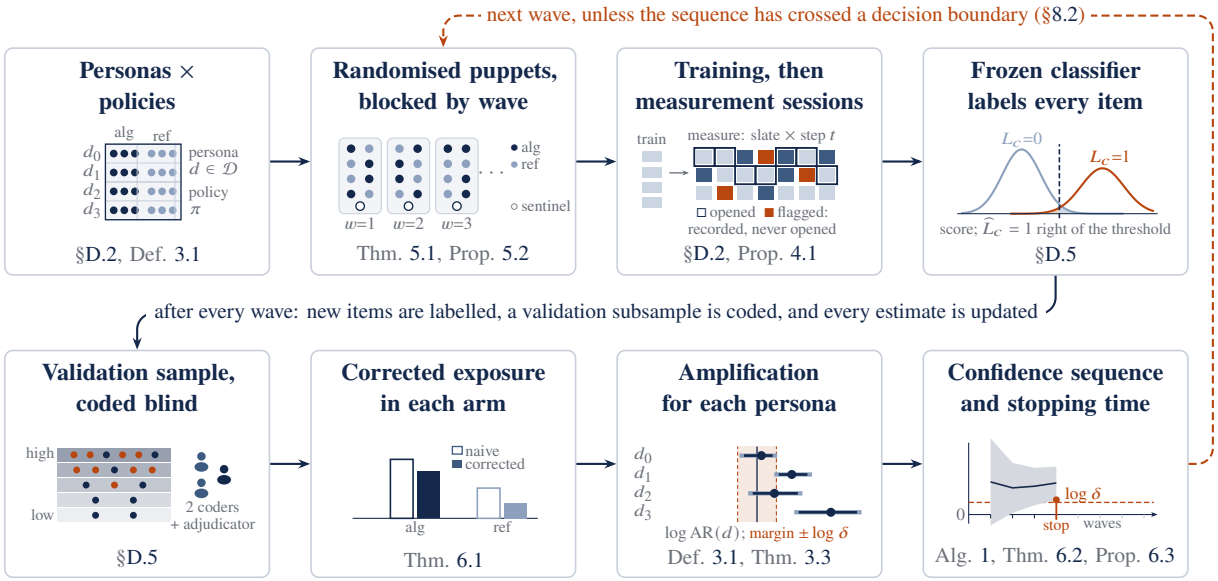


Figure 1. The audit design, from personas to a stopping time. Top row: each persona d (a fixed demand profile) is crossed with the algorithmic and a reference policy; puppets are assigned to the $|\mathcal{D}| \times 2$ cells completely at random within waves, which are blocks, and untrained sentinels monitor interference and drift. Every puppet is trained in the same way; the policy contrast starts with the measurement sessions, in which each slate is recorded, every item is scored by a classifier frozen at registration, and the puppet opens one item per step but never a flagged one. Bottom row: a stratified, oversampled validation sample is coded blind to arm, persona and score, and gives the sensitivity and specificity of the classifier in each arm; exposure is corrected arm by arm, because error rates can differ when the content mix differs; the amplification ratio is estimated for each persona and compared with a registered margin $[1/\delta, \delta]$; and a confidence sequence for $\mathrm{AR}(d)$, built from a puppet-level outcome (in the protocol, each puppet's double-sampled exposure) and approximately valid whenever the audit stops, decides after each wave whether to stop. Here, on the log scale, it stops after wave 4, when it lies entirely above $\log \delta$ (rule S1). Labels under each tile point to the corresponding results. **Schematic:** all glyph values are invented for legibility.

while within a T-item session the effect is capped at $(T-1)/4$ times the change (Theorem 4.2); these bounds are new in this form.

C3 Identification. Randomising puppets to persona–policy cells within waves identifies $\mathrm{AR}(d)$, $\mathrm{AA}(d)$ and a persona-contrast decomposition under four stated conditions, with the puppet as the unit of variance (Theorem 5.1). Treating waves as groups of partial interference (Hudgens and Halloran, 2008) and the puppets' share of engagement as an exposure mapping (Aronow and Samii, 2017), we bound the bias by a constant ξ times that share, show that the direction of the bias depends on how much puppet engagement reaches false negatives, so no sign can be assumed, and give a randomised-saturation design and a breakdown value for sensitivity analysis (Proposition 5.2).

C4 Misclassification-corrected measurement. The naive ratio of flagged shares is attenuated towards one through the classifier's specificity, and when error rates depend on content type and the arms show different mixtures of types, the naive and even a pooled-corrected ratio can lie on the wrong side of one (Theorem 6.1). Validation by arm, as double sampling within puppets provides, is consistent in either case (Neyman, 1938; Rogan and Gladen, 1978).

C5 Sequential auditing and its cost. A confidence sequence from a puppet-level outcome is approximately valid at any data-dependent stopping time, and a betting variant exactly (Algorithm 1

Table 1. Notation. Symbols used throughout. Vectors are columns; $\mathbf{1}$ is the vector of ones and $\mathbf{1}_{\mathcal{A}}$ the indicator of the antisemitic states.

Accounts, demand and policies	
$u = 1, \dots, U$	puppet (synthetic account)
$d \in \mathcal{D}$, d_0	persona (fixed demand profile); baseline persona
alg; ρ	algorithmic policy; reference (chron, rand, pop)
$\mathcal{P}_t$	eligible pool of items when step t is served
$t = 1, \dots, T$; k; S	step of a session; slate positions recorded; sessions per puppet
Design and inference	
$w = 1, \dots, W$; ω_w	wave (block); wave weight
ℓ_w, $\bar{\ell}$	load: puppets' share of the engagement signal; its average
ξ_π, ξ_Δ	change in exposure, and in AA, per unit of load (Proposition 5.2)
n; N; n^*	puppet pairs so far; pairs at stopping; planned pairs
C_n; [LB, UB]	confidence sequence after n pairs; its running intersection (Algorithm 1)
δ; α; β	registered margin; error level; one minus power
V; V_{val}	per-pair variance of $\log \widehat{\text{AR}}$; validation term (Proposition 6.3)
Measurement	
L_c, $\widehat{L}_c$	true label of item c (1: antisemitic); classifier label
$\text{Se}_\pi(d)$, $\text{Sp}_\pi(d)$	sensitivity and specificity on the items shown in cell (d, π)
J; λ_S	Youden's index $\text{Se} + \text{Sp} - 1$; ratio $\text{Se}_{\text{alg}}/\text{Se}_{\text{ref}}$
$q_\pi(d)$; w_0	flagged share; attenuation weight of the naive ratio
$\widetilde{E}_u$; Y_u	double-sampled exposure; confirmed-flagged exposure
m, m'; n_0, n_1	coded flagged, unflagged positions per puppet; coded items with $L_c = 0, 1$
Exposure and amplification	
$E_u(\pi)$	share of antisemitic items in a session under policy π
$\mu_\pi(d)$, $\mu_\pi(\Lambda)$	mean exposure at persona d; averaged over a distribution Λ of personas
AR_ρ, AA_ρ	amplification ratio; additive amplification (Definition 3.1)
F_b	real users' demand, formed under policy b
τ; τ_{sup}, τ_{dem}, ι	total effect; supply, demand-shaping and interaction terms (Theorem 3.3)
ϕ_{sup}, ϕ_{dem}	Shapley shares of supply and demand
Markov model of exposure	
$\mathcal{X} \supseteq \mathcal{A}$; N, B, A	content states; antisemitic states; neutral, borderline, antisemitic
P_{alg}, P_{ref}; ν_π	transition matrices; initial distribution
ϖ; $\varpi(\mathcal{A})$	stationary distribution; long-run exposure
$e_T(P, \nu)$	expected exposure in a T-item session
Z; $H = Z - \mathbf{1}\varpi^\top$	fundamental matrix $(I - P + \mathbf{1}\varpi^\top)^{-1}$; group inverse of $I - P$
$\boldsymbol{g}_{\mathcal{A}} = H\mathbf{1}_{\mathcal{A}}$; $\kappa_{\mathcal{A}}$	exposure potential; sensitivity coefficient $\frac{1}{2}\,\text{osc}(\boldsymbol{g}_{\mathcal{A}})$
$T_{\mathcal{S}}$; M_{xy}; $K(P)$	items until the session reaches $\mathcal{S}$; mean first-passage time; Kemeny's constant

and Theorem 6.2; Howard et al., 2021; Ramdas et al., 2023; Wald, 1945). A closed-form power calculation (Proposition 6.3) shows the cost: under plausible inputs a materiality audit of four personas needs about 4,900 accounts and 170,000 coded exposures.

C6 A simulation study on a synthetic recommender with known amplification evaluates these tools (Section 7). Simulating the registered analysis itself, we find its normal-mixture sequences conservative, its closed-form variance 1.4–2.0 times too large, and its exact fallback nearly powerless; slate-state Markov chains need a correction identifiable only for the two-state chain, so Markov summaries are exploratory.

C7 A pre-registered protocol with a feasibility gate. We fix in advance the access route, personas, randomisation, interference controls, validation, estimands and decision rules (Section 8), with ethical safeguards (Section 9): puppets never post, interact or open flagged items. Because the materiality audit is infeasible for a small lab, the protocol commits only to a pilot and a registered feasibility gate, which under illustrative caps selects the detection of amplification for a single persona.

What audits can and cannot show. An audit with puppets of fixed demand identifies supply effects at the personas it audits. It says nothing about how the algorithm shapes demand: without outside information on how demand forms, the identified set for the algorithm's total effect on real users is as wide as the range of exposures under the reference policy (Proposition 3.4). An audit can establish that, for a stated service, surface, period and persona, the algorithm supplies more, or less, antisemitic content than a stated alternative would supply to the same demand. It cannot establish that the algorithm made anyone want such content. Stating this limit is part of the method, not a disclaimer added to it.

Related work and the series. Section 2 reviews published audits of recommender systems (Guess et al., 2023; Haroon et al., 2023; Hosseinmardi et al., 2024; Huszár et al., 2022; Ribeiro et al., 2020; Sandvig et al., 2014), the supply-and-demand debate (Munger and Phillips, 2022), feedback loops (Chaney et al., 2018; Jiang et al., 2019), the regulatory setting, and classifier-based measurement. Our estimands build on mediation analysis (Pearl, 2001; Robins and Greenland, 1992; VanderWeele, 2014) and on the decomposition of Kitagawa (1955). This is the fourth paper in a series. The first estimates how many antisemitic hate crimes go unrecorded (Kornstein Lab for Antisemitism Research, 2026c), the second models surges in incidents as self-exciting processes (Kornstein Lab for Antisemitism Research, 2026b), and the third maps antisemitic attitudes (Kornstein Lab for Antisemitism Research, 2026a). The first three study what is recorded and what is believed; this one studies what is shown.

Roadmap. Section 3 defines the estimands and the decomposition, Section 4 develops the Markov model, Section 5 gives identification under interference and drift, and Section 6 the misclassification correction, the sequential procedure and the power calculation. Section 7 reports the simulation study, Section 8 summarises the registered protocol, Section 9 sets out the ethical and legal conditions, and Section 10 discusses limitations and extensions. Proofs are in Appendices A and B, simulation details in Appendix C, and the full registered protocol in Appendix D.

2 Background: amplification, audits and platform governance

Every finding below is reported as its authors state it, with the scope they give it, and none is an input to the simulation study. Almost none of the studies measures antisemitic content as such, so they bear on our question by analogy.

2.1 Algorithm audits and what they have found

Sandvig et al. (2014) set out the designs available to outsiders, among them *sock-puppet* audits, in which researchers create accounts that behave in controlled ways and record what the system shows them; it is the design closest to an experiment (Section 5). Table 2 summarises the studies we draw on. Non-personalised crawls of one video service in 2019 reached different conclusions about radicalisation pathways from data of a similar kind (Ledwich and Zaitsev, 2020; Ribeiro et al., 2020). Audits with trained accounts report congenial, extreme or "problematic" recommendations for some profiles (Haroon et al., 2023; Whittaker et al., 2021), and an NGO report, one of the few audits that targets antisemitic content directly, found

Table 2. Published studies of recommendation and exposure discussed in Section 2.1, as reported by their authors. Entries summarise design and scope only; see the text for each study's own statement of its findings. None measures antisemitic content with a validated classifier, and only ADL and Tech Transparency Project (2023) targets antisemitic content.

Study	Service(s)	Design and comparison	Content measure
Ribeiro et al. (2020)	YouTube	Non-personalised recommendation crawl (May–Jul 2019); comment trajectories	Channel categories (Media, Alt-lite, I.D.W., Alt-right)
Ledwich and Zaitsev (2020)	YouTube	Logged-out recommendation crawl	Channel categories
Whittaker et al. (2021)	YouTube, Reddit, Gab	Experimental accounts with personalisation	Extreme and fringe content
Haroon et al. (2023)	YouTube	100,000 trained sock puppets; up-next trails	Ideology; "problematic" channel list
ADL and Tech Transparency Project (2023)	Four platforms	Six test personas; no reference policy (NGO report)	Antisemitic and extremist content
Hosseinmardi et al. (2024)	YouTube	Counterfactual bots primed with real histories	Partisanship of content
Hosseinmardi et al. (2021)	YouTube	Real-user browsing panel, 2016–2019	Channel categories (incl. far right)
Chen et al. (2023)	YouTube	Survey plus browser data, n = 1,181, 2020	Alternative and extremist channels
Robertson et al. (2023)	Google Search	Exposure versus engagement of participants	Partisan and unreliable news sources
Huszár et al. (2022)	Twitter	Randomised chronological holdout	Reach of legislators and news sources
Guess et al. (2023)	Facebook, Instagram	Randomised assignment to chronological feed, 3 months, 2020	Political, untrustworthy, uncivil and slur content (classifiers)

antisemitic recommendations to six test personas on three of four platforms, without a reference policy against which to measure how much more often they occurred (ADL and Tech Transparency Project, 2023). Counterfactual bots that follow rule-based trajectories after replaying real users' histories led Hosseinmardi et al. (2024) to conclude that, at least since the service's 2019 changes, recommendations play, if anything, a moderating role. Real-user studies report little evidence that the service drives users to more radical content (Hosseinmardi et al., 2021), and exposure to extremist channels concentrated among a small group of subscribers with high prior resentment (Chen et al., 2023); on a search service engagement with unreliable sources exceeded exposure to them (Robertson et al., 2023). Two studies randomised the feed itself. Huszár et al. (2022) report, from a long-running holdout with a reverse-chronological timeline, that algorithmic amplification of legislators' tweets favoured the mainstream right in six of seven countries. Guess et al. (2023) report that a chronological feed for three months during the 2020 United States election changed the mix of content users saw in both directions, including, on one platform, less content classified as uncivil or containing slurs; two later errata qualify some estimates for one platform, among them the classifier-based proportions of uncivil content and slurs, so classifier coverage was part of the result.

What the literature does and does not establish. These results are less in conflict with one another than answers to different questions. They differ in the estimand (what is recommended, consumed or believed),

in the comparison (none, a logged-out crawl, counterfactual bots or a randomised chronological feed), in the content category and in the period. A single audit cannot settle whether "recommender systems amplify antisemitism"; it can estimate a well-defined contrast for a stated service, surface, period and demand profile, which Section 3 defines. Audits also observe a system that learns from the behaviour it induces: training on data generated under earlier recommendations homogenises behaviour (Chaney et al., 2018), and a recommender and its users' interests can reinforce one another (Jiang et al., 2019). Exposure therefore depends on a persona's history, which is why Section 4 models sessions as Markov chains, and puppets that feed one learning system can change what it shows to one another (Proposition 5.2).

2.2 Methodological debates

Real-user data show what people saw and chose, but cannot by themselves separate the algorithm's contribution from users' intentions; sock puppets supply the counterfactual by fixing behaviour, at the price of realism (Hosseinmardi et al., 2024). Our estimands embrace this: amplification is defined *for a fixed demand profile*, a statement about supply conditional on demand. A logged-out crawl (Ledwich and Zaitsev, 2020; Ribeiro et al., 2020) is, in our framework, not a flawed personalised audit but a possible *reference policy*, whose contrast with a trained puppet's feed isolates profile-based personalisation. The supply-and-demand framework of Munger and Phillips (2022), and the real-user findings that consumption tracks prior preferences, concern which side of the market accounts for observed consumption. They are compatible with an algorithm that supplies more antisemitic content to some demand profiles than a reference would, and with one that supplies less. Theorem 3.3 turns the dichotomy into a set of contrasts, and Proposition 3.4 states which an audit can identify.

2.3 Platform governance and research access

The Digital Services Act (DSA), Regulation (EU) 2022/2065, requires providers of very large online platforms to assess systemic risks, taking into account the design of their recommender systems (Article 34), to mitigate them (Article 35), and to offer for each recommender system at least one option not based on profiling (Article 38) (European Parliament and Council of the European Union, 2022). Such an option is a candidate reference policy that the service itself supplies. Article 40 opens two routes to data: vetted access through a Digital Services Coordinator (Article 40(4)), with procedures set by Delegated Regulation (EU) 2025/2050 and applications open from 29 October 2025 (European Commission, 2025a,c), and access to publicly accessible data (Article 40(12)); by 19 May 2026 coordinators had received 49 applications (European Commission, 2026b), so the route is operational in law but its practical reach is still being established. In its first non-compliance decision under the DSA, in December 2025, the Commission found among other things that the terms of X prohibited eligible researchers from independently accessing its public data, including by scraping (European Commission, 2025b), and in July 2026 it accepted the provider's action plan, which includes amending those terms (European Commission, 2026a). Transparency reports under Article 42 (European Commission, 2024) report moderation and risk assessment at the level of the service, not exposure by demand profile under alternative policies, which is what an amplification audit estimates.

2.4 Classifier-based measurement of antisemitic content

Every estimand is defined by the true label L_c, but an audit observes a classifier's label $\widehat{L}_c$, whose evaluation is a contextual problem (Kornstein Lab for Antisemitism Research, 2026d,e). Three difficulties matter. Coded terms emerge and evolve (Zannettou et al., 2020), so sensitivity depends on time and content mix.

Detectors confuse quotation and counter-speech with endorsement (Röttger et al., 2021), so a policy that recommends news coverage or counter-speech changes the mix of types and biases the naive amplification ratio even when every type-level error rate is the same in both arms (Theorem 6.1). And ground truth is a coding decision: datasets that keep only items on which annotators fully agreed (Jikeli et al., 2023) lose the borderline items a validation sample must cover, so the protocol adjudicates disagreements and reports scope variants under alternative definitions side by side (International Holocaust Remembrance Alliance, 2016; Jerusalem Declaration on Antisemitism, 2021), as the companion paper does (Kornstein Lab for Antisemitism Research, 2026c).

3 Estimands: what does it mean to amplify?

To say that a recommender *amplifies* antisemitic content is to say that exposure under the algorithm is larger than exposure under something else, for someone. The claim has no truth value until both are named. This section names them: exposure as a potential outcome (Section 3.1), amplification at fixed demand relative to a named reference (Definition 3.1 and Section 3.3), the decomposition of real users' exposure (Theorem 3.3), and what an audit with sock puppets of fixed demand can identify (Proposition 3.4).

The framework is that of potential outcomes (Imbens and Rubin, 2015; Rubin, 1974). The amplification ratio has a precedent in the randomised chronological holdout of Huszár et al. (2022), and feedback between recommendations and preferences has been studied by simulation (Chaney et al., 2018), as a dynamical system (Jiang et al., 2019) and by estimating induced preference shifts (Carroll et al., 2022). The decomposition is a mediation analysis with demand as the mediator (Pearl, 2001; Robins and Greenland, 1992; VanderWeele, 2014; VanderWeele et al., 2014), and its symmetric version is the rate–composition decomposition of Kitagawa (1955), the Shapley value (Shapley, 1953) of a two-player game. What is new is the bilinear representation (3.3), the bounds on the interaction in Theorem 3.3(d), and Proposition 3.4. Proofs are in Appendix A.

3.1 Policies, demand profiles and exposure

An *account* u is shown content in *sessions*. In a session it is exposed to a sequence of items $c_1, \dots, c_T$. Whether "exposed" means an impression or a consumption is fixed by the protocol (Section 8); the theory is the same. Each item c has a true label $L_c \in \{0, 1\}$, with $L_c = 1$ for antisemitic content. Classifier labels and their errors enter in Section 6. The *exposure* of u in a session is the share of antisemitic items,

$$E_u = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{L_{c_t} = 1\} . \tag{3.1}$$

Two rules determine which items an account sees. The *policy* π is the platform's rule for selecting and ordering a slate of items, given the account's history. The *demand profile* $d \in \mathcal{D}$ is the account's rule for acting on what it is shown: which accounts it follows, what it searches for, which items it selects from a slate and how long it dwells on them. For a sock puppet, d is a script fixed by the researcher, and we call such a fixed demand profile a *persona*. For a real user, d summarises preferences and habits at a given time, and it can change. Items are drawn from an *eligible pool* $\mathcal{P}_t$: the items that exist when step t is served, have survived moderation and may be shown to the account. The *algorithmic policy* alg is the platform's production recommender. A *reference policy* ρ is a policy with which it is compared. Three references recur in the audit literature and in this paper:

- *reverse-chronological* (chron): items posted by the accounts that d follows, newest first;

- *random* (rand): items drawn uniformly at random from $\mathcal{P}_t$, or from a declared sub-pool such as the items in the account's language;
- *popularity-only* (pop): items from $\mathcal{P}_t$ ranked by an engagement score that is not personalised.

Let $\Pi = \{\text{alg}\} \cup \Pi_{\text{ref}}$, where Π_{ref} is the set of references in use. Let $E_u(\pi; d)$ be the exposure of account u in a session served by policy π while the account acts according to d. For a puppet with persona d_u, the exposure $E_u(\pi)$ of the notation table is $E_u(\pi; d_u)$. The *mean exposure*

$$\mu_\pi(d) = \mathbb{E}\{E_u(\pi; d)\}, \qquad \pi \in \Pi,\ d \in \mathcal{D}, \tag{3.2}$$

averages over everything that is random given (π, d): the policy's randomisation, the pool, the account's choices and the starting state. Writing μ_π as a function of d alone presumes no interference between accounts and a stable policy, conditions that Section 5 states and relaxes.

3.2 Amplification at fixed demand

Definition 3.1 (Amplification). Fix a reference policy $\rho \in \Pi_{\text{ref}}$. For a persona d with $\mu_\rho(d) > 0$, the *amplification ratio* and the *additive amplification* of antisemitic content by the algorithmic policy, relative to ρ, are

$$\text{AR}_\rho(d) = \frac{\mu_{\text{alg}}(d)}{\mu_\rho(d)}, \qquad \text{AA}_\rho(d) = \mu_{\text{alg}}(d) - \mu_\rho(d).$$

For a probability distribution Λ on $\mathcal{D}$, write $\mu_\pi(\Lambda) = \int \mu_\pi \, d\Lambda$. The *population-weighted* amplification ratio and additive amplification are

$$\text{AR}_\rho(\Lambda) = \frac{\mu_{\text{alg}}(\Lambda)}{\mu_\rho(\Lambda)}, \qquad \text{AA}_\rho(\Lambda) = \mu_{\text{alg}}(\Lambda) - \mu_\rho(\Lambda) = \int \text{AA}_\rho \, d\Lambda.$$

The algorithm *amplifies* relative to ρ (for d, or for Λ) if $\text{AR}_\rho > 1$, equivalently $\text{AA}_\rho > 0$, and *de-amplifies* if $\text{AR}_\rho < 1$.

The distribution Λ is either the persona mix of an audit or the distribution of a target population over $\mathcal{D}$. When a single reference is in play we write ref for it. AA is always defined and is the scale on which exposure burdens add up across users; AR is the natural scale when exposure is rare, and it is undefined when the reference delivers no antisemitic content. The population ratio is a ratio of means, not a mean of ratios.

Lemma 3.2 (Aggregation and change of reference). *Let $\mu_\rho > 0$ on a set of Λ-probability one.*

(a) $\text{AR}_\rho(\Lambda) = \int \text{AR}_\rho \, w_\Lambda \, d\Lambda$, *with weights $w_\Lambda = \mu_\rho / \mu_\rho(\Lambda)$ that integrate to one.*

(b) If $\int \text{AR}_\rho \, d\Lambda < \infty$, *then* $\int \text{AR}_\rho \, d\Lambda - \text{AR}_\rho(\Lambda) = -\operatorname{Cov}_\Lambda(\text{AR}_\rho, \mu_\rho)/\mu_\rho(\Lambda)$.

(c) For two references, $\text{AR}_\rho(d) = \text{AR}_{\rho'}(d)\, \mu_{\rho'}(d)/\mu_\rho(d)$. *For a ladder of policies $\rho_0, \rho_1, \ldots, \rho_k = \text{alg}$ with positive means,* $\text{AR}_{\rho_0}(d) = \prod_{j=1}^{k} \mu_{\rho_j}(d)/\mu_{\rho_{j-1}}(d)$.

Proof. (a) Integrate $\mu_{\text{alg}} = \text{AR}_\rho \, \mu_\rho$ against Λ. (b) Write (a) as $\mathbb{E}_\Lambda[\text{AR}_\rho \, \mu_\rho]/\mathbb{E}_\Lambda[\mu_\rho]$. (c) The products telescope. □

Part (a) says that the population ratio weights each persona's ratio by its exposure under the reference. A persona's contribution to $\text{AR}_\rho(\Lambda)$ is $\Lambda(d)\, \mu_{\text{alg}}(d)/\mu_\rho(\Lambda)$, its share of exposure under the algorithm, so an enormous persona ratio produced by a tiny denominator cannot by itself make $\text{AR}_\rho(\Lambda)$ enormous. Part (b)

says that the unweighted average of persona ratios overstates the population ratio when the personas with the lowest reference exposure have the largest ratios, the pattern to expect if an algorithm leads users to content they would not have found by themselves. A population ratio is therefore meaningful only for a declared Λ, and an average of persona ratios is never a population quantity. The protocol of Section 8 declares no population weighting: an audit supplies no weights over demand profiles, so it reports persona-level estimands only.

3.3 The reference policy is part of the estimand

By Lemma 3.2(c), amplification relative to ρ and relative to ρ' differ by the factor $\mu_{\rho'}(d)/\mu_{\rho}(d)$, which does not involve the algorithm, so the same algorithm can amplify relative to one reference and de-amplify relative to another. The choice of reference is a choice of question.

- $\mathrm{AR}_{\mathrm{chron}}$ asks how ranking, including items from accounts the persona does not follow, changes exposure relative to its own subscriptions. If d follows no source of antisemitic content, the ratio is undefined and $\mathrm{AA}_{\mathrm{chron}}(d) = \mu_{\mathrm{alg}}(d)$.
- $\mathrm{AR}_{\mathrm{rand}}$ asks how selective the algorithm is relative to the base rate in the eligible pool; with exposure counted in impressions it is a selection ratio.
- $\mathrm{AR}_{\mathrm{pop}}$ asks what personalisation adds to a ranking by global engagement.

Applied to the ladder rand $\to$ pop $\to$ alg, Lemma 3.2(c) gives $\mathrm{AR}_{\mathrm{rand}}(d) = \{\mu_{\mathrm{pop}}(d)/\mu_{\mathrm{rand}}(d)\}\,\mathrm{AR}_{\mathrm{pop}}(d)$: selection by popularity times personalisation. A ladder of references thus decomposes amplification by mechanism, exactly and multiplicatively. Three further choices belong to the estimand. *The pool*: every policy draws from the same pool $\mathcal{P}_t$, so moderation that removes items before ranking is not amplification, and producers who respond to what the algorithm rewards, central to the account of Munger and Phillips (2022), change the content supply, which lies outside every estimand here. *The normalisation*: E_u is a share, so if the algorithm also lengthens sessions, total exposure changes by the ratio $\bar{T}_{\mathrm{alg}}/\bar{T}_{\mathrm{ref}}$ of mean session lengths in addition to AR, which is reported separately. *Implementability*: a random draw from the eligible pool requires platform-side access, and a reference that cannot be run is identified only under a model.

3.4 Dynamic demand: supply, demand shaping and feedback

A puppet's demand is fixed by design. A real user's is not: what a user is shown today can change what the user seeks tomorrow (Carroll et al., 2022; Jiang et al., 2019), and a recommender trained on behaviour it has shaped feeds the change back (Chaney et al., 2018).

We formalise this with a *policy schedule*. A formation period precedes a shorter evaluation window. Policy b serves the user during the formation period, and policy a during the evaluation window. Let $E_u(a; b)$ be the user's exposure in the evaluation window, with $E_u(a) = E_u(a; a)$. For a population $\mathcal{U}$ of real users and a user U drawn from it, with population weights if needed, write

$$\bar{\mu}(a; b) = \mathbb{E}\{E_U(a; b)\}, \qquad a, b \in \{\mathrm{alg}, \mathrm{ref}\}.$$

Each of the four cells is the outcome of a feasible intervention on the platform's policy schedule, not a cross-world counterfactual. The diagonal cells are the worlds run by one policy throughout; the off-diagonal cells switch the policy at the start of the evaluation window, so that $\bar{\mu}(\mathrm{ref}; \mathrm{alg})$ is what the reference would deliver to users whose demand the algorithm has formed. Let $D_U(b) \in \mathcal{D}$ be the user's demand profile at the start of the evaluation window after formation under b, and let F_b be its distribution over the population.

Assumption A1 (Demand sufficiency). For $a, b \in \{\text{alg}, \text{ref}\}$, $\mathbb{E}\{E_U(a;b) \mid D_U(b) = d\} = \mu_a(d)$ for F_b-almost every d, where μ_a is the persona-level mean exposure (3.2).

Assumption A1 presumes that demand does not change materially during the short evaluation window, and it has two parts. First, the formation regime affects exposure only through the demand profile; this fails if the policy carries per-user state, such as a learned embedding, beyond what d records, which then becomes a third factor in the game below. Second, a user with demand profile d is, as far as the policy is concerned, exchangeable with a puppet of persona d: the transport assumption behind every sock-puppet audit (Assumption A7). Under Assumption A1, iterated expectations give

$$\bar{\mu}(a;b) = \int \mu_a \, dF_b = \mu_a(F_b), \tag{3.3}$$

which is bilinear in the *supply response* μ_a and the *demand distribution* F_b.

Define the total algorithmic effect and its components:

$$\begin{aligned} \tau &= \bar{\mu}(\text{alg};\text{alg}) - \bar{\mu}(\text{ref};\text{ref}), & \tau_{\text{sup}} &= \bar{\mu}(\text{alg};\text{ref}) - \bar{\mu}(\text{ref};\text{ref}), \\ \iota &= \bar{\mu}(\text{alg};\text{alg}) - \bar{\mu}(\text{alg};\text{ref}) - \bar{\mu}(\text{ref};\text{alg}) + \bar{\mu}(\text{ref};\text{ref}), & \tau_{\text{dem}} &= \bar{\mu}(\text{ref};\text{alg}) - \bar{\mu}(\text{ref};\text{ref}). \end{aligned} \tag{3.4}$$

The *supply effect* τ_{sup} is the change in exposure from switching the policy while demand stays as the reference formed it. The *demand-shaping effect* τ_{dem} is the change caused by the algorithm's effect on demand, seen through the reference. The *feedback interaction* ι is the rest. On the ratio scale, write $\text{TR} = \bar{\mu}(\text{alg};\text{alg})/\bar{\mu}(\text{ref};\text{ref})$ for the total ratio, $\text{AR}(F_b) = \bar{\mu}(\text{alg};b)/\bar{\mu}(\text{ref};b)$ for the amplification ratio at the demand formed under b, and $\text{DR}(a) = \bar{\mu}(a;\text{alg})/\bar{\mu}(a;\text{ref})$ for the demand ratio under policy a. Under Assumption A1, $\text{AR}(F_b) = \text{AR}_{\text{ref}}(F_b)$ in the sense of Definition 3.1.

The attribution problem is a two-player game. The players are *supply* and *demand*. For a coalition $J \subseteq \{\text{sup}, \text{dem}\}$ let $a_J = \text{alg}$ if $\text{sup} \in J$ and $a_J = \text{ref}$ otherwise, define b_J in the same way from dem, and let $v(J) = \bar{\mu}(a_J; b_J) - \bar{\mu}(\text{ref};\text{ref})$. Let $\text{osc}_S f = \sup_S f - \inf_S f$, let d_{TV} be the total-variation distance, and let d_{W} be the Wasserstein-1 distance.

Theorem 3.3 (Supply/demand decomposition).

(a) Two paths. *Without further assumptions,* $\bar{\mu}(\text{alg};\text{alg}) = \bar{\mu}(\text{ref};\text{ref}) + \tau_{\text{sup}} + \tau_{\text{dem}} + \iota$, *and*

$$\tau = \tau_{\text{sup}} + (\tau_{\text{dem}} + \iota) = (\tau_{\text{sup}} + \iota) + \tau_{\text{dem}}.$$

The supply contribution is τ_{sup} *if the policy is switched first, at demand formed by the reference, and* $\tau_{\text{sup}} + \iota$ *if demand is switched first, so that the policy acts on demand formed by the algorithm. The two path attributions differ by* ι, *and so do the two demand contributions.*

(b) Shapley attribution. *The Shapley value of* v *is*

$$\phi_{\text{sup}} = \tau_{\text{sup}} + \tfrac{1}{2}\iota, \qquad \phi_{\text{dem}} = \tau_{\text{dem}} + \tfrac{1}{2}\iota, \qquad \phi_{\text{sup}} + \phi_{\text{dem}} = \tau.$$

It is the average of the two path attributions in (a), and it is the only allocation that is efficient, symmetric and additive and gives nothing to a null player. If all four cells are positive, the Shapley value of the game $\log \bar{\mu}(a_J; b_J) - \log \bar{\mu}(\text{ref};\text{ref})$ *gives the exact factorisation* $\text{TR} = \Phi_{\text{sup}}\Phi_{\text{dem}}$, *with*

$$\Phi_{\text{sup}} = \{\text{AR}(F_{\text{ref}})\,\text{AR}(F_{\text{alg}})\}^{1/2}, \qquad \Phi_{\text{dem}} = \{\text{DR}(\text{ref})\,\text{DR}(\text{alg})\}^{1/2},$$

and multiplicative interaction $\iota^{\times} = \text{AR}(F_{\text{alg}})/\text{AR}(F_{\text{ref}}) = \text{DR}(\text{alg})/\text{DR}(\text{ref})$.

(c) Bilinear form. *Under Assumption A1, let* $\Delta\mu = \mu_{\text{alg}} - \mu_{\text{ref}}$, $\mu_{\text{mid}} = \frac{1}{2}(\mu_{\text{alg}} + \mu_{\text{ref}})$ *and* $F_{\text{mid}} = \frac{1}{2}(F_{\text{alg}} + F_{\text{ref}})$. *Then*

$$\tau_{\text{sup}} = \text{AA}_{\text{ref}}(F_{\text{ref}}), \qquad \tau_{\text{sup}} + \iota = \text{AA}_{\text{ref}}(F_{\text{alg}}), \qquad \phi_{\text{sup}} = \text{AA}_{\text{ref}}(F_{\text{mid}}),$$
$$\tau_{\text{dem}} = \int \mu_{\text{ref}}\, d(F_{\text{alg}} - F_{\text{ref}}), \qquad \iota = \int \Delta\mu\, d(F_{\text{alg}} - F_{\text{ref}}), \quad \phi_{\text{dem}} = \int \mu_{\text{mid}}\, d(F_{\text{alg}} - F_{\text{ref}}).$$

(d) When the interaction vanishes. *Under Assumption A1,* $\iota = 0$ *if and only if* $\text{AA}_{\text{ref}}(F_{\text{alg}}) = \text{AA}_{\text{ref}}(F_{\text{ref}})$, *and* $\iota^{\times} = 1$ *if and only if* $\text{AR}_{\text{ref}}(F_{\text{alg}}) = \text{AR}_{\text{ref}}(F_{\text{ref}})$. *In particular,* $\iota = 0$ *in either of two cases.*

(i) No demand shaping*:* $F_{\text{alg}} = F_{\text{ref}} = F$. *Then also* $\tau_{\text{dem}} = 0$, *the two paths in (a) coincide, and observed exposure splits uniquely into a demand-driven part and an algorithm-driven part,* $\bar{\mu}(\text{alg}; \text{alg}) = \mu_{\text{ref}}(F) + \text{AA}_{\text{ref}}(F)$.

(ii) Homogeneous supply effect*:* $\Delta\mu$ *is constant on a set* S *with* $F_{\text{alg}}(S) = F_{\text{ref}}(S) = 1$.

In general, for any such S,

$$|\iota| \le \text{osc}_S(\Delta\mu)\, d_{\text{TV}}(F_{\text{alg}}, F_{\text{ref}}), \qquad |\tau_{\text{dem}}| \le \text{osc}_S(\mu_{\text{ref}})\, d_{\text{TV}}(F_{\text{alg}}, F_{\text{ref}}).$$

If $\mathcal{D}$ *is a metric space and* $\Delta\mu$ *and* μ_{ref} *are Lipschitz with constants* L_Δ *and* L_{ref}, *then* $|\iota| \le L_\Delta d_{\text{W}}(F_{\text{alg}}, F_{\text{ref}})$ *and* $|\tau_{\text{dem}}| \le L_{\text{ref}} d_{\text{W}}(F_{\text{alg}}, F_{\text{ref}})$.

Part (a) is why decompositions of observed exposure are path dependent. Taking users as they are and asking what the reference would deliver to them (the demand-first path) assigns the whole interaction to supply; switching the policy first assigns it to demand. Neither is wrong: they answer different counterfactual questions and disagree by exactly ι. Part (b) resolves the ambiguity by symmetry. By part (c), the Shapley supply share is the additive amplification at the average of the two demand distributions, Kitagawa's (1955) rate component with μ_a as the rates and F_b as the composition. The Shapley value is a convention with axiomatic support, not an identification result: when exposure and mediator interact, no allocation of ι is "the true one" (Robins and Greenland, 1992), so we report τ_{sup}, τ_{dem} and ι alongside it. Part (d) bounds the interaction by the product of the heterogeneity of the supply effect and the distance by which the algorithm has moved demand. It is of second order when both are small, and large when the algorithm amplifies most for the kinds of demand it has itself produced, the signature of a feedback loop.

In the language of mediation analysis, τ_{sup} and $\tau_{\text{sup}} + \iota$ are the pure and total direct effects, τ_{dem} and $\tau_{\text{dem}} + \iota$ the pure and total indirect effects (Pearl, 2001; Robins and Greenland, 1992), and ι the mediated interaction (VanderWeele, 2014); unlike the interventional analogues of VanderWeele et al. (2014), every cell of our schedule can be produced by an experiment.

3.5 What a fixed-demand audit identifies

An audit whose puppets never update their demand observes, at best, the persona-level means. Suppose the conditions of Section 5 hold, so that the audit identifies $\mu_\pi(d)$ for every $\pi \in \Pi$ and every persona d in its design $\mathcal{D}_A \subseteq \mathcal{D}$.

Proposition 3.4 (Scope of fixed-demand audits). *Suppose that a fixed-demand audit identifies* $\mu_\pi(d)$ *for* $\pi \in \Pi$ *and* $d \in \mathcal{D}_A$, *and that Assumption A1 holds.*

(a) The audit identifies $\text{AR}_\rho(\Lambda)$ *and* $\text{AA}_\rho(\Lambda)$ *for every reference* ρ *and every* Λ *concentrated on* $\mathcal{D}_A$. *If* F_{ref}, F_{alg} *or* F_{mid} *is known and concentrated on* $\mathcal{D}_A$, *then* $\Lambda = F_{\text{ref}}$ *identifies* τ_{sup}, $\Lambda = F_{\text{alg}}$

identifies $\tau_{\text{sup}} + \iota$*, and* $\Lambda = F_{\text{mid}}$ *identifies* ϕ_{sup}*.*

(b) The law of the audit data does not depend on how demand forms, so the audit data do not identify τ_{dem}*,* ι*,* τ *or* ϕ_{dem}*. Even with oracle knowledge of* μ_{alg} *and* μ_{ref} *on all of* $\mathcal{D}$ *and of* F_{alg}*, the identified set for* τ *is an interval with endpoints* $\bar{\mu}(\text{alg};\text{alg}) - \sup_{\mathcal{D}} \mu_{\text{ref}}$ *and* $\bar{\mu}(\text{alg};\text{alg}) - \inf_{\mathcal{D}} \mu_{\text{ref}}$*, of length* $\operatorname{osc}_{\mathcal{D}}(\mu_{\text{ref}})$*. The same holds for* τ_{dem}*, with* $\mu_{\text{ref}}(F_{\text{alg}})$ *in place of* $\bar{\mu}(\text{alg};\text{alg})$*. With that oracle knowledge,* τ *and* τ_{dem} *are identified if and only if* μ_{ref} *is constant on* $\mathcal{D}$*,* ι *if and only if* $\Delta\mu$ *is, and* ϕ_{dem} *if and only if* μ_{mid} *is. An audit knows these functions only on* $\mathcal{D}_A$ *and does not know* F_{alg}*, so its identified sets are wider still.*

(c) Sensitivity. *$\tau - \text{AA}_{\text{ref}}(F_{\text{alg}}) = \tau_{\text{dem}}$ and $\phi_{\text{sup}} - \text{AA}_{\text{ref}}(F_{\text{alg}}) = -\iota/2$. If $\mathcal{D}$ is a metric space, $d_{\text{W}}(F_{\text{alg}}, F_{\text{ref}}) \le \eta$, and μ_{ref} and $\Delta\mu$ are Lipschitz with constants L_{ref} and L_Δ, then $|\tau - \text{AA}_{\text{ref}}(F_{\text{alg}})| \le L_{\text{ref}}\eta$ and $|\phi_{\text{sup}} - \text{AA}_{\text{ref}}(F_{\text{alg}})| \le L_\Delta\eta/2$.*

A fixed-demand audit therefore identifies supply-side effects at the personas it audits, for the population fixed by Λ. Speaking for present-day users requires personas that represent F_{alg}; speaking for a world without the algorithm requires F_{ref}, which no audit of the present platform observes. Part (b) is the best case: even with every persona-level mean and present-day demand known, information on demand formation is needed to narrow the total effect beyond the range of reference exposures. Part (c) replaces that information with a sensitivity parameter η, the distance by which the algorithm has moved demand, which must come from outside the audit.

Four designs reach further, each at a cost. A long-running randomised holdout (Huszár et al., 2022) identifies τ for real users; a randomised switch to the reference (Guess et al., 2023) starts at the cell $\bar{\mu}(\text{ref};\text{alg})$, and the full 2×2 schedule needs switches in both directions, which only a platform can run. Replay designs (Hosseinmardi et al., 2024), like fixed-demand audits, say nothing by themselves about F_{ref}, and adaptive personas identify the full table only for synthetic users whose scripted dynamics no audit can validate against people.

Persona contrasts. With point masses at a baseline persona d_0 and a persona d in place of F_{ref} and F_{alg}, parts (a)–(c) of Theorem 3.3 split $\mu_{\text{alg}}(d) - \mu_{\text{ref}}(d_0)$ into a supply contrast $\text{AA}_{\text{ref}}(d_0)$, a demand contrast $\mu_{\text{ref}}(d) - \mu_{\text{ref}}(d_0)$ and an interaction $\text{AA}_{\text{ref}}(d) - \text{AA}_{\text{ref}}(d_0)$, with the same Shapley shares. This persona-contrast decomposition is identified by a fixed-demand audit, and it is the version registered for the four-persona rung of the protocol. Its demand contrast compares personas chosen by the researcher, not demand formed by the algorithm.

4 A Markov model of exposure

Definition 3.1 is model-free. A model is needed to extrapolate from the audited session length to other horizons and the long run, to compare policies whose sessions start differently, and to understand why a small change in a recommender can produce a large change in exposure. We model a session as a Markov chain on content states, with the classical tools of finite chains: the fundamental matrix (Kemeny and Snell, 1960), the group inverse (Meyer, 1975) and perturbation theory (Cho and Meyer, 2000, 2001; Meyer, 1980; Schweitzer, 1968; Seneta, 1988). Proposition 4.1 collects the exposure functionals and Theorem 4.2 bounds their sensitivity. Parts (a) and (b) of Theorem 4.2 restate classical results; the closed form in part (c) is personalised PageRank with teleportation to a single state (Jeh and Widom, 2003; Page et al., 1999). We have not found the hitting-time form of the derivative in part (c), or parts (d) and (e), stated in this form, although each follows in a few lines from the classical theory.

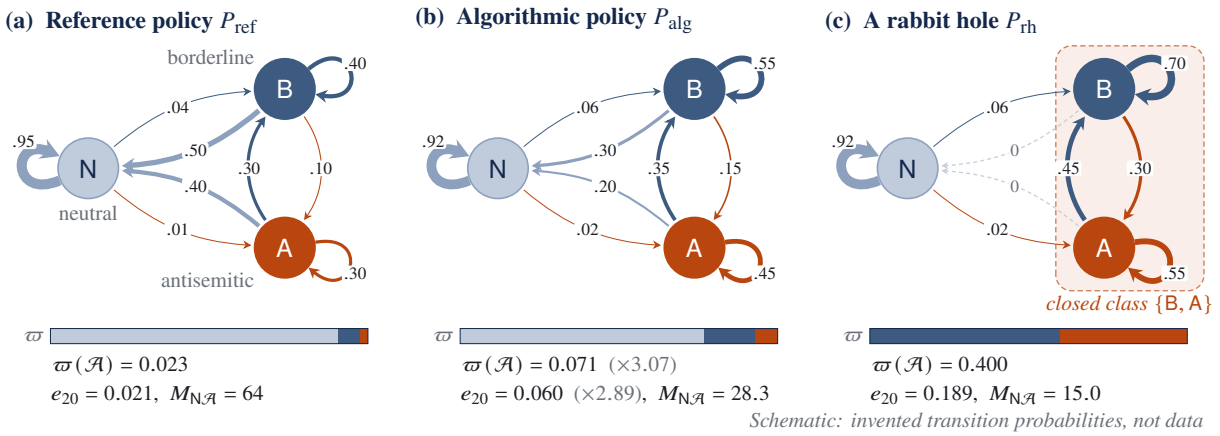


Figure 2. A session as a Markov chain over content states, under two policies. Each node is a content state, neutral (N), borderline (B) or antisemitic (A); each arrow is the probability that an item in one state is followed by an item in another, and its width is proportional to that probability (zero transitions are dashed). Arrows are coloured by the state they lead to. *(a)* A reference policy. *(b)* The algorithmic policy moves at most 0.2 of each row's probability, yet long-run antisemitic exposure $\varpi(\mathcal{A})$ triples: the extra mass sent from N is small, but borderline and antisemitic content now lead back to neutral content less often, so sessions that reach them stay longer (Theorem 4.2). *(c)* A rabbit-hole variant: borderline and antisemitic content never lead back to neutral content, so {B, A} is a closed class, a rabbit hole. A session that starts in N enters it after 12.5 items on average and never leaves, and long-run exposure is $\varpi(\mathcal{A}) = 0.4$ whatever the entry probability (Proposition 4.1(d)). Bars show the stationary distribution ϖ; e_{20} is the expected share of antisemitic items in a 20-item session that starts in N, and $M_{\mathsf{N}\mathcal{A}}$ the expected number of further items before the first antisemitic one; ratios in (b) are relative to (a). **Schematic:** the transition matrices are the invented ones of Section 4.5, and the summaries are exact computations from them (Table 3). They describe no platform.

4.1 Sessions as Markov chains

Let $\mathcal{X}$ be a finite set of content states, partitioned into neutral states $\mathcal{X}_N$, borderline or adjacent states $\mathcal{X}_B$, and antisemitic states $\mathcal{A}$. The coarsest useful choice has one state of each kind, and finer choices cross severity with topic. Only the target set $\mathcal{A}$ matters for the theory. A session is a sequence of states $X_1, \dots, X_T$, the states of the items $c_1, \dots, c_T$, so that $\mathbf{1}\{L_{c_t} = 1\} = \mathbf{1}\{X_t \in \mathcal{A}\}$.

> **Assumption A2** (Markov sessions). Given the policy π and the persona d, the states of a session form a time-homogeneous Markov chain with initial distribution $\nu_\pi(d)$ and transition matrix $P_\pi(d)$, where $P_\pi(d)_{xy}$ is the probability that an item in state x is followed by an item in state y.

The transition matrix combines supply and demand: the policy determines what is offered after an item in state x, and the persona what is taken. We write P_{alg} and P_{ref} when comparing policies at the same persona. Assumption A2 is an approximation, which can be made to hold more nearly by enlarging the state, and whose order can be tested by likelihood-ratio tests (Anderson and Goodman, 1957). The expected exposure (3.1) of a T-item session is

$$e_T(P, \nu) = \mathbb{E}\Big\{\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{X_t \in \mathcal{A}\}\Big\} = \frac{1}{T}\sum_{t=0}^{T-1} \nu^\top P^t \mathbf{1}_{\mathcal{A}}, \tag{4.1}$$

and $\mu_\pi(d) = e_T\{P_\pi(d), \nu_\pi(d)\}$.

For a stochastic matrix P with stationary distribution ϖ (a column vector), let

$$Z = (I - P + \mathbf{1}\varpi^\top)^{-1}, \qquad H = Z - \mathbf{1}\varpi^\top, \qquad \boldsymbol{g}_{\mathcal{A}} = H\mathbf{1}_{\mathcal{A}}.$$

Z is the fundamental matrix of Kemeny and Snell (1960), and H is the group inverse $(I-P)^{\#}$ of Meyer (1975), also called the deviation matrix. We call $\boldsymbol{g}_{\mathcal{A}}$ the *exposure potential*. It solves the Poisson equation $(I-P)\boldsymbol{g}_{\mathcal{A}} = \mathbf{1}_{\mathcal{A}} - \varpi(\mathcal{A})\mathbf{1}$, and for aperiodic P, $g_{\mathcal{A}}(y) = \sum_{t\geq 0}\{\mathbb{P}_y(X_{1+t}\in\mathcal{A}) - \varpi(\mathcal{A})\}$ is the excess antisemitic exposure, summed over the rest of the session, of a session that passes through state y. For a set $\mathcal{S}\subseteq\mathcal{X}$ and a chain started at $X_1 = x$, let $T_{\mathcal{S}} = \inf\{n\geq 0 : X_{1+n}\in\mathcal{S}\}$ be the number of further items until the session first reaches $\mathcal{S}$, let $M_{x\mathcal{S}} = \mathbb{E}_x T_{\mathcal{S}}$, and write $M_{xy} = M_{x\{y\}}$, so that $M_{yy} = 0$. Finally, $\|B\|_\infty = \max_x \sum_y |B_{xy}|$ is the maximum absolute row sum, $\operatorname{erg}(B) = \frac{1}{2}\max_{x,y}\sum_z |B_{xz} - B_{yz}|$ is the ergodicity coefficient of Seneta (2006), and $\operatorname{osc}(\boldsymbol{v}) = \max_x v_x - \min_x v_x$.

4.2 Exposure functionals

A stochastic matrix is *unichain* if it has exactly one closed communicating class. It then has a unique stationary distribution, and states outside the class are transient.

Proposition 4.1 (Exposure functionals). *Let P be a stochastic matrix. In parts (a)–(c), P is unichain with stationary distribution ϖ, and Z, H and $\boldsymbol{g}_{\mathcal{A}}$ are as above. Part (d) makes no such assumption.*

(a) Long run. *Z exists, H is the group inverse of $I-P$, and $\varpi^\top = \mathbf{1}^\top(I - P + \mathbf{1}\mathbf{1}^\top)^{-1}$. For every initial distribution, the session exposure $E = T^{-1}\sum_{t=1}^{T}\mathbf{1}\{X_t\in\mathcal{A}\}$ converges to $\varpi(\mathcal{A})$ almost surely, and $\sqrt{T}\{E - \varpi(\mathcal{A})\}$ converges in distribution to $\mathrm{N}(0,\sigma^2_{\mathcal{A}})$, where, with $\tilde{\boldsymbol{f}} = \mathbf{1}_{\mathcal{A}} - \varpi(\mathcal{A})\mathbf{1}$,*

$$\sigma^2_{\mathcal{A}} = 2\sum_x \varpi_x \tilde{f}_x (H\tilde{\boldsymbol{f}})_x - \sum_x \varpi_x \tilde{f}_x^2 = \sum_x \varpi_x \operatorname{Var}_{y\sim P(x,\cdot)}\{g_{\mathcal{A}}(y)\}.$$

(b) Finite horizon. *For every ν and $T\geq 1$,*

$$e_T(P,\nu) = \varpi(\mathcal{A}) + \frac{1}{T}\nu^\top(I-P^T)\boldsymbol{g}_{\mathcal{A}} = \varpi(\mathcal{A}) + \frac{1}{T}\nu^\top(I-P^T)Z\mathbf{1}_{\mathcal{A}}.$$

Hence $e_T(P,\nu) = \varpi(\mathcal{A}) + b(\nu)/T + r_T/T$, where $b(\nu) = \nu^\top\boldsymbol{g}_{\mathcal{A}} = (\nu-\varpi)^\top Z\mathbf{1}_{\mathcal{A}}$ and $|r_T|\leq \frac{1}{2}\operatorname{osc}(\boldsymbol{g}_{\mathcal{A}})\,\|\nu^\top P^T - \varpi^\top\|_1$, which tends to zero if P is aperiodic.

(c) Hitting. *Let $\bar{\mathcal{A}} = \mathcal{X}\setminus\mathcal{A}$. The probability that a T-item session contains no antisemitic item is $\nu_{\bar{\mathcal{A}}}^\top P_{\bar{\mathcal{A}}\bar{\mathcal{A}}}^{T-1}\mathbf{1}$. If $\mathcal{A}$ is accessible from every state, then $I - P_{\bar{\mathcal{A}}\bar{\mathcal{A}}}$ is invertible and $M_{x\mathcal{A}} = \{(I-P_{\bar{\mathcal{A}}\bar{\mathcal{A}}})^{-1}\mathbf{1}\}_x$ for $x\in\bar{\mathcal{A}}$. If P is irreducible, then $M_{xy} = (H_{yy} - H_{xy})/\varpi_y$ for all x, y, and*

$$K(P) = \sum_y \varpi_y M_{xy} = \operatorname{tr} H = \sum_{k=2}^{|\mathcal{X}|}\frac{1}{1-\lambda_k}$$

does not depend on x. Here $\lambda_2,\dots,\lambda_{|\mathcal{X}|}$ are the eigenvalues of P other than the simple eigenvalue 1, counted with algebraic multiplicity.

(d) Rabbit holes. *Let P be any stochastic matrix, with closed classes $C_1,\dots,C_r$ ($r\geq 1$) and transient set $\mathcal{T}$, and let $\varpi^{(k)}$ be the stationary distribution of P restricted to C_k. Then $Z_{\mathcal{T}} = (I - P_{\mathcal{T}\mathcal{T}})^{-1}$ exists. From $x\in\mathcal{T}$ the session enters C_k with probability $B_{xk} = (Z_{\mathcal{T}}P_{\mathcal{T}C_k}\mathbf{1})_x$, after an expected $(Z_{\mathcal{T}}\mathbf{1})_x$ items, and for every ν,*

$$\lim_{T\to\infty} e_T(P,\nu) = \sum_{k=1}^{r}\left\{\nu(C_k) + \nu_{\mathcal{T}}^\top Z_{\mathcal{T}}P_{\mathcal{T}C_k}\mathbf{1}\right\}\varpi^{(k)}(\mathcal{A}).$$

Parts (a), (c) and (d) are classical (Kemeny and Snell, 1960, Chs. 3–4), with the group-inverse forms of Meyer (1975); $\sigma^2_{\mathcal{A}}$ is a quadratic form of the limiting covariance of occupation counts (Kemeny and Snell, 1960, Thm. 4.6.1). $K(P)$ is Kemeny's constant, the expected number of items needed to reach a state drawn from ϖ, from any starting state. Part (b) is exact for every T: the $1/T$ term $b(\nu)$ is the transient excess of a session that starts at ν rather than in equilibrium, negative when sessions start in states of below-average exposure potential, as neutral starting states typically are, so that short sessions understate long-run exposure.

Part (d) gives the formal meaning of a *rabbit hole*: a closed class C that contains an antisemitic state and no neutral state. Once a session enters it, its long-run exposure is $\varpi^{(C)}(\mathcal{A})$, however rarely the class is entered. A strict rabbit hole is an idealisation; the relevant case is a *nearly* closed class, which Theorem 4.2(d) addresses.

4.3 Sensitivity: small nudges, large exposure

Let the algorithm differ from the reference by a change $P_{\text{alg}} - P_{\text{ref}}$ in recommendation probabilities. Along the path $P(\varepsilon) = P_{\text{ref}} + \varepsilon\Delta$, with $\Delta\mathbf{1} = \mathbf{0}$, the parameter ε measures the size of the nudge and Δ its direction. Any algorithm can be reached this way, with $\Delta = (P_{\text{alg}} - P_{\text{ref}})/\varepsilon$. The question is how much long-run and finite-horizon exposure can change per unit of nudge. Define the *sensitivity coefficient* of $\mathcal{A}$ under a chain P as

$$\kappa_{\mathcal{A}}(P) = \tfrac{1}{2}\operatorname{osc}(\boldsymbol{g}_{\mathcal{A}}), \qquad \kappa_y = \kappa_{\{y\}}.$$

Theorem 4.2 (Sensitivity of exposure). *Let P_{ref} be irreducible, with stationary distribution ϖ and with Z, H, $\boldsymbol{g}_{\mathcal{A}}$, $\kappa_{\mathcal{A}}$ and M computed from P_{ref}. Let P_{alg} be unichain with stationary distribution ϖ_{alg}.*

(a) Exact change and derivative. *$\varpi_{\text{alg}}^{\top} - \varpi^{\top} = \varpi_{\text{alg}}^{\top}(P_{\text{alg}} - P_{\text{ref}})H$. The matrix $I - (P_{\text{alg}} - P_{\text{ref}})H$ is invertible, and $\varpi_{\text{alg}}^{\top} = \varpi^{\top}\{I - (P_{\text{alg}} - P_{\text{ref}})H\}^{-1}$. Along a path $P(\varepsilon)$ of unichain matrices, $\varpi(\varepsilon)$ is a rational function of ε with*

$$\frac{d\varpi(\varepsilon)^{\top}}{d\varepsilon} = \varpi(\varepsilon)^{\top}\Delta H(\varepsilon) = \varpi(\varepsilon)^{\top}\Delta Z(\varepsilon), \qquad \frac{d\varpi_{\mathcal{A}}}{d\varepsilon}\Big|_{\varepsilon=0} = \sum_x \varpi_x \sum_y \Delta_{xy}\, g_{\mathcal{A}}(y).$$

(b) Global bounds. *$|\varpi_{\text{alg}}(\mathcal{A}) - \varpi(\mathcal{A})| \le \kappa_{\mathcal{A}}\,\|P_{\text{alg}} - P_{\text{ref}}\|_\infty$ and $\|\varpi_{\text{alg}} - \varpi\|_1 \le \operatorname{erg}(H)\,\|P_{\text{alg}} - P_{\text{ref}}\|_\infty$. Moreover*

$$\kappa_{\mathcal{A}} = \tfrac{1}{2}\operatorname{osc}_x\Big(\sum_{y\in\mathcal{A}} \varpi_y M_{xy}\Big) \le \tfrac{1}{2}\min\{\operatorname{erg}(H), K(P_{\text{ref}})\}, \qquad \kappa_y = \tfrac{1}{2}\varpi_y \max_x M_{xy}.$$

The first bound is sharp to first order (Remark A.3).

(c) Nudges toward an antisemitic state. *Let $j \in \mathcal{A}$ and $P(\varepsilon) = (1-\varepsilon)P_{\text{ref}} + \varepsilon\mathbf{1}\boldsymbol{e}_j^{\top}$: with probability ε, the next item is replaced by an item in state j. Then $\varpi(\varepsilon)^{\top} = \varepsilon\,\boldsymbol{e}_j^{\top}\{I - (1-\varepsilon)P_{\text{ref}}\}^{-1}$ for $0 < \varepsilon \le 1$, and*

$$\frac{d\log\varpi_{\mathcal{A}}(\varepsilon)}{d\varepsilon}\Big|_{\varepsilon=0} = \frac{g_{\mathcal{A}}(j)}{\varpi(\mathcal{A})} = \frac{1}{\varpi(\mathcal{A})}\sum_{y\in\mathcal{A}} \varpi_y\big(\mathbb{E}_{\varpi}[T_y] - M_{jy}\big),$$

where $\mathbb{E}_{\varpi}[T_y] = \sum_x \varpi_x M_{xy}$ is the mean number of items needed to reach y from equilibrium. If $\mathcal{A} = \{j\}$, the derivative equals $\mathbb{E}_{\varpi}[T_j]$. For a memoryless stream, $P_{\text{ref}} = \mathbf{1}\varpi^{\top}$, it equals $\{1 - \varpi(\mathcal{A})\}/\varpi(\mathcal{A})$.

(d) Stickiness. *Let $j \in \mathcal{X}$, let $S \subseteq \mathcal{X} \setminus \{j\}$ be nonempty, and let $q(S) = \max_{x \in S} P_{\mathrm{ref}}(x, \mathcal{X} \setminus S)$ be the largest one-step probability of leaving S. Then $M_{xj} \geq 1/q(S)$ for every $x \in S$, so that*

$$\kappa_j \geq \frac{\varpi_j}{2q(S)}, \qquad \mathbb{E}_{\varpi}[T_j] \geq \frac{\varpi(S)}{q(S)}.$$

(e) Finite horizon. *For any stochastic matrices P_{ref} and P_{alg}, every ν and $T \geq 2$,*

$$|e_T(P_{\mathrm{alg}}, \nu) - e_T(P_{\mathrm{ref}}, \nu)| \leq \frac{\|P_{\mathrm{alg}} - P_{\mathrm{ref}}\|_\infty}{2T} \sum_{r=0}^{T-2} (T - 1 - r)\ \mathrm{erg}(P_{\mathrm{ref}}^r) \leq \frac{T-1}{4}\, \|P_{\mathrm{alg}} - P_{\mathrm{ref}}\|_\infty,$$

and along $P(\varepsilon)$, $de_T/d\varepsilon|_{\varepsilon=0} = T^{-1} \sum_{t=1}^{T-1} \sum_{s=0}^{t-1} \nu^\top P_{\mathrm{ref}}^s \Delta P_{\mathrm{ref}}^{t-1-s} \mathbf{1}_{\mathcal{A}}$.

The identity in (a) is due to Schweitzer (1968) and, in group-inverse form, Meyer (1975, 1980). It is exact, so long-run exposure under the algorithm can be computed from the reference chain and the change alone. The derivative has a direct reading: moving recommendation mass Δ_{xy} from x, visited with frequency ϖ_x, to y is worth $g_{\mathcal{A}}(y)$ units of long-run exposure per unit moved.

Part (b) turns this into a condition number. The ℓ_1 bound is Seneta's (1988), the single-state form of the first bound is the mean-first-passage bound of Cho and Meyer (2000), and Cho and Meyer (2001), Hunter (2006) and Ortner (2024) compare such bounds. For exposure, the useful form is the middle expression: $\kappa_{\mathcal{A}}$ is large only if some state is far, in expected items, from the antisemitic states, and it is at most half of Kemeny's constant. Part (d) gives a partial converse: if a region that excludes an antisemitic state j is hard to leave and j still carries non-negligible long-run mass, κ_j is large. The chain then has two nearly closed regions joined by a bottleneck (Simon and Ando, 1961), which in the limit becomes a rabbit hole.

Part (c) answers the question in the section title. When a fraction ε of recommendations is redirected to an antisemitic state j, long-run exposure rises by the relative amount $\varepsilon\, g_{\mathcal{A}}(j)/\varpi(\mathcal{A})$ to first order, where $g_{\mathcal{A}}(j)$ is, for aperiodic P_{ref}, the excess antisemitic exposure that one injected item in state j generates over the rest of the session. With a single antisemitic state the multiplier is $\mathbb{E}_{\varpi}[T_j]$, the mean time to reach j from equilibrium. For a memoryless stream it is $\{1 - \varpi(\mathcal{A})\}/\varpi(\mathcal{A})$, so the ratio $g_{\mathcal{A}}(j)/\{1 - \varpi(\mathcal{A})\}$ measures how much stickiness magnifies a nudge.

Example 4.3 (Two regions). Consider a two-state chain on $\{\mathsf{N}, \mathsf{R}\}$, where R collects borderline and antisemitic content, with entry probability $p = P(\mathsf{N}, \mathsf{R})$ and exit probability $q = P(\mathsf{R}, \mathsf{N})$, so that $q = q(\{\mathsf{R}\})$ in the notation of part (d). Then $\varpi_{\mathsf{R}} = p/(p+q)$, $K = 1/(p+q)$, $\kappa_{\mathsf{R}} = 1/\{2(p+q)\}$ and $d\varpi_{\mathsf{R}}/dp = q/(p+q)^2 = (1 - \varpi_{\mathsf{R}})^2/q$. Take $p = 0.001$ and $q = 0.02$, so that $\varpi_{\mathsf{R}} = 0.048$. Raising the entry probability to 0.002, so that one more neutral item in a thousand is followed by a borderline one, raises long-run exposure to $\varpi_{\mathsf{R}} = 0.091$. The bound in part (b), with $\|P_{\mathrm{alg}} - P_{\mathrm{ref}}\|_\infty = 0.002$, is 0.048, against an exact change of 0.043. The derivative is $(1 - \varpi_{\mathsf{R}})^2/q = 45.4$, and the secant over this change is 43.3. The magnification is the dwell time $1/q = 50$ items per visit to R, discounted by $(1 - \varpi_{\mathsf{R}})^2 = 0.91$.

Part (e) shows that sessions cap the effect: in a session of T items, a nudge changes expected exposure by at most $(T-1)/4$ times its size, while in the long run the change can reach $\kappa_{\mathcal{A}}$ times its size, and $\kappa_{\mathcal{A}}$ can be as large as $K/2$. An audit measures e_T, not ϖ, so when sessions are short relative to the time needed to reach the antisemitic states, the additive effect on audited sessions can understate the long-run effect by a large factor, while the ratio can lie on either side of its long-run value (Appendix A.6). Extrapolating across horizons is the job of the model.

4.4 Estimation from audit sessions

Within one arm of an audit, the maximum-likelihood estimators of ν and P are the empirical frequencies of starting states and of transitions, and every smooth exposure functional of (ν, P) is asymptotically normal by the delta method, with the gradients supplied by Theorem 4.2(a) and (e). Proposition A.4 in Appendix A.5 states this, with the asymptotic variances, for the design of an audit: many independent sessions of fixed length (Anderson and Goodman, 1957; Billingsley, 1961). Two observations guide the use of these estimators.

First, the model buys extrapolation more than precision. For e_T at the audited horizon, the plug-in $e_T(\hat{\nu}, \hat{P})$ is asymptotically efficient if Assumption A2 holds, and so at least as precise as the session average $n^{-1}\sum_s E_s$, but the gain is often modest, and the plug-in is biased if the model is wrong. The session average is therefore the primary estimator at the audited horizon, and the chain is used for what the session average cannot do: the long run, other horizons, other starting distributions, and the sensitivity analysis of Theorem 4.2.

Second, the states are observed through a classifier. If classification errors are independent across items given the true states, with confusion matrix Γ (Γ_{xy} is the probability that an item in true state x is classified as y), the observed label sequence is a hidden Markov chain, not a Markov chain, and $\hat{P}$ computed from labels is biased. Its expected transition counts per session are $\Gamma^\top \operatorname{diag}(\bar{\boldsymbol{n}})P\Gamma$, where $\bar{n}_x$ is the expected number of departures from x per session. When Γ is invertible, $\operatorname{diag}(\bar{\boldsymbol{n}})P = (\Gamma^\top)^{-1}\,\mathbb{E}(\text{observed counts}/n)\,\Gamma^{-1}$, which yields a corrected estimator of P by row normalisation. This is the transition-matrix analogue of the prevalence correction of Rogan and Gladen (1978), and Section 6 treats the estimation of error rates and differential error.

4.5 An illustrative example

Table 3 applies Proposition 4.1 and Theorem 4.2 to a three-state chain with states N (neutral), B (borderline) and A (antisemitic), $\mathcal{A} = \{\mathsf{A}\}$, and sessions that start in N. **All matrices are invented for exposition.** They describe no platform, and the example makes no empirical claim. Rows and columns are ordered (N, B, A):

$$P_{\text{ref}} = \begin{pmatrix} .95 & .04 & .01 \\ .50 & .40 & .10 \\ .40 & .30 & .30 \end{pmatrix}, \quad P_{\text{alg}} = \begin{pmatrix} .92 & .06 & .02 \\ .30 & .55 & .15 \\ .20 & .35 & .45 \end{pmatrix}, \quad P_{\text{rh}} = \begin{pmatrix} .92 & .06 & .02 \\ 0 & .70 & .30 \\ 0 & .45 & .55 \end{pmatrix}.$$

P_{alg} shifts at most 0.2 of each row's probability, $\|P_{\text{alg}} - P_{\text{ref}}\|_\infty = 0.4$. P_{rh} is a rabbit-hole variant in which borderline and antisemitic content never lead back to neutral content, so that $\{\mathsf{B}, \mathsf{A}\}$ is a closed class. A sticky reference P_{stk}, with almost the same long-run exposure as P_{ref}, is analysed in Appendix A.6.

The algorithm's moderate changes compound. No row of P_{alg} moves more than 0.2 of probability, yet long-run exposure triples, from 2.3% to 7.1%, which the exact identity of Theorem 4.2(a) reproduces from P_{ref} and the change alone; the first-order prediction, 0.030 against an exact change of 0.048, falls short because the change is not small, and the global bound $\kappa_{\mathcal{A}}\|P_{\text{alg}} - P_{\text{ref}}\|_\infty = 0.297$ holds with room to spare. The ratio at $T = 20$, 2.89, is below the long-run ratio because sessions start in neutral territory (Proposition 4.1(b)). In the rabbit-hole variant every session ends in $\{\mathsf{B}, \mathsf{A}\}$, after 12.5 items on average, and long-run exposure is 40% whatever the entry probability, while a 20-item session reaches only $e_{20} = 0.189$.

Stickiness magnifies nudges. In Appendix A.6, two references with nearly the same long-run exposure, about 2.3%, respond to the redirection of one recommendation in a hundred to A by 1.4 and 5.6 percentage points, because the mean time to reach A from equilibrium is 62 items under one and 262 under the other (Theorem 4.2(c)–(d)).

Table 3. Exposure functionals and sensitivity in a three-state chain (Proposition 4.1 and Theorem 4.2). **Illustrative, invented matrices; not data.** Sessions start in N; $\mathcal{A} = \{\mathsf{A}\}$. The ratio column is alg/ref. All values are exact computations from the matrices in the text.

	P_{ref}	P_{alg}	ratio	P_{rh}
Long-run exposure $\varpi(\mathcal{A})$	0.0232	0.0711	3.07	0.400
Session exposure e_{20}	0.0207	0.0599	2.89	0.189
Session exposure e_{100}	0.0227	0.0689	3.03	0.348
$\mathbb{P}$(at least one A item in 20)	0.254	0.489	1.93	0.739
Items until first A item, from N, $M_{\mathsf{N}\mathcal{A}}$	64.0	28.3	0.44	15.0
Asymptotic variance $\sigma^2_{\mathcal{A}}$	0.044	0.189		0.400
Kemeny's constant K	3.13	4.27		
Sensitivity coefficient $\kappa_{\mathcal{A}}$	0.742	1.008		

5 Identification from sock-puppet audits

Section 3 defined amplification at a fixed demand profile. This section asks when a sock-puppet audit (Sandvig et al., 2014), run as a randomised experiment in which scripted accounts are assigned at random to personas and policies, identifies it. (The counterfactual bots of Hosseinmardi et al. (2024) instead replay real users' histories and then follow a scripted rule.) Under four stated conditions, corrected contrasts of mean exposures identify $\mathrm{AA}(d)$ and $\mathrm{AR}(d)$, and a blocked design extends this to a drifting policy (Theorem 5.1). Under bounded interference, the bias is at most a constant times the puppets' share of the engagement signal (Proposition 5.2). The framework is that of potential outcomes (Imbens and Rubin, 2015; Rubin, 1974). Interference is treated with the partial-interference framework of Hudgens and Halloran (2008) and the exposure mappings of Aronow and Samii (2017). The variance estimator is the cluster-robust estimator of Liang and Zeger (1986), and the design effect is that of Kish (1965). Measurement error is introduced here only as far as identification requires; Section 6 treats it in full. Proofs are in Appendix B.

5.1 The audit design

Puppets, cells and waves. An audit has puppets $u = 1, \dots, U$, personas $d \in \mathcal{D}_A$ and two policies $\pi \in \{\text{alg}, \text{ref}\}$. It runs in waves $w = 1, \dots, W$. Within wave w, puppets are assigned completely at random to the cells (d, π), with $n_{wd\pi} \geq 1$ puppets in cell (d, π); write (D_u, Π_u) for the cell of puppet u and $n_{d\pi} = \sum_w n_{wd\pi}$. Every puppet is trained in the same way for its persona, and the policy contrast starts with the measurement phase.

Sessions and exposure. In the measurement phase, puppet u runs S sessions. Each session has T steps, and at each step the first k positions of the slate are recorded and classified. The *exposure* E_{us} of session s is the share of these kT recorded items with $L_c = 1$. It is (3.1) with the recorded positions counted as items. The *flagged share* Q_{us} is the share with $\widehat{L}_c = 1$. Write $\bar{E}_u = S^{-1} \sum_s E_{us}$ and $\bar{Q}_u = S^{-1} \sum_s Q_{us}$. Only $\bar{Q}_u$ is observed for every puppet. The labels L_c are observed only on a validation sample (Appendix D.5).

Potential outcomes. Let $z = (z_1, \dots, z_U)$ be an assignment of all puppets, and let $\mathcal{R}_u(z)$ be the record that puppet u would produce under it: its slates, the items it opened, and their true and classifier labels. Everything about the service that the audit does not control during the window (the versions of its policies, the eligible pool, and the behaviour of real users) is the *environment* of the window. Our statements are

conditional on the environment. An audit therefore estimates amplification by the policies that actually ran during the window. It does not estimate amplification by some time-invariant "algorithm".

Exposure-weighted error rates. For cell (d, π) define the *exposure-weighted* sensitivity and specificity

$$\mathrm{Se}_\pi(d) = \frac{\mathbb{E}\sum \mathbf{1}\left\{L_c = 1, \widehat{L}_c = 1\right\}}{\mathbb{E}\sum \mathbf{1}\{L_c = 1\}}, \qquad \mathrm{Sp}_\pi(d) = \frac{\mathbb{E}\sum \mathbf{1}\left\{L_c = 0, \widehat{L}_c = 0\right\}}{\mathbb{E}\sum \mathbf{1}\{L_c = 0\}}, \tag{5.1}$$

where each sum runs over the kTS recorded positions of one puppet of the cell, and an item recorded twice counts twice. They are the error rates of the classifier on the content that the cell is actually shown. Let $J_\pi(d) = \mathrm{Se}_\pi(d) + \mathrm{Sp}_\pi(d) - 1$ be Youden's index, $\mu_\pi(d) = \mathbb{E}\bar{E}_u$ the mean exposure of the cell, and $q_\pi(d) = \mathbb{E}\bar{Q}_u$ its mean flagged share. Counting the recorded positions by true and classifier label gives the identity

$$q_\pi(d) = \mathrm{Se}_\pi(d)\,\mu_\pi(d) + \{1 - \mathrm{Sp}_\pi(d)\}\{1 - \mu_\pi(d)\} = \{1 - \mathrm{Sp}_\pi(d)\} + J_\pi(d)\,\mu_\pi(d), \tag{5.2}$$

which holds without assumptions (Lemma B.1).

Conditions. The four conditions of Theorem 5.1 follow.

> **Assumption A3** (No interference between puppets). Given the environment, $\mathcal{R}_u(\boldsymbol{z}) = \mathcal{R}_u(z_u)$ for every puppet u, and the records of distinct puppets are independent.

> **Assumption A4** (Stable policies, or a blocked design). Either (a) *stability*: the law of $\mathcal{R}_u(d, \pi)$, given the environment, is the same for every puppet of the window, whatever its wave; or (b) *blocking*: puppets of the same wave and cell have records with a common law $F_{wd\pi}$, and within each wave the calendar times of the measurement sessions have the same distribution in every cell.

> **Assumption A5** (Classifier error does not depend on the arm). For each audited persona d, $\mathrm{Se}_{\mathrm{alg}}(d) = \mathrm{Se}_{\mathrm{ref}}(d) = \mathrm{Se}(d)$ and $\mathrm{Sp}_{\mathrm{alg}}(d) = \mathrm{Sp}_{\mathrm{ref}}(d) = \mathrm{Sp}(d)$, with $J(d) = \mathrm{Se}(d) + \mathrm{Sp}(d) - 1 > 0$, and $(\mathrm{Se}(d), \mathrm{Sp}(d))$ are identified by the validation sample. When d is fixed we write Se, Sp and J.

> **Assumption A6** (Scripted demand). A puppet's actions are a fixed function of its persona, of what it has been shown, and of randomisation that is independent of the assignment, and this function is the same under both policies. In particular, nobody intervenes in a session by hand, and the choice rule of Appendix D.2 is applied identically in both arms. Consequently $\bar{E}_u$ under $z_u = (d, \pi)$ is a draw of $E_u(\pi; d)$ averaged over sessions, and $\mathbb{E}\bar{E}_u(d, \pi) = \mu_\pi(d)$ of (3.2).

Assumption A3 is the stable-unit condition of Rubin (1974) for a population that shares a learning system; Proposition 5.2 relaxes it. Assumption A4(b) is what the waves of the protocol deliver. Assumption A5 is *not* implied by error rates that depend only on the content: when error rates differ across content types and the arms show different mixtures of types, (5.1) differs by arm (Theorem 6.1(c)). Assumption A6 holds by construction when the script is executed by software that is blind to the arm, and fails if an operator steers sessions.

A fifth condition is needed to read the estimands as statements about people. It is the second part of Assumption A1.

> **Assumption A7** (Transport from puppets to users). For a real user whose demand profile is d, the mean exposure under policy π equals $\mu_\pi(d)$, the mean exposure of a puppet of persona d.

Theorem 5.1 does not use Assumption A7. Without it, the theorem identifies amplification for the scripted personas, a property of the service's policies evaluated at those scripts; with it, the numbers

describe users who behave like the scripts. The audit cannot test Assumption A7, which fails if the service recognises automated accounts, serves new accounts differently, or uses signals that puppets lack. We therefore report estimates as properties of the audited personas, and treat their transport to real users as a separate claim.

5.2 Identification and estimation

Let $\widehat{q}_{d\pi} = n_{d\pi}^{-1} \sum_{u:(D_u,\Pi_u)=(d,\pi)} \bar{Q}_u$ be the mean flagged share of cell (d, π), and let

$$\widehat{\mu}_{d\pi} = \frac{\widehat{q}_{d\pi} + \mathrm{Sp} - 1}{J}, \qquad \widehat{\mathrm{AA}}(d) = \widehat{\mu}_{d,\mathrm{alg}} - \widehat{\mu}_{d,\mathrm{ref}}, \qquad \widehat{\mathrm{AR}}(d) = \frac{\widehat{\mu}_{d,\mathrm{alg}}}{\widehat{\mu}_{d,\mathrm{ref}}}. \tag{5.3}$$

These are the mean *corrected* exposures of Rogan and Gladen (1978) and their difference and ratio. Here (Se, Sp) are treated as known; Theorem 6.1 adds the uncertainty of their estimates. Let $\sigma^2_{d\pi} = \mathrm{Var}(\bar{Q}_u)$ for a puppet of cell (d, π), let $\sigma^2_{\mathrm{s},d\pi} = S^{-1} \sum_s \mathbb{E}\{Q_{us} - q_\pi(d)\}^2$ be the mean squared deviation of a single session from the cell mean, and let $\mathrm{deff}_{d\pi} = S\sigma^2_{d\pi}/\sigma^2_{\mathrm{s},d\pi}$ be the design effect of averaging S correlated sessions. If the sessions of a puppet are exchangeable, with pairwise correlation $\gamma_{d\pi}$, then $\mathrm{deff}_{d\pi} = 1 + (S-1)\gamma_{d\pi}$ (Kish, 1965).

Theorem 5.1 (Identification by randomised persona–policy audits). *Let puppets be randomised to cells within waves as above, and let Assumptions A3 and A6 hold.*

(a) Identification. *Under Assumption A4(a), $q_\pi(d) = \mathbb{E}\{\bar{Q}_u \mid (D_u, \Pi_u) = (d, \pi)\}$ is a functional of the law of the observed data. Under Assumption A5 in addition,*

$$\mu_\pi(d) = \frac{q_\pi(d) + \mathrm{Sp} - 1}{J}, \qquad \mathrm{AA}(d) = \frac{q_{\mathrm{alg}}(d) - q_{\mathrm{ref}}(d)}{J}, \qquad \mathrm{AR}(d) = \frac{q_{\mathrm{alg}}(d) + \mathrm{Sp} - 1}{q_{\mathrm{ref}}(d) + \mathrm{Sp} - 1}. \tag{5.4}$$

Thus, within a persona, $\mathrm{AA}(d)$ depends on the classifier only through $J(d)$, and $\mathrm{AR}(d)$ only through $\mathrm{Sp}(d)$. Every linear combination of cell means is identified, as $\sum_{d,\pi} c_{d\pi}\mu_\pi(d) = \sum_{d,\pi} c_{d\pi}\{q_\pi(d) + \mathrm{Sp}(d) - 1\}/J(d)$. This includes the persona-contrast decomposition of Section 3.5: the supply contrast $\mathrm{AA}(d_0)$, the demand contrast $\mu_{\mathrm{ref}}(d) - \mu_{\mathrm{ref}}(d_0)$ and the interaction $\mathrm{AA}(d) - \mathrm{AA}(d_0)$. Contrasts across personas need the error rates of each persona. The exposure-weighted rates generally differ between personas, because personas are shown different mixtures of content (Theorem 6.1(c)). The demand contrast therefore needs both Se *and* Sp *of both personas, and the interaction needs $J(d)$ and $J(d_0)$.*

(b) Asymptotic normality. *Under Assumption A4(a) and Assumption A5, let $n = \sum_{d,\pi} n_{d\pi} \to \infty$ with $n_{d\pi}/n \to p_{d\pi} > 0$, while S, T and k stay fixed. If $\mu_{\mathrm{alg}}(d) > 0$ and $\mu_{\mathrm{ref}}(d) > 0$, then*

$$\sqrt{n}\{\widehat{\mathrm{AA}}(d) - \mathrm{AA}(d)\} \Rightarrow \mathrm{N}\Big(0,\ \frac{1}{J^2}\sum_\pi \frac{\sigma^2_{d\pi}}{p_{d\pi}}\Big),$$

$$\sqrt{n}\{\log \widehat{\mathrm{AR}}(d) - \log \mathrm{AR}(d)\} \Rightarrow \mathrm{N}\Big(0,\ \frac{1}{J^2}\sum_\pi \frac{\sigma^2_{d\pi}}{p_{d\pi}\,\mu_\pi(d)^2}\Big),$$

with the sums over $\pi \in \{\mathrm{alg}, \mathrm{ref}\}$. The two cell means are asymptotically independent.

(c) Cluster-robust variance. *With puppets as clusters and the small-sample factor $n_{d\pi}/(n_{d\pi} - 1)$, the cluster-robust variance estimator of Liang and Zeger (1986) for the regression of Q_{us} on cell indicators equals $s^2_{d\pi}/n_{d\pi}$, with $s^2_{d\pi}$ the sample variance of the $\bar{Q}_u$ in the cell, when every puppet*

has S sessions; it is consistent, and gives intervals with asymptotic coverage $1-\alpha$*. The estimator that treats sessions as independent converges, after multiplication by* $n_{d\pi}$*, to* $\sigma^2_{\mathrm{s},d\pi}/S = \sigma^2_{d\pi}/\mathrm{deff}_{d\pi}$*, and is anticonservative when sessions are positively correlated.*

(d) Blocked design. *Under Assumption A4(b) instead of (a), let* $\mu_{w\pi}(d)$ *and* $q_{w\pi}(d)$ *be the means under* $F_{wd\pi}$*, and let* ω_w *be fixed design weights with* $\sum_w \omega_w = 1$*, for example* $\omega_w = \sum_\pi n_{wd\pi}/\sum_\pi n_{d\pi}$*. Let* $\bar{\mu}_\pi(d) = \sum_w \omega_w \mu_{w\pi}(d)$ *be the* window-averaged *mean exposure, with contrasts* $\overline{\mathrm{AA}}(d)$ *and* $\overline{\mathrm{AR}}(d)$*. Assume Assumption A5 with error rates that may differ by wave but not by arm within a wave. Then (a) holds wave by wave, the blocked estimator* $\widehat{\bar{\mu}}_{d\pi} = \sum_w \omega_w \widehat{\mu}_{wd\pi}$ *is unbiased for* $\bar{\mu}_\pi(d)$*, and with W fixed and* $n_{wd\pi} \to \infty$ *in fixed proportions, (b) holds for* $\overline{\mathrm{AA}}(d)$ *and* $\overline{\mathrm{AR}}(d)$ *with* $J^{-2}\sigma^2_{d\pi}/p_{d\pi}$ *replaced by* $\sum_w \omega_w^2 \sigma^2_{wd\pi}/(J_w^2 p_{wd\pi})$*, where* J_w *is the Youden index of wave w, and with* $\mu_\pi(d)$ *replaced by* $\bar{\mu}_\pi(d)$*. The stratified variance estimator* $\sum_w \omega_w^2 s^2_{wd\pi}/(J_w^2 n_{wd\pi})$ *is consistent.*

Proof. See Appendix B.2. Randomisation makes each observed puppet record a draw from its cell's potential law, identity (5.2) under Assumption A5 is solved for $\mu_\pi(d)$, and the cluster-robust sandwich collapses to the between-puppet variance when every puppet has the same number of sessions. □

The puppet is the unit. The sessions of one puppet share its trained profile, so they are usually positively correlated. Treating sessions as independent then understates the variance by the factor $\mathrm{deff}_{d\pi}$, which is 3.7 for $S = 10$ sessions with $\gamma = 0.3$, so that a nominal 95% interval covers with probability $2\Phi(1.96/3.7^{1/2}) - 1 \approx 0.69$. The cluster-robust estimator is correct whatever the sign of the correlation.

The classifier enters differently on the two scales. A naive difference of flagged shares estimates $J(d)\,\mathrm{AA}(d)$, so a test of $\mathrm{AA}(d) = 0$ needs no validation data under Assumption A5, whereas the naive ratio is biased towards one and its correction needs $\mathrm{Sp}(d)$. Both simplifications hold within a persona only: a persona contrast corrected with common error rates can be wrong even in sign.

Unequal sessions and persona contrasts. If puppets complete different numbers of sessions, we analyse puppet means, which remain unbiased if attrition is unrelated to exposure. Part (a) identifies the persona-contrast decomposition of Section 3.5 for any two audited personas; its demand contrast compares two scripts that the researcher chose.

5.3 Interference through the shared recommender

Puppets share the system they audit. A puppet's engagement can enter an item's popularity, item-to-item associations or the next model update, and through these change what other puppets are shown. Assumption A3 then fails, as it does in experiments on online marketplaces, where treated and control units compete for the same inventory (Bajari et al., 2023; Johari et al., 2022). Standard estimators remain consistent for an expected average effect under limited interference (Sävje et al., 2021); we need more, a bound on the distance between that effect and the effect without the audit, and model the failure with an exposure mapping (Aronow and Samii, 2017).

The load. Let waves be separated by gaps longer than the memory of the learning system, so that they are the groups of partial interference (Hudgens and Halloran, 2008). Within wave w, let $\ell_w \in [0, 1)$ be the *load*: the share of the engagement signal, in the region of content that the wave's personas visit, that the wave's puppets generate. A service with millions of users can still have a substantial load in a niche.

Assumption A8 (Bounded interference). Waves do not interfere with one another. Within a wave, given its load ℓ_w, the records of distinct puppets are independent, and the record of a puppet with assignment

(d, π) depends on the other puppets' assignments only through ℓ_w, with mean exposure $m_{d\pi}(\ell_w)$ and classifier error rates that do not depend on the arm. Moreover, there are constants $\xi_\pi \geq 0$ and $\xi_\Delta \geq 0$ such that for all $\ell \in [0, \ell_{\max}]$,

$$|m_{d\pi}(\ell) - m_{d\pi}(0)| \leq \xi_\pi \, \ell, \qquad \left|\{m_{d,\mathrm{alg}}(\ell) - m_{d,\mathrm{ref}}(\ell)\} - \{m_{d,\mathrm{alg}}(0) - m_{d,\mathrm{ref}}(0)\}\right| \leq \xi_\Delta \, \ell.$$

The no-audit quantities are $\mu_\pi(d) = m_{d\pi}(0)$, the mean exposure that a single puppet, too small to move the system, would see, and their contrasts $\mathrm{AA}^0(d)$ and $\mathrm{AR}^0(d)$ are the estimands of Definition 3.1. With the smallest admissible constants, $\xi_\Delta \leq \xi_{\mathrm{alg}} + \xi_{\mathrm{ref}}$, and a reference that does not learn from engagement has $\xi_{\mathrm{ref}} = 0$.

Proposition 5.2 (Interference bias, design remedies and sensitivity). *Let Assumption A4(b), Assumptions A5, A6 and A8 hold, with waves randomised as in Theorem 5.1(d), wave weights ω_w and loads $\ell_w \leq \ell_{\max}$ fixed by the design of the waves. Let $\bar{\ell} = \sum_w \omega_w \ell_w$ be the average load, and $\bar{m}_\pi(d) = \sum_w \omega_w m_{d\pi}(\ell_w)$. All statements are for the given waves; the corrected cell means are linear in the data, so they are exact statements about expectations.*

(a) Bias bound. *The blocked estimator $\widehat{\mathrm{AA}}(d)$ of Theorem 5.1(d) has expectation $\bar{m}_{\mathrm{alg}}(d) - \bar{m}_{\mathrm{ref}}(d)$, and*

$$\left|\mathbb{E}\widehat{\mathrm{AA}}(d) - \mathrm{AA}^0(d)\right| \leq \xi_\Delta \, \bar{\ell} \leq (\xi_{\mathrm{alg}} + \xi_{\mathrm{ref}}) \, \bar{\ell}.$$

If $\bar{m}_\pi(d) > \xi_\pi \bar{\ell}$ for both π, the ratio of expected corrected means satisfies $|\log\{\bar{m}_{\mathrm{alg}}(d)/\bar{m}_{\mathrm{ref}}(d)\} - \log \mathrm{AR}^0(d)| \leq \sum_\pi \xi_\pi \bar{\ell}/\{\bar{m}_\pi(d) - \xi_\pi \bar{\ell}\}$. Each inequality for $\widehat{\mathrm{AA}}$ is attained by some m satisfying Assumption A8.

(b) Direction. *The design does not fix the sign of the bias. For a policy that fills each position with probability proportional to a relevance factor times a power of the item's popularity, the share of positions that show antisemitic items, at a fixed state of the puppet, cannot rise with the load if the puppets' engagement never reaches antisemitic items, and it rises with the load whenever that engagement, weighted by relevance, is more concentrated on antisemitic items than the baseline exposure (Proposition B.3 in Appendix B.3).*

(c) Randomised saturation. *If waves are randomised to two loads $\ell_1 < \ell_2$ and $\mathrm{AA}(\ell) = m_{d,\mathrm{alg}}(\ell) - m_{d,\mathrm{ref}}(\ell)$ has $|\mathrm{AA}''| \leq \zeta$ on $[0, \ell_2]$, linear extrapolation of the blocked estimators to zero load has bias at most $\frac{1}{2}\zeta\ell_1\ell_2$, and the slope between the two loads gives a test of differential interference (Proposition B.4).*

(d) Sensitivity analysis. *For a postulated ξ_Δ, the interval $\widehat{\mathrm{AA}}(d) \pm \{\xi_\Delta \bar{\ell} + z_{1-\alpha/2}\widehat{\mathrm{se}}\}$ covers $\mathrm{AA}^0(d)$ with probability at least $1-\alpha$, to the accuracy of the normal approximation to $\widehat{\mathrm{AA}}(d)$ given in Theorem 5.1. The sign of $\mathrm{AA}^0(d)$ is established at level α for every $\xi_\Delta < \xi^* = (|\widehat{\mathrm{AA}}(d)| - z_{1-\alpha/2}\widehat{\mathrm{se}})_+/\bar{\ell}$, the* breakdown value *of the conclusion.*

Proof. See Appendix B.3; (a) is the triangle inequality applied wave by wave, and (d) is the union of Wald intervals over the identified set of (a). □

Design remedies and reporting. Proposition 5.2 turns interference into a problem of keeping $\bar{\ell}$ small and measuring what remains. Puppets that record slates, open one item per step with fixed dwell time and never like, follow or share contribute the least engagement per session, which is also the footprint rule of Section 9.2. Washouts longer than the system's memory make waves independent, and staggering puppets lowers the instantaneous load; assigning whole waves to policies, a switchback design (Bojinov et al., 2023),

trades interference between arms for comparisons across waves. Randomising wave sizes between two levels (Baird et al., 2018) identifies the slope in part (c), but with $\ell_2 = 2\ell_1$ the extrapolated estimator has about ten times the variance of the blocked one, so it is a robustness analysis. Untrained sentinels give evidence about ξ_π for their own profile, not a bound for the audited personas. Part (b) is why the sign of the bias cannot be taken for granted: the avoid-on-flag rule still opens false negatives, and when exposure is rare a small share of engagement on a few relevant false negatives can raise it; part (b) also holds only at a fixed state. We therefore make no claim about the sign, and report $\bar{\ell}$, the sentinel and saturation slopes, and the breakdown value ξ^* of part (d). Because exposure is rare, the ratio bound in part (a) is informative only if $\xi_\pi \bar{\ell}$ is small relative to $\bar{m}_\pi(d)$. Section 7 measures the interference bias, and the effect of the remedies, on the synthetic recommender.

5.4 Drift: what you audit is a moving target

Services change their policies continually, by retraining, by experiments on subsets of users, and by announced changes. Under drift, Assumption A4(a) fails, and the question is what an audit spread over weeks estimates.

The blocked design. Under Assumption A4(b), the blocked estimator targets $\overline{\mathrm{AA}}(d) = \sum_w \omega_w \mathrm{AA}_w(d)$, the additive amplification averaged over the waves of the window, and the ratio of window-averaged means $\overline{\mathrm{AR}}(d)$, which is not the average of the wave ratios. Blocking does not remove drift from the estimand. It ensures that drift affects both arms equally, so that it cannot be mistaken for amplification, whereas a design that runs the arms at different times confounds the two.

Diagnosing drift. The wave-level estimates are independent under Assumption A4(b), and $X^2_{\text{drift}} = \sum_w (\widehat{\mathrm{AA}}_w - \widehat{\mathrm{AA}}^{\text{iv}})^2 / \widehat{v}_w$, with $\widehat{v}_w$ the estimated variance of $\widehat{\mathrm{AA}}_w$ and $\widehat{\mathrm{AA}}^{\text{iv}}$ the inverse-variance-weighted mean, is asymptotically χ^2_{W-1} when AA_w does not vary: a test of drift in amplification, as distinct from drift common to both arms, which sentinels track. If capacity does not allow every cell in every wave, cells can be rotated in a balanced incomplete block design (Appendix B.4), but the two arms of a persona always share a wave.

The moving target. An audit identifies amplification by the policies that ran during its window, and every conclusion is indexed by the window. The Markov model extrapolates across session lengths, not across policy versions. Sequential monitoring shortens the window and so narrows the target; under drift, its guarantee holds for the window-averaged estimand only approximately (Section 6.2).

5.5 What audits cannot identify

Theorem 5.1 identifies $\mu_\pi(d)$ for the audited personas and both policies, and hence, by Proposition 3.4, amplification for any declared mixture Λ of them. It does not identify the demand-shaping effect, the feedback interaction or the total effect of the algorithm on real users, because a puppet's demand does not respond to what it is shown, and the "demand" in the persona-contrast decomposition is a difference between two scripts. Statements about real users require Assumption A7 in addition, and statements about the algorithm's total effect on them require the designs listed after Proposition 3.4 or its sensitivity analysis.

6 Measurement: misclassification and sequential inference

An audit sees exposure through a classifier. This section corrects for the classifier's errors (Section 6.1), turns the corrected estimators into an audit that can stop as soon as the evidence is decisive (Section 6.2),

and sizes the audit (Section 6.3). Theorem 6.1 gives the bias of naive estimators and shows that error rates that differ by arm can push the naive ratio in either direction. Algorithm 1 and Theorem 6.2 give a confidence sequence for the amplification ratio that remains approximately valid when the audit stops early, and Proposition 6.3 gives the numbers of puppets, sessions and codes in closed form. Throughout, the persona d is fixed and suppressed, the arms are $\pi \in \{a, r\}$ with a = alg and r = ref, and the notation is that of Section 5.1. In particular $q_\pi = (1 - \mathrm{Sp}_\pi) + J_\pi \mu_\pi$ by (5.2).

6.1 Misclassification

The correction of a prevalence for a test with known sensitivity and specificity is due to Rogan and Gladen (1978). Double sampling, in which a second-phase subsample of the first-phase units is measured more accurately, is due to Neyman (1938), and Tenenbein (1970) applied it to binomial data subject to misclassification. The double-sampling estimator below is also the simplest case of recent methods that combine many machine labels with a few gold-standard labels, prediction-powered inference (Angelopoulos et al., 2023) and design-based supervised learning (Egami et al., 2023). Our contribution is to work out what these imply for a *ratio* of two prevalences, measured on two arms that show different content.

Validation designs. In an *external* design, the error rates are estimated from coded items that are independent of the audit's measurement data: $\widehat{\mathrm{Se}}$ is the share flagged among n_1 coded items with $L_c = 1$, and $\widehat{\mathrm{Sp}}$ the share unflagged among n_0 coded items with $L_c = 0$. In an *internal* design, which the protocol uses, coded items are drawn from each arm's own recorded exposures, stratified by classifier output. Either way, the target is the exposure-weighted error rate (5.1) of each arm, so a coded item must represent exposures, not distinct items: an item that appeared in many recorded positions is weighted by its number of appearances divided by its inclusion probability.

Theorem 6.1 (Misclassification: attenuation, correction and differential error). *Let $\mu_r > 0$ and $J_\pi > 0$ for both arms.*

(a) Naive estimands under non-differential error. *Under Assumption A5, the naive additive contrast is $q_a - q_r = J\,\mathrm{AA}$, and the naive ratio is*

$$\mathrm{AR}^{\mathrm{nv}} = \frac{q_a}{q_r} = w_0 + (1 - w_0)\,\mathrm{AR}, \qquad w_0 = \frac{1 - \mathrm{Sp}}{q_r} \in [0, 1). \tag{6.1}$$

The naive ratio is a convex combination of 1 *and* AR*, so it is attenuated towards one. The weight w_0 depends on the classifier only through* Sp *and q_r, and $w_0 = 0$ if and only if* $\mathrm{Sp} = 1$.

(b) Corrected estimators. *Under Assumption A5, with $(\widehat{\mathrm{Se}}, \widehat{\mathrm{Sp}})$ from an external design, the corrected $\widehat{\mathrm{AA}} = \widehat{\mu}_a - \widehat{\mu}_r$ and* $\log \widehat{\mathrm{AR}}$*, with $\widehat{\mu}_\pi = (\widehat{q}_\pi + \widehat{\mathrm{Sp}} - 1)/\widehat{J}$, are consistent and asymptotically normal. Their delta-method variances are given in Proposition B.5 (Appendix B.5); the estimation error of $\widehat{\mathrm{Se}}$ does not enter the variance of* $\log \widehat{\mathrm{AR}}$.

(c) Differential error. *Without Assumption A5, let $r_\pi = q_\pi/\mu_\pi = \mathrm{Se}_\pi + (1 - \mathrm{Sp}_\pi)(1 - \mu_\pi)/\mu_\pi$. Then, exactly,*

$$\mathrm{AR}^{\mathrm{nv}} = \mathrm{AR} \cdot \frac{r_a}{r_r}, \qquad \mathrm{AR}^{\mathrm{pool}} = \frac{J_a \mu_a + \overline{\mathrm{Sp}} - \mathrm{Sp}_a}{J_r \mu_r + \overline{\mathrm{Sp}} - \mathrm{Sp}_r}, \tag{6.2}$$

where $\mathrm{AR}^{\mathrm{pool}}$ is the limit of the corrected ratio computed with error rates $(\overline{\mathrm{Se}}, \overline{\mathrm{Sp}})$ pooled over the two arms. Either can lie on the other side of one from AR*, and Lemma B.6 gives the range of $\mathrm{AR}^{\mathrm{nv}}$ compatible with bounds on the error rates. If content falls into types k with labels L_k and*

type-level flag rates q_k common to both arms, and arm π shows type k with exposure share $e_{\pi k}$, then $\mathrm{Se}_\pi = \mu_\pi^{-1} \sum_{k:L_k=1} e_{\pi k} q_k$ *and* $1 - \mathrm{Sp}_\pi = (1-\mu_\pi)^{-1} \sum_{k:L_k=0} e_{\pi k} q_k$. *Assumption A5 then fails unless, within each label, the two arms' mixtures of types have the same average flag rate, which fails generically when the arms mix types with different q_k in different proportions.*

(d) Stratified validation. *Let the error rates be estimated separately in each arm, by an external design per arm or by an internal design, with estimators consistent for* $(\mathrm{Se}_\pi, \mathrm{Sp}_\pi)$. *Then* $\widehat{\mu}_\pi^{\mathrm{st}} = (\widehat{q}_\pi + \widehat{\mathrm{Sp}}_\pi - 1)/\widehat{J}_\pi$ *is consistent for μ_π, whether or not Assumption A5 holds, and so are the resulting* $\widehat{\mathrm{AA}}^{\mathrm{st}}$ *and* $\widehat{\mathrm{AR}}^{\mathrm{st}}$. *In the internal design that codes $m_{\pi 1}$ flagged and $m_{\pi 0}$ unflagged exposures of arm π, and weights them by the arm's recorded flagged and unflagged counts, the corrected estimator equals the double-sampling estimator exactly,*

$$\frac{\widehat{q}_\pi + \widehat{\mathrm{Sp}}_\pi - 1}{\widehat{\mathrm{Se}}_\pi + \widehat{\mathrm{Sp}}_\pi - 1} = \widehat{q}_\pi \, \widehat{\mathrm{PPV}}_\pi + (1 - \widehat{q}_\pi) \, \widehat{\mathrm{FOR}}_\pi, \tag{6.3}$$

where $\widehat{\mathrm{PPV}}_\pi$ *and* $\widehat{\mathrm{FOR}}_\pi$ *are the shares with $L_c = 1$ among the coded flagged and coded unflagged exposures. If the coding is done within each puppet, the puppet-level version of the right side, the* double-sampled exposure $\widetilde{E}_u$, *is conditionally unbiased for the puppet's true exposure $\bar{E}_u$.*

(e) Confirmed-positive form. *Let* PPV_π *be the exposure-weighted share of true positives among flagged exposures of arm π. Then* $q_\pi \mathrm{PPV}_\pi = \mathrm{Se}_\pi \mu_\pi$ *exactly, and hence*

$$\mathrm{AR} = \frac{1}{\lambda_S} \cdot \frac{q_a \mathrm{PPV}_a}{q_r \mathrm{PPV}_r}, \qquad \lambda_S = \frac{\mathrm{Se}_a}{\mathrm{Se}_r}. \tag{6.4}$$

If the sensitivities are equal, AR *is identified by the flagged shares and coded flagged exposures alone, whatever the specificities of the two arms.*

Proof. See Appendix B.5; (c) follows from dividing q_π by μ_π, and (6.3) is algebra on weighted counts. □

Rare content makes specificity decisive. When antisemitic content is rare, even a highly specific classifier makes w_0 large. With $\mu_r = 0.02$, Se = 0.85 and Sp = 0.98, $w_0 = 0.55$: a true ratio of 1.5 appears as 1.23, while the naive difference is attenuated only by $J = 0.83$. In the variance of $\log \widehat{\mathrm{AR}}$ (B.2), the specificity term is multiplied by $(1/\mu_a - 1/\mu_r)^2$: with AR = 1.5 and $n_0 = 1{,}500$ coded negatives it alone contributes 0.073 to the standard deviation. The delta method is also less accurate than usual, being good only if $\mathrm{cv}_{\mathrm{Sp}} = v_{\mathrm{Sp}}^{1/2}/(J\mu_r)$, the coefficient of variation that estimating Sp gives the corrected reference exposure, is small; here it is 0.22. We recommend reporting $\mathrm{cv}_{\mathrm{Sp}}$ and a bootstrap alongside the delta-method interval (Section 7 reports the understatement).

Differential error is the rule, not the exception. Part (c) formalises the problem raised in Section 2.4: if the algorithm shifts the mix of content types, arm-level error rates differ, even though no type-level error rate does. Table 4 gives two invented examples. In scenario I, the algorithm shows more coded antisemitic content, which the classifier misses: the true ratio is 1.8, but the naive ratio is 0.97 and the corrected ratio with pooled error rates is 0.94. In scenario II, the algorithm shows less antisemitic content and much more counter-speech: the true ratio is 0.57, but the naive ratio is 1.24 and the pooled correction makes things worse, at 1.69. Correcting with the wrong error rates is not a safe default. By part (d), validation stratified by arm recovers the true ratio in both cases.

Table 4. Differential misclassification can reverse the direction of the naive and the pooled-corrected amplification ratio (Theorem 6.1(c)). **Illustrative: invented exposure mixes and error rates; not data.** Types and flag rates q_k: explicit antisemitic (0.95), coded antisemitic (0.40), quotation or counter-speech (0.30), borderline (0.05), neutral (0.005). Exposure shares (ref; alg) are given for the first four types; the rest is neutral. $\mathrm{AR}^{\mathrm{nv}}$: naive ratio; $\mathrm{AR}^{\mathrm{pool}}$: limit of the correction with error rates pooled over the arms; $\lambda_S\mathrm{AR}$: limit of the confirmed-positive ratio (6.4) computed with $\lambda_S = 1$. The arm-stratified correction of Theorem 6.1(d) converges to AR. All entries are exact computations.

Scenario (shares ref; alg)	μ_r	μ_a	Se_r; Se_a	Sp_r; Sp_a	AR	$\mathrm{AR}^{\mathrm{nv}}$	$\mathrm{AR}^{\mathrm{pool}}$	$\lambda_S\mathrm{AR}$
I: more coded content (explicit .008; .004, coded .002; .014, quoting .010; .002, borderline .03; .05)	.010	.018	.84; .52	.991; .992	1.80	0.97	0.94	1.12
II: more counter-speech (explicit .010; .006, coded .004; .002, quoting .004; .030, borderline .03; .06)	.014	.008	.79; .81	.992; .983	0.57	1.24	1.69	0.59

Validation by arm, and where to spend it. Part (d) is why the protocol stratifies validation by arm. The internal design needs no error rates that transfer from elsewhere, and done within puppets it yields corrected puppet-level exposures whose cluster-robust variance contains the validation uncertainty. Its cost lies in the unflagged stratum, where false negatives are rare, so score bands with the allocation of Neyman (1934) concentrate the coding where they are. Part (e) exploits the asymmetry between the strata: the ratio needs the flagged stratum of each arm and one number about the unflagged strata, λ_S. Differential specificity is then corrected automatically (scenario II: $\lambda_S = 1.02$, and (6.4) with $\lambda_S = 1$ gives 0.59 against 0.57), while differential sensitivity enters as a single factor (scenario I: $\lambda_S = 0.62$, and setting it to one gives 1.12).

6.2 Sequential, anytime-valid audits

An audit that stops as soon as its evidence is decisive uses fewer accounts and sessions, which matters for its footprint (Section 9.2) and for the moving target of Section 5.4. Stopping on the data invalidates a fixed-sample interval, however, because a monitored fixed-sample analysis performs repeated significance tests. Sequential tests go back to Wald (1945). A *confidence sequence* (Darling and Robbins, 1967) is a sequence of sets C_n, indexed here by the number n of puppet pairs, with $\mathbb{P}(\theta \in C_n \text{ for all } n \geq 1) \geq 1 - \alpha$. It remains valid at any stopping time. Modern constructions use nonnegative supermartingales and Ville's inequality (Ville, 1939), through normal mixtures (Robbins, 1970), nonparametric boundaries (Howard et al., 2021) or betting (Waudby-Smith and Ramdas, 2024). Waudby-Smith et al. (2024) give the asymptotic analogue of the central-limit interval, and Ramdas et al. (2023) review the field.

The puppet outcome. The sequence is built from one number per puppet, an *outcome* $O_u \in [0, 1]$ computed from the puppet's own record and codes, whose mean in arm π is proportional to μ_π. Let $\vartheta = \mathbb{E}O_a/\mathbb{E}O_r = \lambda\,\mathrm{AR}$, where the factor λ is declared to lie in a registered range $[\lambda_-, \lambda_+]$. Two outcomes qualify.

(i) The *double-sampled exposure* $O_u = \widetilde{E}_u$ of Theorem 6.1(d), computed from m coded flagged and m' coded unflagged positions of the puppet. It is conditionally unbiased for $\bar{E}_u$, so $\lambda = 1$ and no assumption about error rates is needed. It is the protocol's primary outcome (Section 8.1).

(ii) The *confirmed-flagged exposure* $O_u = Y_u = \bar{Q}_u\widehat{P}_u$, where $\widehat{P}_u$ is the share confirmed as antisemitic among m flagged positions of the puppet drawn uniformly without replacement (all of them if it has fewer). Given the record, Y_u is unbiased for the share of the puppet's recorded positions that are flagged and truly antisemitic, so $\mathbb{E}Y_u = \mathrm{Se}_\pi\mu_\pi$ (Lemma B.2) and $\lambda = \lambda_S$ by Theorem 6.1(e). It needs no unflagged codes, but its range for λ_S is an assumption.

Algorithm 1 Anytime-valid audit of $\mathrm{AR}(d)$ for one persona, from a puppet outcome O_u

Require: level α; margin $\delta > 1$; wave budget $W_{\max}$; burn-in waves $b \geq 1$; planned pairs n^* and tuning constant a; outcome $O_u \in [0,1]$ with range $[\lambda_-, \lambda_+]$ for λ

LB $\leftarrow -\infty$; UB $\leftarrow +\infty$
for $w = 1, 2, \ldots, W_{\max}$ **do**
  run wave w; code each puppet of persona d as its outcome requires, and compute O_u
  pair each alg puppet with a distinct ref puppet of the wave, at random; append $(O_{a,i}, O_{r,i})$; $n \leftarrow$ number of pairs
  $\bar{O}_a \leftarrow n^{-1} \sum_{i \leq n} O_{a,i}$; $\bar{O}_r \leftarrow n^{-1} \sum_{i \leq n} O_{r,i}$; $\widehat{\vartheta}_n \leftarrow \bar{O}_a / \bar{O}_r$
  $\widehat{\sigma}_n^2 \leftarrow n^{-1} \sum_{i \leq n} \big(O_{a,i}/\bar{O}_a - O_{r,i}/\bar{O}_r\big)^2$ ▷ n times the variance of $\log \widehat{\vartheta}_n$
  $h_n \leftarrow \widehat{\sigma}_n \, (n^*)^{1/2} \, h(n/n^*)/n$ ▷ normal-mixture radius
  $C_n \leftarrow \big[\log \widehat{\vartheta}_n - \log \lambda_+ - h_n,\ \log \widehat{\vartheta}_n - \log \lambda_- + h_n\big]$
  if $w > b$ **then** LB $\leftarrow \max(\mathrm{LB}, \min C_n)$; UB $\leftarrow \min(\mathrm{UB}, \max C_n)$ ▷ after burn-in
  if LB $> \log \delta$ **or** UB $< -\log \delta$ **or** $-\log \delta <$ LB $\leq$ UB $< \log \delta$ **then stop** ▷ S1–S2
return [LB, UB] for $\log \mathrm{AR}(d)$ and the conclusion of Table 7

Pairs i consist of one puppet per arm from the same wave and persona, with outcomes $(O_{a,i}, O_{r,i})$. For $s \geq 0$ and a fixed tuning constant $a > 0$, let

$$h(s) = \big\{2(1+as)\, a^{-1} \log\big((1+as)^{1/2}/\alpha\big)\big\}^{1/2},$$

the normal-mixture boundary of Robbins (1970) in units of the planned horizon.

Algorithm 1 is the delta-method confidence sequence of Waudby-Smith et al. (2024) for the log of a ratio of two means, widened by the range for λ. The running intersection [LB, UB] uses only the sets after the burn-in, and the protocol registers $b = 1$: the first wave updates the estimates but can neither narrow [LB, UB] nor stop the audit. The tuning constant is fixed before the audit as $a = -2\log\alpha + \log(1 - 2\log\alpha)$, which nearly minimises the radius at the planned number of pairs n^*. The protocol runs one copy per persona, at level $\alpha/|\mathcal{D}_A|$, and the same construction applied to the pair differences $O_{a,i} - O_{r,i}$ gives a confidence sequence for $\mathrm{AA}(d)$ when $\lambda = 1$.

Theorem 6.2 (Validity of Algorithm 1). *Let Assumption A3, Assumption A4(a) and Assumption A6 hold, so that the pairs $(O_{a,i}, O_{r,i})$ are independent and identically distributed. Let $\mathbb{E}O_a, \mathbb{E}O_r > 0$, $\lambda = \vartheta/\mathrm{AR} \in [\lambda_-, \lambda_+]$, and let $\psi_i = O_{a,i}/\mathbb{E}O_a - O_{r,i}/\mathbb{E}O_r$ have variance $\sigma_\psi^2 \in (0, \infty)$.*

(a) Approximate time-uniform coverage. *Let the burn-in end after $n_{\mathrm{b}} = \lceil cn^* \rceil$ pairs, for a fixed $c > 0$. Then, as the planned number of pairs $n^* \to \infty$,*

$$\limsup_{n^* \to \infty} \mathbb{P}(\exists\, n \geq n_{\mathrm{b}} : \log \mathrm{AR} \notin C_n) \leq \bar{\alpha}(c) = \mathbb{P}\big(\exists\, s \geq c : |\mathbb{B}_s| \geq h(s)\big) \leq \alpha,$$

where $\mathbb{B}$ is a standard Brownian motion. Moreover $\bar{\alpha}(c) \to \alpha$ as $c \to 0$, so the bound is not vacuous. Consequently, the probability that the audit misses $\log \mathrm{AR}$ *at a random number of pairs $N \geq n_{\mathrm{b}}$, stopping or not, and the probability that rule S1 or S2 states a conclusion that the true* AR *contradicts, are at most $\bar{\alpha}(c) + o(1)$.*

(b) Exact variant. *Let $\mathrm{Cap}_n(\vartheta)$ be the hedged capital process of Waudby-Smith and Ramdas (2024), with predictable bets, for the null hypothesis that $O_{a,i} - \vartheta O_{r,i} \in [-\vartheta, 1]$ has mean zero. Then*

$C_n^{\mathrm{bet}} = \{\log \vartheta - \log \lambda' : \max_{n' \le n} \mathrm{Cap}_{n'}(\vartheta) < 1/\alpha,\ \lambda' \in [\lambda_-, \lambda_+]\}$ *satisfies* $\mathbb{P}(\exists n \ge 1 : \log \mathrm{AR} \notin C_n^{\mathrm{bet}}) \le \alpha$ *for every sample size, without asymptotics.*

Proof. See Appendix B.6. Part (a) couples the linearised log ratio with a Brownian motion (Komlós et al., 1975, 1976), rescales time by n^* and applies Ville's inequality to the normal mixture of exponential martingales (Robbins, 1970); tying the mixture to n^* is what keeps the limit informative about the level. Part (b) is Ville's inequality for the hedged capital process. □

What is exact and what is approximate. We state the approximations plainly.

(1) Part (a) is a large-sample statement, with the planned horizon and the burn-in growing together, so the guarantee is approximate at the protocol's sizes. The normal approximation is poorest in the first wave, when only a few pairs are available and the outcome is skewed. That is why one wave of burn-in ($b = 1$) is registered. Even with it, at five pairs per wave the miss rate exceeds its nominal level at $\alpha = 0.0125$ (coverage 98.0–98.8% against 98.75%), though not at $\alpha = 0.05$ (95.0–96.2%) (Section 7.7). If the time-uniform coverage falls below the nominal level at the registered sizes, the exact sequence of part (b) is the fallback.

(2) The double-sampled outcome is harder for the normal approximation than the confirmed-flagged one. At the protocol's sizes each coded unflagged position stands for about 16 recorded positions, so a single false negative found among the codes adds about 0.04 to $\widetilde{E}_u$, against a mean exposure near 0.02: the distribution of $\widetilde{E}_u$ is heavy-tailed, and a few puppets can dominate a wave. In the simulation of the registered design, with waves of 20 pairs and one burn-in wave, the excess kurtosis of $\widetilde{E}_u$ reaches about 13 for the persona with the lowest exposure, yet the sequences for AR and AA are conservative, with time-uniform coverage of 98.4–99.4% at $\alpha = 0.05$, null cases included, because a boundary built for continuous monitoring is inspected only five times after the burn-in (Section 7.6). That is evidence at stated inputs, not a guarantee: tails heavier than the simulation produced, or smaller waves, could break the approximation.

(3) For the confirmed-flagged outcome, false negatives enter only through λ_S; a confidence sequence for λ_S at level α_v from coded unflagged positions can replace the declared range, at level $\alpha + \alpha_v$.

(4) Under drift, a time-uniform central limit theorem of Lindeberg type (Waudby-Smith et al., 2024) makes the sequence target the ratio of window-averaged means, $\lambda\overline{\mathrm{AR}}$ if λ does not drift, and nothing beyond that.

(5) The exact variant (b) pays for the range of its observations: O_u can lie anywhere in $[0, 1]$ while its typical value is near 0.02, so the capital grows slowly. It is a robustness check.

Peeking. For comparison, consider the analyst who recomputes a fixed-sample 95% interval after every wave and stops when it excludes the null. By the law of the iterated logarithm, the standardised estimate leaves $[-1.96, 1.96]$ with probability one if monitoring continues indefinitely, and with eight equally spaced looks at a Gaussian random walk the probability that some look rejects is 17.6% rather than 5% (Armitage et al., 1969). Algorithm 1 pays for the right to look by a wider radius, quantified in Proposition 6.3(c). Section 7 compares the two on the synthetic recommender.

6.3 Power and sample size

The design question is how many puppets, sessions per puppet and codes are needed to conclude that $\mathrm{AR} = r_1$ lies outside the null region bounded by r_0. For detection, $r_0 = 1$; for the materiality rule S1, $r_0 = \delta$;

and the registered alternative is $r_1 = \delta^2$ in both cases (Section 8.2). Let n be the number of puppets per arm, which is also the number of pairs, and let $\sigma^2_{\mathrm{s},\pi}$ and γ_π be the variance and the intra-puppet correlation of the *true* session exposures. Write

$$\sigma^2_{E,\pi} = \frac{\sigma^2_{\mathrm{s},\pi}\{1 + (S-1)\gamma_\pi\}}{S}, \qquad \sigma^2_{\mathrm{c},\pi} = \frac{\varsigma^2_\pi}{kTS}, \qquad \varsigma^2_\pi = \mu_\pi \mathrm{Se}(1-\mathrm{Se}) + (1-\mu_\pi)\mathrm{Sp}(1-\mathrm{Sp}). \quad (6.5)$$

The first is the variance of a puppet's true exposure, with the design effect of Kish (1965). The second is the variance that the classifier adds to $\bar{Q}_u$ when its errors are independent across positions given the labels, and ς^2_π is the corresponding variance per position. Let

$$V = \sum_\pi \frac{\sigma^2_{E,\pi} + \sigma^2_{\mathrm{c},\pi}/J^2}{\mu^2_\pi}, \qquad V_0 = \sum_\pi \frac{\sigma^2_{E,\pi}}{\mu^2_\pi},$$

let $\Delta = |\log r_1 - \log r_0|$, and let $z = z_{1-\alpha/2} + z_{1-\beta}$. The validation term depends on the design. With error rates pooled over the arms (external design under Assumption A5), with error rates stratified by arm (external design per arm, $n_{1\pi}$ positives and $n_{0\pi}$ negatives), and with internal double sampling ($m_{1\pi}$ flagged and $m_{0\pi}$ unflagged codes per arm), it is

$$V^{\mathrm{pool}}_{\mathrm{val}} = \frac{(1/\mu_a - 1/\mu_r)^2\,\mathrm{Sp}(1-\mathrm{Sp})}{J^2 n_0}, \qquad V^{\mathrm{st}}_{\mathrm{val}} = \sum_\pi \frac{1}{J^2_\pi}\left\{\frac{\mathrm{Se}_\pi(1-\mathrm{Se}_\pi)}{n_{1\pi}} + \left(\frac{1-\mu_\pi}{\mu_\pi}\right)^2 \frac{\mathrm{Sp}_\pi(1-\mathrm{Sp}_\pi)}{n_{0\pi}}\right\},$$
$$V^{\mathrm{ds}}_{\mathrm{val}} = \sum_\pi \frac{1}{\mu^2_\pi}\left\{\frac{q^2_\pi \mathrm{PPV}_\pi(1-\mathrm{PPV}_\pi)}{m_{1\pi}} + \frac{(1-q_\pi)^2 \mathrm{FOR}_\pi(1-\mathrm{FOR}_\pi)}{m_{0\pi}}\right\}. \quad (6.6)$$

In $V^{\mathrm{pool}}_{\mathrm{val}}$ the specificity enters through the difference $1/\mu_a - 1/\mu_r$, because a common error in $\widehat{\mathrm{Sp}}$ shifts both arms. In $V^{\mathrm{st}}_{\mathrm{val}}$ the two arms' errors are independent and nothing cancels.

Proposition 6.3 (Power and sample size). *Let* $\log \mathrm{AR} = \log r_1$, *with n puppets per arm and S sessions per puppet.*

(a) Fixed sample. *Under the conditions of Theorem 6.1(b) or (d), with* V_{val} *the validation term (6.6) of the design used, the normal approximation to the probability that the interval for* $\log \mathrm{AR}$ *excludes* $\log r_0$ *is at least* $1-\beta$ *if and only if*

$$n \;\geq\; n_{\mathrm{fix}} = \frac{z^2 V}{\Delta^2 - z^2 V_{\mathrm{val}}}, \qquad \textit{which requires} \qquad \Delta^2 > z^2 V_{\mathrm{val}}. \quad (6.7)$$

With the other stratum of validation unlimited and equal sizes in the two arms, the requirement is $n_0 > n_0^{\min}$, $n_{0\pi} > n_{0,\mathrm{st}}^{\min}$ *or* $m_{0\pi} > m_0^{\min}$, *where*

$$n_0^{\min} = \frac{z^2(1/\mu_a - 1/\mu_r)^2 \mathrm{Sp}(1-\mathrm{Sp})}{J^2\Delta^2}, \qquad n_{0,\mathrm{st}}^{\min} = \frac{z^2}{\Delta^2}\sum_\pi \left(\frac{1-\mu_\pi}{\mu_\pi}\right)^2 \frac{\mathrm{Sp}_\pi(1-\mathrm{Sp}_\pi)}{J^2_\pi},$$
$$m_0^{\min} = \frac{z^2}{\Delta^2}\sum_\pi \frac{(1-q_\pi)^2 \mathrm{FOR}_\pi(1-\mathrm{FOR}_\pi)}{\mu^2_\pi}.$$

Below these sizes no number of puppets achieves the target power. The ratio V/V_0 *is the variance inflation due to classifier noise, and* V_{val} *is the inflation due to validation. For double sampling, the factor* J_π^{-2} *in the classifier part of V becomes* $(\mathrm{PPV}_\pi - \mathrm{FOR}_\pi)^2$*; we write* V^{ds} *for the result.*

(b) Sessions per puppet. *Write* $V = v_{\mathrm{p}} + v_{\mathrm{s}}/S$, *with* $v_{\mathrm{p}} = \sum_\pi \sigma^2_{\mathrm{s},\pi}\gamma_\pi/\mu^2_\pi$ *and* $v_{\mathrm{s}} = \sum_\pi\{\sigma^2_{\mathrm{s},\pi}(1-\gamma_\pi) + \varsigma^2_\pi/(kTJ^2)\}/\mu^2_\pi$. *If a puppet costs* c_p *and each of its sessions* c_s, *the cost* $n(c_p + Sc_s)$ *of meeting* (6.7) *with* $V_{\mathrm{val}} = 0$ *is minimised at* $S^* = \{(c_p/c_s)\, v_{\mathrm{s}}/v_{\mathrm{p}}\}^{1/2}$.

(c) Sequential. *For a normal-mixture confidence sequence tuned to* n^* *pairs, the radius at* n^* *is* $c_\alpha(V/n^*)^{1/2}$, *where*

$$c_\alpha = h(1) = \left\{2(1+a^{-1})\log\!\left((1+a)^{1/2}/\alpha\right)\right\}^{1/2}$$

does not depend on n^*. *A horizon of* $n_{\mathrm{seq}} = (c_\alpha + z_{1-\beta})^2 V/\Delta^2$ *pairs gives power at least* $1-\beta$ *by the horizon, to the same approximation, and the expected number of pairs used is smaller because the audit can stop early.*

(d) Confirmed-flagged outcome. *For* $O_u = Y_u$ *with* m *codes per puppet,* V *is replaced by*

$$V_Y = \sum_\pi \frac{1}{\mu^2_\pi}\left\{\sigma^2_{E,\pi} + \frac{\mu_\pi(1-\mathrm{Se}_\pi)}{\mathrm{Se}_\pi kTS} + \frac{q^2_\pi \mathrm{PPV}_\pi(1-\mathrm{PPV}_\pi)}{m\,\mathrm{Se}^2_\pi}\right\},$$

up to the finite-population correction of the coding and the variation of PPV *across puppets. With the declared range* $[\lambda_-, \lambda_+]$, *the horizon becomes* $n_{\mathrm{seq}} = (c_\alpha + z_{1-\beta})^2 V_Y/\{\Delta - \log(\lambda_+/\lambda_-)\}^2$, *which requires* $\Delta > \log(\lambda_+/\lambda_-)$.

Proof. See Appendix B.7. □

For $\alpha = 0.05$, $c_\alpha = 3.04$, against $z_{0.975} = 1.96$. For the per-persona level $\alpha = 0.0125$ of a four-persona audit, $c_\alpha = 3.50$, against $z_{0.99375} = 2.50$. Being allowed to look after every wave therefore costs a horizon $\{(c_\alpha + z_{1-\beta})/(z_{1-\alpha/2} + z_{1-\beta})\}^2$ times the fixed sample size, which is 1.9 at $\alpha = 0.05$ and 1.7 at $\alpha = 0.0125$ for power 0.8. Early stopping recovers part of this, and Section 7 reports the distribution of stopping times.

Table 8 in Section 8.3 applies Proposition 6.3 to invented but plausible inputs at the protocol's design. Three conclusions hold across its rows. First, materiality is expensive: with four personas it needs about five times as many puppets as detection, because Δ falls from log 1.5625 to log 1.25 and the per-persona level is stricter, and its horizon, about 610 pairs per persona, is an order of magnitude beyond a conventional small-lab design with 40 puppets per cell over eight waves of five pairs. Second, validation can bind before puppets do: with error rates stratified by arm, as protection against differential error requires, about 21,400 coded negatives are needed per arm and persona before any number of puppets suffices, whereas double sampling's coding grows with the number of puppets, and the confirmed-flagged outcome replaces the floor by the range for λ_S, a range of ±5% more than tripling its four-persona horizon. Third, the optimal number of sessions per puppet is small, $S^* \approx 3$–7, because sessions of one puppet are correlated, and fewer sessions also mean a smaller footprint.

7 Simulation study

This section tests the tools of Sections 3 to 6, and the registered analysis of Section 8, on a synthetic recommender whose amplification is known. Every number comes from `results/sim_results.json`, which the code in `code/` produces. No real platform, item or user is involved, nothing here is a finding about any service, and every figure is stamped accordingly. E1 evaluates the correction for misclassification (Theorem 6.1) and E1b the variance estimators of Theorem 5.1; E2 evaluates interference (Proposition 5.2) and E3 the decomposition of Theorem 3.3. E4 evaluates the Markov model (Proposition 4.1 and Theorem 4.2) on consumed items, and E4b on the slate states that the protocol registers. E5b simulates the registered

primary analysis (Appendix D.7) and the pilot check that guards it, and E5 evaluates Algorithm 1 in a conventional small-lab design, kept as a benchmark. Appendix C gives the configurations, seeds, versions, runtimes, reproduction commands and further results.

7.1 The synthetic world

Catalogue and policies. The catalogue has 5,000 items in six invented topics, embedded in eight dimensions, one of which is a direction on which antisemitic items load. Items are neutral (4,401), borderline (399), coded antisemitic (100) or explicit antisemitic (100), with $L_c = 1$ for the last two; the antisemitic share ranges from 0.6% of items in two topics to 11.5% in a third. Borderline and antisemitic items also have a higher engagement prior ("bait"), popularity and posting rate. These are assumptions, not estimates (Table 10). Each slate has $k = 4$ items. The algorithm ranks an eligible pool of 50 items, redrawn at every step from the topics that the account follows, together with 15 neighbours of the current item and 15 trending items, by $2\langle v_u, x_c\rangle + 0.3\log(\text{popularity}_c) + \text{bait}_c$ plus Gumbel noise, and moves the user embedding toward each opened item, $v_u \leftarrow 0.9\, v_u + 0.1\, x_c$. The references of Section 3.3 show the pool's four newest, four random, or four most popular items.

Classifier. An impression is flagged when a type-specific mean plus normal noise is positive. Under ND, specificity is 0.98 on neutral and borderline items and sensitivity 0.85 on both antisemitic types. DU flags 40% of borderline items, and DD has Se = 0.35 on coded and 0.95 on explicit items. NDI has the rates of ND, but its noise is fixed per item. Error never depends on the arm, so arm-level rates differ only when the arms show different mixtures of types (Theorem 6.1(c)). Scores fall into five bands, of which the top two are flagged.

Accounts and personas. Protocol puppets (Appendix D.2) are trained by opening 30 seed items from the 200 neutral items closest to their persona. They then run 10 sessions of $T = 20$ steps (800 recorded impressions), of which E4b and E5b use four, as the protocol registers. At each step they open the highest-ranked item that is similar to the persona and scores below $s_{\text{safe}} = -0.5$ (avoid-on-flag), so the true $\text{AR}(d)$ depends slightly on the classifier. Naturalistic accounts (E3, E4) choose by a multinomial logit in taste, appetite for each content state, bait, position and similarity to the current item. The four personas are coded P1 to P4; P1 is the baseline d_0, and Table 11 maps each code to the candidate persona of the protocol for which it stands in. They are topic weights in the invented catalogue, with appetites for naturalistic accounts, and describe no real group of users.

Interference and demand. Global popularity, which the algorithm and the popularity-only feed use, is frozen except in E2. There, puppets' openings feed it, with a half-life of one day. In E3, users' taste drifts toward what they open, their appetite for borderline and antisemitic items rises with the share of such openings, and a growing share of their pool comes from such sources that they now follow. The Monte Carlo standard error (MCSE) of a coverage near 95% is 0.7 points with $R = 1{,}000$ audits, 0.5 with 2,000, and 4.4 with E2's 24 replications.

7.2 Misclassification (E1)

For each regime we simulated pools of 1,000 protocol puppets per persona and policy, without interference. An audit draws $U \in \{5, 10, 20, 40, 80\}$ puppets per cell, $R = 1{,}000$ times, and validates 500 impressions per arm (100 per score band, weighted by inverse inclusion probabilities). The estimators are the oracle with true labels, the naive ratio of flagged shares, the Rogan–Gladen correction with error rates pooled over

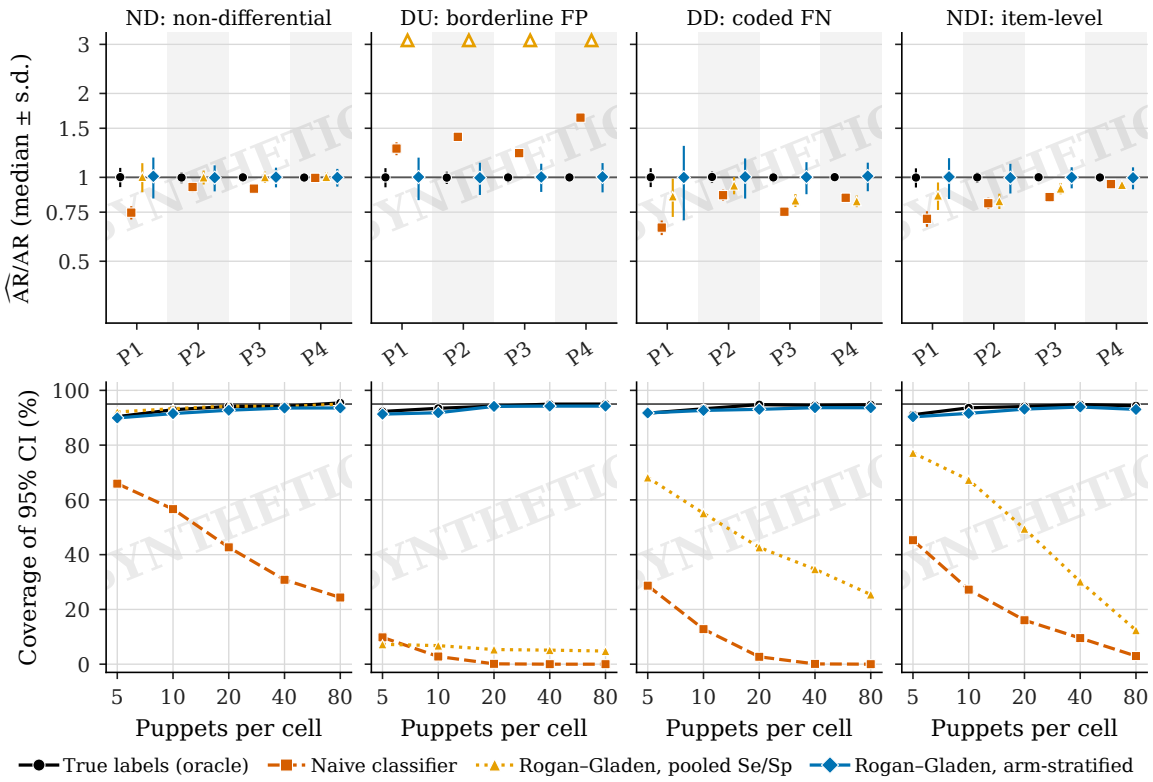


Figure 3. Synthetic data, E1: estimation of AR relative to the reverse-chronological reference (R = 1,000 audits per configuration; personas P1–P4 of Table 11). Top: estimate divided by the true AR, median ± one Monte Carlo s.d., with 20 puppets per persona–arm cell; open triangles on the upper edge mark medians beyond the axis (pooled correction under DU, medians 3.4 to 31). Bottom: coverage of nominal 95% intervals, averaged over personas, against puppets per cell. ND: non-differential; DU: borderline false positives; DD: coded false negatives; NDI: rates of ND, deterministic per item.

the arms, and the correction with error rates by arm, which is the double-sampling estimator (6.3) (Figure 3 and Table 5).

The reference is part of the estimand. Under ND, AR is 1.79, 1.33, 1.67 and 1.04 for P1 to P4 relative to the reverse-chronological feed, 2.64, 1.76, 2.17 and 1.29 relative to the random feed, and 2.76, 2.05, 1.17 and 1.16 relative to the popularity-only feed, so the ranking of personas changes with the reference (Section 3.3). With true labels, intervals cover in 93.2–95.3% of audits with 20 puppets per cell and in 89.8–93.1% with five.

Non-differential error. The naive ratio is attenuated toward one by 25.3%, 7.7%, 8.9% and 0.4%, as Theorem 6.1(a) predicts. The bias does not shrink with more puppets, so coverage falls as the audit grows (P1: 25.5% with five puppets, 0.0% with 20). Both corrections have median biases below 1% and coverage of 91.8–95.6%.

Differential error. For P1 the algorithm shows more borderline content than the reverse-chronological feed (11.2% against 3.0% of impressions) and a larger coded share of antisemitic content (63% against 48%), so its specificity is lower under DU and its sensitivity under DD, although no type is classified differently by arm. Under DU the naive ratio is biased upward by 22–64%, and for P4 it turns a true 0.89

Table 5. Synthetic data, E1: amplification ratio relative to the reverse-chronological reference, with 20 puppets per persona–arm cell (R = 1,000; validation of 500 impressions per arm). Truth: AR of the pools; it differs between regimes because the avoid-on-flag rule uses the classifier. AR: median estimate. Bias: median relative bias (%). Cov.: coverage of the nominal 95% interval (%; MCSE at most 1.6). RG: Rogan–Gladen correction, error rates pooled over arms or stratified by arm. For P1 under DU the pooled correction is defined in only 421 of 1,000 audits, because the corrected reference exposure is otherwise not positive; its median is over those, and the others count as misses.

		P1			P2			P3			P4		
Classifier error	Estimator	AR	Bias	Cov.	AR	Bias	Cov.	AR	Bias	Cov.	AR	Bias	Cov.
Non-differential	Truth		1.79			1.33			1.67			1.04	
	True	1.80	+0.2	94	1.33	−0.2	94	1.67	+0.0	95	1.04	−0.2	93
	Naive	1.34	−25.3	0	1.23	−7.7	50	1.52	−8.9	27	1.04	−0.4	94
	RG pooled	1.80	+0.4	96	1.33	+0.0	94	1.67	+0.2	94	1.05	+0.3	93
	RG stratified	1.81	+0.9	92	1.33	−0.3	92	1.67	+0.2	93	1.04	−0.1	94
Differential: borderline FP	Truth		1.62			1.15			1.38			0.89	
	True	1.62	−0.0	95	1.15	−0.3	94	1.38	−0.1	94	0.88	−0.1	94
	Naive	2.05	+26.8	0	1.61	+39.7	0	1.69	+22.2	0	1.45	+63.8	0
	RG pooled	31.35	$> +10^3$	21	4.39	+281.6	0	4.72	+241.8	0	3.39	+282.8	0
	RG stratified	1.62	+0.3	93	1.15	−0.4	94	1.38	+0.3	95	0.89	+0.4	95
Differential: coded FN	Truth		1.82			1.36			1.78			1.07	
	True	1.82	+0.1	95	1.36	+0.3	94	1.78	−0.1	94	1.07	+0.2	95
	Naive	1.20	−34.0	0	1.17	−13.7	10	1.34	−24.7	0	0.90	−15.5	1
	RG pooled	1.55	−14.5	80	1.27	−6.5	83	1.47	−17.3	6	0.88	−17.9	2
	RG stratified	1.82	−0.1	91	1.36	+0.4	93	1.78	+0.2	93	1.08	+1.0	95
Non-diff. by type, item-level	Truth		1.82			1.30			1.69			1.03	
	True	1.82	−0.2	94	1.30	−0.0	95	1.69	+0.1	94	1.03	−0.1	94
	Naive	1.29	−29.0	0	1.05	−19.3	1	1.43	−15.1	2	0.97	−5.5	61
	RG pooled	1.57	−13.9	70	1.07	−17.8	12	1.54	−8.8	49	0.97	−6.0	67
	RG stratified	1.83	+0.4	93	1.30	−0.3	92	1.69	−0.0	94	1.02	−0.4	93

into 1.45. The pooled correction is worse: it subtracts a false-positive rate that is too large for the reference arm. For P1 the corrected reference exposure was not positive in 579 of 1,000 audits, and elsewhere the median ratio was 31.3 against a truth of 1.62. Under DD the naive ratio is biased by −14 to −34%, and for P4 the naive (0.90) and pooled (0.88) ratios put a true 1.07 below one, as in scenario I of Table 4. Under NDI each item is always classified in the same way, so an arm's sensitivity depends on which antisemitic items it shows (0.69 against 0.79 for P2), and the naive and pooled ratios are biased by up to −29.0% and −17.8%. In all three regimes, correction by arm has median biases within 1% and coverage of 90.7–95.2%; with five puppets per cell, its coverage is 88.1–94.8% and its median bias within 2.5%. Validation by arm, as in the protocol, is the safe default; the pooled correction is not.

Variance estimators (E1b). When accounts sit in different experiment buckets, the algorithmic arm's design effect rises to 2.75–5.65, and session-level intervals for AA cover in only 62.9–82.5% of audits, against 93.0–93.2% for cluster-robust ones (Theorem 5.1(c)); with external validation, the delta-method s.d. of $\log \widehat{\mathrm{AR}}$ understates by up to 33% when $\mathrm{cv}_{\mathrm{Sp}}$ is large, as Section 6.1 warns (Table 13 and further results in Appendix C.6).

7.3 Interference (E2)

Each audit has 15 protocol puppets per persona and arm (algorithmic, reverse-chronological, popularity-only) and 8 untrained sentinels, whose openings feed global popularity, with background volumes that give the puppets 0.02 to 0.3 of all openings, a load $\bar{\ell}$ (Section 5.3) of 0.023 to 0.331 ($R = 24$ paired replications with common random numbers; Figure 6 and Table 14 in Appendix C.6). The bias of $\widehat{\mathrm{AA}}_{\mathrm{chron}}$ is negative for every persona and grows with the load. At share 0.3 it is −19, −45, −23 and −247% of the no-audit value, so for P4 interference reverses the sign of the estimate, and Wald intervals cover in only 25.0–70.8% of replications. Avoid-on-flag puppets rarely open antisemitic items, so the popularity they add goes to other items, and exposure under the learning feeds falls with the load: case (i) of Proposition 5.2(b). Staggered waves lower the load to 0.118 and the bias by 64–78%, and extrapolating to zero load from shares 0.1 and 0.3 (Proposition 5.2(c)) leaves biases within 1.2 SE of zero, at 2.3–2.8 times the variance of one audit. The sensitivity interval of Proposition 5.2(d), computed with the *oracle* ξ_Δ, which no audit observes, restores coverage to 95.8–100%; it shows what a correct postulate buys, not what an audit can achieve. Sentinel exposure falls by 0.0063 per unit of load, against 0.0075–0.0355 for the trained algorithmic arms, so sentinels are evidence about ξ, not a bound on it. Further results are in Appendix C.6.

7.4 Supply, demand and the fixed-demand audit (E3)

The population has 3,000 synthetic users with naturalistic choice and new accounts, whose initial demand F_0 is drawn from the four personas with individual noise. Two copies are formed for $H \in \{3, 10, 20, 40\}$ sessions, one under each policy, while demand evolves, and each copy is then evaluated under both policies with demand held fixed, which gives the four cells $\bar{\mu}(a; b)$ of Theorem 3.3 (Figure 7 in Appendix C.6). After 40 formation sessions, the total effect $\tau = 2.58$ percentage points (SE 0.08) splits into supply $\tau_{\mathrm{sup}} = 1.70$, demand shaping $\tau_{\mathrm{dem}} = 1.25$ and an interaction $\iota = -0.37$ (SE 0.07): the algorithm adds less to demand that it has formed. The fixed-demand audit, which holds each user's initial demand fixed in a new account (Proposition 3.4 at $\Lambda = F_0$), gives AA = 1.64 points, close to the supply effect, and falls short of the total by 0.23 points at $H = 3$ and 0.94 (SE 0.07) at $H = 40$, as Proposition 3.4 says it must. The rates at which demand changes are invented, so E3 shows what a fixed-demand audit targets, not how real demand evolves.

7.5 Markov predictions (E4, E4b)

Consumed items (E4). For each persona and policy, 250 naturalistic accounts with fixed demand, which always open an item, run three burn-in sessions; a chain over the states N, B and A of the opened items is estimated from 10 sessions of 20 steps (Proposition A.4) and compared with one further session of 200 steps (Figure 4(a)). Out of sample, e_{200} is predicted within 7% in six of eight cells, and errs by −22% and +26% in the algorithmic arms of P1 and P2, where consumed exposure is rarest (about 0.3%). The chains mix fast (Kemeny's constant 1.98–2.17, $\kappa_{\mathcal{A}}$ 0.50–0.52), so this world lacks the sticky regions in which Theorem 4.2 predicts the largest magnifications. The exact identity of Theorem 4.2(a) tracks the effect of a score tilt toward antisemitic items, the bound of part (b) holds but is loose, and the closed form of part (c), from the base chain alone, predicts the effect of replacing the next item by an antisemitic one within 9% (Appendix C.6). But consumed exposure is not exposure. For the same accounts, impression exposure is 3.3–23.7 times consumed exposure in the algorithmic arm, and the ratio of the arms differs in direction: consumed items give 0.87, 0.86 and 0.62 for P1, P2 and P4, impressions 1.60, 1.14 and 1.06.

Slate states (E4b). The protocol therefore applies the chain to slates: the state of step t is the most severe true state among its k recorded positions, observed through the classifier (A if any position is flagged,

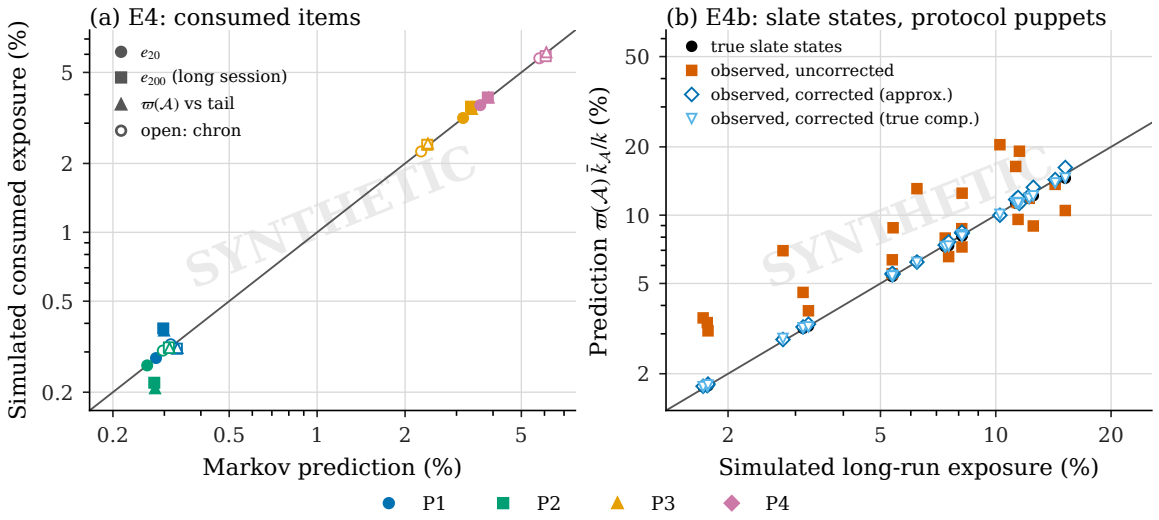


Figure 4. Synthetic data, E4 and E4b: Markov predictions against simulation (log axes). (a) E4, chains of consumed items of naturalistic accounts (250 per persona and policy): e_{20}, e_{200}, and $\varpi(\mathcal{A})$ against the second half of a 200-item session; open markers: reverse-chronological. (b) E4b, slate states of protocol puppets (1,000 per persona, policy and regime ND, DU, DD): long-run impression exposure predicted by the chain, $\varpi(\mathcal{A})\bar{k}_{\mathcal{A}}/k$, against the second half of a 200-step session, for true slate states, observed states, and observed states corrected in the two-state chain {A, not A} under the protocol's approximation or with the true composition of the slates. The perturbation results of E4 are in Figure 8.

else B if any lies in the score band just below the threshold), and the observed transitions are corrected with a slate-level confusion matrix (Appendix D.6). E4b re-simulates the E1 pools of regimes ND, DU and DD with the same seeds, recording slate states, plus one session of 200 steps per puppet. Chains are estimated from the 10 sessions of all 1,000 puppets per cell and corrected with the cell's known item-level error rates, under the protocol's approximation (a slate not in A has k non-antisemitic items, a slate in A one antisemitic item) or with the simulated composition. The approximation misreads a slate as A with probability $1 - \mathrm{Sp}^k = 0.076$–$0.079$ under ND and DD and 0.084–0.587 under DU, against true shares of A slates of 0.069–0.456. (1) With true states, the chain predicts long-run impression exposure within −4.2 to +2.5% in all 24 cells (Figure 4(b)), but the test is weak: slates here have almost no memory, and long-run exposure differs from the 20-step session average by only −2.0 to +5.6%. (2) Uncorrected, the observed chain errs by −31.0 to +150.6%, and can reverse the comparison of the arms: for P4 its ratio is 0.94 under DD and 1.25 under DU, against 1.10 and 0.91. (3) The three-state correction is not identifiable under ND and DD, where the classifier scores borderline and neutral items alike (condition numbers of the slate confusion matrix 394 to 56,587); under DU it is identifiable but biased by +7.0 to +91.6%. Collapsed to {A, not A}, the correction errs by −2.5 to +6.8% in the long run and by up to +11.5% at 20 steps, and by −4.1 to +2.9% with the true composition. (4) At the registered size, 120 puppets per arm with four sessions each ($R = 500$), the corrected chain's s.d. of the log ratio of the arms is 1.09–2.33 times that of the session average, its median is off target by up to 0.057, and negative entries had to be clipped in up to 4.4% of replications. At the audited horizon the session average is unbiased and more precise, and the chain's distinctive outputs rest on a corrected two-state chain whose error is small here only because this world barely tests extrapolation. The Markov summaries of the protocol are therefore best treated as exploratory, computed on the two-state chain, with the three-state chain reported only if the pilot shows that the borderline score band separates B from N.

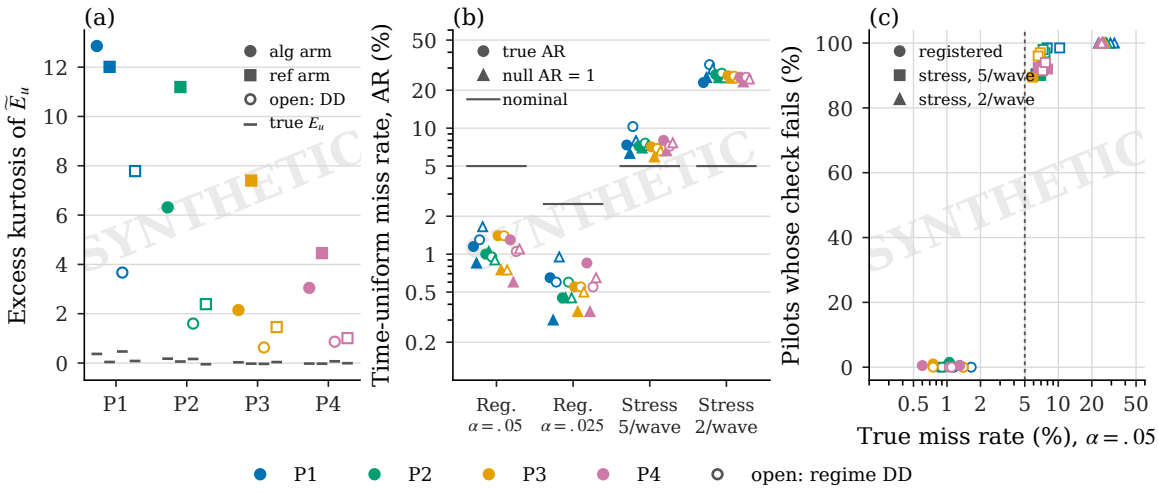


Figure 5. Synthetic data, E5b: the registered primary analysis ($\widetilde{E}_u$; waves of 20 pairs, $W_{\max} = 6$, $b = 1$; $R = 2{,}000$ audits; filled: regime ND, open: DD). (a) Excess kurtosis of $\widetilde{E}_u$ in each arm, against that of the puppet's true exposure E_u (dashes). (b) Time-uniform miss rate of the sequence for AR: registered design at $\alpha = 0.05$ and 0.025, and two stress designs at $\alpha = 0.05$ with no burn-in and the mixture tuned to 40 pairs, waves of 5 and of 2 pairs (log scale; bars: nominal level). (c) The pilot check: share of 200 simulated pilots (10 puppets per cell) in which the resampling estimate of the miss rate exceeds α by more than its Monte Carlo error, against the true miss rate.

7.6 The registered primary analysis (E5b)

E5b simulates the primary analysis of Appendix D.7. Pairs of one algorithmic and one reverse-chronological puppet are resampled from the detailed pools, each puppet contributing four of its ten sessions, drawn at random (320 recorded impressions). Each puppet has $m = 10$ flagged, 5 opened and 20 other unflagged positions coded, the last allocated over the three unflagged score bands in proportion to $n_h s_h$ (Neyman, 1934), at least one per non-empty band, with s_h from an ideal pilot (puppets average 33.8–35.0 codes). Waves have 20 pairs, $W_{\max} = 6$ (120 pairs, about rung R4), $b = 1$, and the mixture is tuned to $n^* = 120$. Algorithm 1 runs on $\widetilde{E}_u$ with $\lambda = 1$ for AR, the same construction on pair differences for AA, and the betting sequences of Theorem 6.2(b) for both, at $\alpha = 0.05$ and 0.025, under ND, DD and a null case in which both arms come from the reference pool ($R = 2{,}000$; Figure 5 and Table 6). The synthetic reference exposures are 1.8–11.3%; P1, at 1.76%, is closest to the protocol's illustrative 0.02.

Tails and variance. $\widetilde{E}_u$ is heavy-tailed where exposure is rare. For P1 under ND its excess kurtosis is 12.9 and 12.0 in the two arms, and up to 47.3 in the null case, and single values reach 22.3 times the mean, against an excess kurtosis below 0.5 and a maximum of 3.0 times the mean for the true exposure. Under DD, where coded items are missed in low score bands, 0.40% of P1's reference puppets exceed five times the mean. The closed form $V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}}$ overstates the simulated variance per pair by a factor of 1.42–2.04, which is conservative: given its record, $\widetilde{E}_u$ is unbiased for the puppet's true exposure, so classifier noise enters only through the coding and the classifier term of V^{ds} counts it a second time, and the score bands make the coding variance smaller than the single-stratum $V^{\mathrm{ds}}_{\mathrm{val}}$.

Coverage and conclusions. The tails do not break the normal approximation at this design. Time-uniform coverage is 98.4–99.4% at $\alpha = 0.05$ and 99.1–99.7% at $\alpha = 0.025$, null cases included, and 98.8–99.4% and 99.2–99.8% for AA; without the burn-in it is 98.0–99.1% at $\alpha = 0.05$. The sequences are conservative because a boundary built for continuous monitoring is inspected only five times after the burn-in, with at

Table 6. Synthetic data, E5b: the registered primary analysis, by persona and classifier regime ($\alpha = 0.05$ unless stated; $R = 2{,}000$, MCSE of a coverage near 99% at most 0.3). Kurt.: excess kurtosis of $\widetilde{E}_u$ in each arm (true exposure: below 0.5). max/μ: largest $\widetilde{E}_u$ over the 240,000 simulated puppets of an arm, divided by the arm's mean. Var. ratio: $V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}}$ of Appendix D.8, evaluated at the pool's inputs, divided by the simulated variance per pair. Coverage: time-uniform over waves 2–6 (after the burn-in), normal-mixture sequences. Excl. 1: the final interval for AR excludes 1. Material: conclusion "material amplification" (Table 7). Null: both arms drawn from the reference pool; share of audits with a conclusion that AR = 1 contradicts.

			Tails of $\widetilde{E}_u$			Coverage (%)			Final interval (%)		Null
Persona	Error	AR	Kurt. alg; ref	max/μ	Var. ratio	AR	AR, .025	AA	Excl. 1	Material	wrong (%)
P1	ND	1.79	12.9; 12.0	12.9	1.99	98.9	99.4	99.2	100.0	99.3	0.85
	DD	1.82	3.7; 7.8	10.0	1.51	98.7	99.4	99.1	100.0	84.9	1.65
P2	ND	1.33	6.3; 11.2	8.5	1.76	99.0	99.6	99.1	99.9	12.2	0.60
	DD	1.36	1.6; 2.4	5.3	1.48	99.1	99.4	99.1	99.0	9.7	0.80
P3	ND	1.67	2.1; 7.4	5.5	1.65	98.6	99.5	99.0	100.0	100.0	0.00
	DD	1.78	0.6; 1.5	3.7	1.42	98.6	99.5	99.0	100.0	100.0	0.60
P4	ND	1.04	3.0; 4.5	4.5	1.67	98.7	99.2	98.8	4.2	0.0	0.00
	DD	1.07	0.9; 1.0	3.4	1.43	99.0	99.5	99.1	11.8	0.0	0.35

least 40 pairs. In the null case wrong conclusions occur in at most 1.65% of audits at $\alpha = 0.05$ and 0.95% at $\alpha = 0.025$, and under the alternatives in at most 0.05%. The audit concludes material amplification for P1 (AR = 1.79, 1.82) in 99.3 and 84.9% of audits, after 2.66 and 3.91 waves on average, and for P3 in all audits. For P2, whose ratio of 1.33–1.36 lies just above $\delta = 1.25$, it detects a difference in 99.0–99.9% of audits but leaves materiality undetermined in 87.8–89.3%, and for P4 it concludes no material difference in 99.8% (ND) and 76.6% (DD) of audits. These rates reflect synthetic exposures that are higher and less variable than the illustrative inputs of Table 8 (the coefficient of variation of a puppet's true exposure is 0.16–0.42); they say nothing about the sizes a real audit needs.

The betting fallback. The exact sequences cover in every audit but are nearly powerless at 120 pairs: they exclude AR = 1 only for P3 (3.6% of audits under ND, 56.0% under DD, at $\alpha = 0.05$), and their final intervals for log AR have median widths of 0.85–3.23, against 0.17–0.55 for the normal-mixture sequence.

The pilot check. The check of Section 8.2 resamples the ten pilot puppets per cell (all flagged and 50 unflagged positions coded), re-draws the registered codes from the pilot's codes, and fails when the estimated miss rate exceeds α by more than its Monte Carlo error ($B = 1{,}000$; 200 simulated pilots per configuration). At the registered design it fails in 0–1.5% of pilots at $\alpha = 0.05$ and 0–3.0% at $\alpha = 0.025$, and its mean estimates of the miss rate, 1.07–1.56%, are close to the true 0.60–1.65%. On two stress designs on which the sequence does fail, waves of 5 pairs with no burn-in and the mixture tuned to 40 pairs (coverage 89.7–94.1% at $\alpha = 0.05$) and waves of 2 pairs (68.1–77.0%), it fails in 89.5–98.5% and in 100% of pilots (Figure 5(c)). Here the failure it must catch comes from early looks with few pairs, which ten pilot puppets reproduce; a failure driven by extremes rarer than ten puppets can show is not represented.

7.7 A conventional small-lab benchmark (E5)

E5 runs Algorithm 1 on the confirmed-flagged statistic Y_u in a conventional small-lab design, kept as a benchmark: 40 pairs per persona in 8 waves of five, ten sessions per puppet, $\alpha = 0.0125$ per persona and $b = 1$ ($R = 2{,}000$; Figure 9 in Appendix C.6). With the burn-in, time-uniform coverage is 95.0–96.2%

at $\alpha = 0.05$ and 98.0–98.8% at $\alpha = 0.0125$ (nominal 98.75%); without it, 92.6–93.5% and 96.2–97.0%. A fixed-sample interval recomputed after each of eight waves covers at every look in only 69.5–71.2% of audits, and under the null rejects AR = 1 at some look in 29.2–33.0%, against 2.1–3.4% for Algorithm 1. Within eight waves, materiality stays undetermined for P2 in 93.8% of audits. Under DD, assuming $\lambda_S = 1$ gives coverage of 2.8, 19.8, 0.0 and 0.0%, because the sequence then covers λ_SAR; the range $[0.8, 1.2]$ restores 99.4–100%, at the cost of most of the power.

7.8 What the simulation does and does not show

Where their conditions hold, the results of Sections 3 to 6 hold in finite samples: corrections by arm, cluster-robust intervals, the closed forms of Theorems 4.2 and 6.1, the scope of fixed-demand audits (Proposition 3.4), and the registered primary analysis, whose sequences are conservative at the registered design despite heavy tails, whose closed-form variance is conservative, and whose pilot check rarely fails when the sequence is valid and almost always fails when it is not; its betting fallback is nearly powerless. The methods also fail where the theory says they will: the naive ratio and the pooled correction can lie on the wrong side of one under content-dependent error, the pooled correction explodes when the corrected reference exposure nears zero, interference can reverse the sign of the additive amplification, and assuming $\lambda_S = 1$ under differential sensitivity gives 0–20% coverage. Approximations show at small sizes: with waves of five pairs and no burn-in, the primary sequence misses in up to 10.3% of audits. Chains of opened items misstate exposure, and observed slate chains need a correction that is identifiable only for the two-state chain.

The simulation does not show how any real recommender behaves. The world, the classifier, the personas and the demand dynamics are invented, and the ratios from 0.89 to 2.76, the interference biases and the decomposition shares follow from our parameters. E5b resamples pairs independently from fixed pools, so it has no drift, interference, attrition or coder error; its Neyman allocation uses an ideal pilot; its personas are more exposed than the illustrative inputs, so it does not test the sizes of the feasibility gate, and rung R1 was not simulated; and tails rarer than ten pilot puppets can reveal would defeat the pilot check. E4b's slates have almost no memory, so the sticky regime in which the Markov summaries matter most is not tested. E2 has one interference channel, 24 replications, and no load below 0.023. E3 rests on dynamics of demand for which we have no evidence. And because puppets and users are served by the same rules by construction, the transport condition (Assumption A7), which a real audit can least check, is never at risk.

8 A pre-registered audit protocol

We will register the protocol, time-stamped and with the commit hash of the analysis code and the classifier version, before any audit account is created or any session is run (Nosek et al., 2018). This section summarises it; the full registered text is Appendix D, and the two are one document. Any departure will be reported as a deviation, with its reason. The protocol is written for a generic service and names no platform. It contains no results: every number in it is a design input, a cap, or an evaluation of a closed form at stated illustrative inputs. No step is taken unless the ethical and legal conditions of Section 9 are met.

The sample-size calculation shapes the whole protocol. Under plausible illustrative inputs, a materiality audit of four personas needs about 4,900 trained accounts and 170,000 coded exposures, and it stays in the thousands of accounts under every sensitivity input of Table 8. That is beyond a small lab, and beyond what the footprint rules of Section 9.2 would justify. The protocol therefore commits unconditionally only to a

pilot wave. A registered feasibility gate then selects the most informative confirmatory audit that the pilot shows to be feasible within registered capacity caps, or stops the audit (Section 8.3). Under the illustrative caps, the gate selects the detection of amplification for one persona.

8.1 Design and analysis at a glance

Scope and access. The unit of audit is one recommendation surface of one service, in one EU Member State, during one audit window (Appendix D.1). A surface is eligible only if a reference policy is available on it: the non-profiling option of Article 38 of the DSA, a reverse-chronological option, or logged-out sessions with the same choice rule. Access routes are used in a registered order: vetted access under Article 40(4) or a provider's research interface, then publicly accessible data under Article 40(12), and a sock-puppet audit only where neither can identify the estimand, after legal review and ethics approval.

Personas, exposure and randomisation. Four candidate personas are ranked in a registered order: Jewish interest, conspiracy-adjacent, Middle East news, and a general-interest baseline d_0 (Appendix D.2); none is trained on antisemitic content. At each step a puppet records the first k positions of the slate, which a frozen classifier scores, and opens the highest-ranked relevant item scoring below a threshold s_{safe} under the decision threshold, so that no flagged item is opened. Exposure counts impressions. Waves, separated by washouts, are blocks within which puppets are assigned completely at random to persona–policy cells, with the policy contrast starting at measurement (Appendix D.3). Interference is limited by isolation, washouts and a load cap, and reported through the breakdown value of Proposition 5.2(d) (Appendix D.4).

Validation. Coding is done within puppets, by two annotators blind to arm, persona and score, with adjudication (Appendix D.5). Each puppet contributes $m = 10$ coded flagged positions and $m' = 25$ coded unflagged positions, five of them among the items it opened and twenty stratified by classifier-score bands with the allocation of Neyman (1934). The codes give each puppet's double-sampled exposure $\widetilde{E}_u$ of Theorem 6.1(d), which assumes nothing about error rates in either arm, and exposure-weighted error rates by cell.

Primary analysis. For each confirmatory persona $d \in \mathcal{D}_A$, the primary estimands are $\mathrm{AR}(d)$ and $\mathrm{AA}(d)$ relative to the registered reference, window-averaged under drift (Theorem 5.1(d)), estimated from the double-sampled exposures blocked by wave, and monitored by Algorithm 1 with $O_u = \widetilde{E}_u$ and $\lambda = 1$ at level $\alpha/|\mathcal{D}_A|$, with one burn-in wave (Appendix D.6). If the pilot's upper confidence bound for $\mu_{\text{ref}}(d)$ is below a registered floor (illustratively 0.002), $\mathrm{AA}(d)$ becomes that persona's primary estimand, with an additive margin δ_{A} (illustratively 0.005) in place of $\log \delta$. No population weighting of personas is registered, so every estimand is persona-level. The coverage of the sequence on $\widetilde{E}_u$ is approximate: in the simulation of the registered design it is conservative despite heavy tails (Section 7.6), and the protocol checks it before the first confirmatory wave (Section 8.2).

Secondary analyses. The secondary analyses are: Algorithm 1 on the confirmed-flagged exposures Y_u, with the registered range $[\lambda_-, \lambda_+] = [0.95, 1.05]$ for the sensitivity ratio and the largest symmetric range $[1/\lambda^\dagger, \lambda^\dagger]$ for which its conclusion still holds; the persona-contrast decomposition of Section 3.5, only if all four personas are audited; and the diagnostics of interference, drift (X^2_{drift} of Section 5.4), attrition and error rates by cell. The Markov summaries of Proposition 4.1 and Theorem 4.2 are exploratory. They are computed on slate states with the slate-level misclassification correction, on the two states $\{\mathsf{A}, \text{not } \mathsf{A}\}$ unless the pilot shows that the borderline score band separates B from N, because in simulation the uncorrected slate chain can reverse the comparison of the arms and the three-state correction can be unidentifiable (Section 7.5 and Appendix D.6).

Table 7. Registered decision rules for persona d, based on the final running interval $[\mathrm{LB}, \mathrm{UB}]$ for $\log \mathrm{AR}(d)$, with margin δ; the first matching row applies. Each conclusion is stated for the audited service, surface, jurisdiction, window and persona only, as a statement about the window-averaged $\overline{\mathrm{AR}}(d)$ if policies drift, and with the breakdown value ξ^* of Proposition 5.2(d).

Final interval for $\mathrm{AR}(d)$	Reported conclusion
Entirely above δ	Material amplification relative to the reference policy
Entirely below $1/\delta$	Material de-amplification relative to the reference policy
Entirely within $(1/\delta, \delta)$	No material difference at margin δ
Excludes 1 but straddles δ or $1/\delta$	Difference detected; materiality undetermined
Contains 1 and extends beyond the margin	Inconclusive

8.2 Stopping and decision rules

For each persona, data collection stops at the first wave after the burn-in at which one of the following holds for the running interval $[\mathrm{LB}, \mathrm{UB}]$ for $\log \mathrm{AR}(d)$:

(S1) $[\mathrm{LB}, \mathrm{UB}]$ lies entirely above $\log \delta$ or entirely below $-\log \delta$ (a material difference in one direction);

(S2) $[\mathrm{LB}, \mathrm{UB}]$ lies entirely within $(-\log \delta, \log \delta)$ (no material difference);

(S3) the wave budget $W_{\max}$ is reached.

A persona that stops receives no puppets in later waves. Every confirmatory persona is reported, whatever its result, with the conclusion of Table 7. Early stopping reduces the footprint of the audit, and the sequences make it compatible with an error rate close to the nominal one (Ramdas et al., 2023; Wald, 1945).

The guarantee is approximate, not exact (Theorem 6.2(a)). Before the first confirmatory wave, the time-uniform miss rate at the registered design is estimated by resampling pilot puppets, with the pilot's ratio of mean exposures as the truth. If it exceeds the per-persona level by more than its Monte Carlo error, the exact betting sequence of Theorem 6.2(b) becomes the primary sequence, and the gate is re-evaluated with that sequence's horizon. Otherwise the betting sequence is reported at the stopping time, and a conclusion that it does not support is labelled as resting on the normal approximation. In simulation, the check passes at the registered design in at least 97% of pilots and fails in 89.5–100% of pilots for designs whose coverage breaks down through early looks with few pairs (Section 7.6). Because ten pilot puppets per cell cannot reveal the extreme upper tail of $\widetilde{E}_u$, it guards against such failures, not against failures driven by extremes rarer than the pilot can show (Appendix D.7).

8.3 Sample size, pilot and feasibility gate

Table 8 evaluates the closed forms of Proposition 6.3 at illustrative inputs, which the pilot replaces: $\mu_{\mathrm{ref}}(d) = 0.02$ for every persona, a session-level coefficient of variation of true exposure of 1, intra-puppet correlation $\gamma = 0.3$, $S = 4$ measurement sessions of $kT = 100$ recorded positions ($k = 5$, $T = 20$), Se = 0.85 and Sp = 0.98 in both arms, and power 0.8 at the registered alternative $\mathrm{AR} = \delta^2$, with $\delta = 1.25$, for detection ($r_0 = 1$) and materiality ($r_0 = \delta$) alike. The candidate $S = 4$ rounds up $S^* = 3.6$ of Proposition 6.3(b): it needs more puppets than $S = 10$ but fewer sessions in all, and sessions are the audit's footprint. For the primary analysis, the double-sampled outcome has variance $V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}}$ per pair, with V^{ds} as in Proposition 6.3(a) and $V^{\mathrm{ds}}_{\mathrm{val}}$ of (6.6) at $m_{1\pi} = nm$ and $m_{0\pi} = nm'$, so that $nV^{\mathrm{ds}}_{\mathrm{val}}$ does not depend on n, and the horizon is that of Proposition 6.3(c) with this variance (Appendix D.8). This closed form is

Table 8. Sample sizes from Proposition 6.3, for the rungs of the feasibility gate and for changes of one input. **Illustrative: invented inputs, to be replaced by pilot estimates; not a result and not a registered design.** Base inputs as in the text (design effect 1.9; $m = 10$, $m' = 25$; $c_p/c_s = 3$); detection and materiality share the alternative AR = $\delta^2 = 1.5625$. α_d: level per persona. Per arm and persona: n_{fix}, fixed sample with known error rates; validation floors of Proposition 6.3(a), in coded negatives with rates stratified by arm ($n_{0,\text{st}}^{\min}$) and in coded unflagged exposures for double sampling ($m_0^{\min}$); sequential horizons in pairs for the confirmed-flagged outcome Y_u with $\lambda_S = 1$ or $\lambda_S \in [0.95, 1.05]$, and for the primary outcome $\widetilde{E}_u$. Per persona: $n_0^{\min}$, coded negatives with error rates pooled over the arms. Totals over arms and personas for the primary outcome: trained puppets, excluding the pilot and sentinels, and coded positions, at most $m + m' = 35$ per puppet. S^*: Proposition 6.3(b). Exact evaluation of the closed forms.

			Validation floor			Horizon (pairs)			Primary, total		
Rung or scenario	α_d	n_{fix}	$n_0^{\min}$	$n_{0,\text{st}}^{\min}$	$m_0^{\min}$	Y_u, λ_S=1	Y_u, ±5%	$\widetilde{E}_u$	puppets	codes	S^*
R1: materiality, 4 personas	.0125	277	2,065	21,422	2,634	446	1,464	609	4,872	170,520	3.6
R2: materiality, 2 personas	.025	236	1,760	18,260	2,245	401	1,317	548	2,192	76,720	3.6
R3: detection, 2 personas	.025	59	440	4,565	562	101	167	137	548	19,180	3.6
R4: detection, 1 persona	.05	49	364	3,770	464	89	148	122	244	8,540	3.6
R1, $\gamma = 0.1$	.0125	209	2,065	21,422	2,634	332	1,090	495	3,960	138,600	6.6
R1, $\mu_{\text{ref}} = 0.05$	.0125	225	331	3,188	1,014	394	1,293	456	3,648	127,680	2.8
R1, $\mu_{\text{ref}} = 0.01$	.0125	452	8,258	87,738	5,334	533	1,750	857	6,856	239,960	5.3
R1, Sp = 0.995	.0125	235	506	5,246	2,674	392	1,288	571	4,568	159,880	3.0
R1, Se = 0.70	.0125	316	3,076	31,915	5,268	481	1,579	776	6,208	217,280	4.0
R1, coefficient of variation 0.5	.0125	117	2,065	21,422	2,634	176	576	339	2,712	94,920	5.4
R1, $S = 10$ sessions	.0125	192	2,065	21,422	2,634	358	1,176	518	4,144	145,040	3.6
R3, $\mu_{\text{ref}} = 0.05$	.025	48	71	680	217	89	147	103	412	14,420	2.8
R3, coefficient of variation 0.5	.025	25	440	4,565	562	40	66	77	308	10,780	5.4

conservative: with coding within puppets, classifier noise enters $\widetilde{E}_u$ only through the codes, so the classifier term of V^{ds} counts it twice, and the score bands reduce the coding variance; in simulation it overstates the variance per pair by a factor of 1.4–2.0. The sizes in Table 8 therefore err on the large side, and the pilot replaces them.

Feasibility, stated plainly. Rung R1, materiality for four personas, needs about 4,900 trained accounts and 170,000 coded exposures, and stays in the thousands of accounts in every sensitivity row; rung R2 needs about 2,200 accounts. Detection is much cheaper: rung R3 needs about 550 trained puppets and rung R4 about 250. A conventional small-lab design, with 40 puppets per cell over eight waves of five pairs, $S = 10$ and 2,000 coded items in all, would have power of about 17% for the four-persona materiality test at a fixed sample with known error rates, and about 1% with the double-sampled outcome and sequential monitoring. With $\lambda_S \in [0.95, 1.05]$, the confirmed-flagged outcome needs more than twice the pairs of the double-sampled one on rung R1 and about a fifth more on R3 and R4, but fewer when sessions vary less (rung R3 with coefficient of variation 0.5: 66 against 77); double sampling is primary because it assumes nothing about the sensitivity ratio, not because it is always more efficient. A horizon sized for detection rarely permits rule S2, so on rungs R3 and R4 "difference detected; materiality undetermined" is the conclusion that the audit is powered to reach (Appendix D.8). The pilot is one wave under the full protocol, with all four personas, ten trained puppets per cell, two sentinels per arm, $S = 4$, and heavier coding (all flagged and 50 unflagged positions per puppet); it estimates the inputs of Table 8 and is not part of the confirmatory analysis.

The feasibility gate. Before the pilot, the registration fixes four capacity caps: the number $U_{\max}$ of accounts that the ethics approval and the budget allow, counting every account the audit creates, pilot puppets and sentinels included; the number of positions that the annotation team can code within the protections of Section 9.3, pilot coding included; the load cap $\ell_{\max}$; and the wave budget $W_{\max}$. After the pilot, the required sizes of each rung are computed from the closed forms with each input at the conservative end of an 80% interval, so that an optimistic pilot cannot license an underpowered audit. For variances, correlations and costs this is a bootstrap interval over pilot puppets. For proportions estimated from coded positions, such as the false-omission rate by score band, it is the Clopper–Pearson bound, because with about 500 coded unflagged positions per cell the estimated false-omission rate is often zero, and a bootstrap interval then collapses to a point. The gate selects the first rung, in the order R1 (materiality, four personas, level $\alpha/4$, with the persona-contrast decomposition), R2 (materiality, two first-ranked personas, $\alpha/2$), R3 (detection, two first-ranked personas, $\alpha/2$), R4 (detection, the Jewish-interest persona, α), that meets four conditions: its accounts fit within $U_{\max}$; its coded positions fit within the coding cap; with its puppets spread over at most $W_{\max}$ waves, the load projected from the pilot stays below $\ell_{\max}$; and the resampling check of Section 8.2 passes, or the betting sequence's horizon also meets the first three conditions. If no rung passes, no further accounts are created; the pilot is reported as a feasibility study, without amplification estimates, and the lab pursues platform-side access under Article 40(4) instead. The selected rung and its n, W, S, m, m', n^* and a are registered as an amendment before the first confirmatory wave.

With the illustrative caps $U_{\max} = 400$ accounts, 15,000 coded positions, $\ell_{\max} = 0.01$ and $W_{\max} = 6$, the inputs of Table 8 select rung R4. Its 122 pairs run in six waves of about 20 pairs. With the pilot (80 trained puppets and four sentinels) and four sentinels per confirmatory wave, it creates 352 accounts, and its 8,540 confirmatory codes and about 5,300 pilot codes stay within the coding cap; rung R3 would need about 660 accounts. The randomised-saturation design of Proposition 5.2(c) is run only on rungs R1 and R2: on R3 and R4 its test would have almost no power, and halving the smaller waves would weaken the normal approximation where it is already poorest, so waves there have equal sizes.

9 Ethics and legal considerations

An amplification audit studies a system that real people use, through accounts that are not real people, in order to measure exposure to content that harms a real community. This section sets out the conditions on which the protocol depends; the audit does not proceed unless they are met. We describe the legal landscape as we understand it from the sources cited. This is not legal advice, and before registration the protocol will be reviewed by an institutional review board (IRB) and by legal counsel in the jurisdictions concerned.

9.1 Terms of service and the legal landscape

Terms of service and United States law. The terms of many services prohibit automated access, accounts that do not represent a real person, or both, and a sock-puppet audit typically involves both. The protocol treats the terms of the chosen service as a constraint to be reviewed before registration, not an obstacle to be engineered around. Breaching them is at least a contractual matter: in *hiQ Labs v. LinkedIn*, the District Court for the Northern District of California held that a user agreement prohibiting scraping and fake profiles was enforceable and that the scraper had breached it (hiQ Labs, Inc. v. LinkedIn Corp., 2022). Two decisions concern the federal Computer Fraud and Abuse Act. In *Sandvig v. Barr*, brought by researchers who wished to study discrimination with fictitious accounts, the District Court for the District of Columbia concluded that violating the terms of service of public websites does not violate the statute's

access provisions (Sandvig v. Barr, 2020). In *Van Buren v. United States*, the Supreme Court read "exceeds authorized access" as a question of whether the gates to areas of a computer are up or down, and left open whether contractual limits can set the gates (Van Buren v. United States, 2021). Neither decision makes a given audit lawful. States have their own computer-crime statutes, such as section 502 of the California Penal Code, whose reach need not match the federal statute's, and other jurisdictions apply their own law.

European Union. Article 40(4) of the DSA provides vetted access to data through a Digital Services Coordinator, and Article 40(12) access to publicly accessible data for researchers who meet its conditions (European Commission, 2025c; European Parliament and Council of the European Union, 2022). Their application to a sock-puppet audit, which creates accounts and receives personalised recommendations, has not to our knowledge been settled. Recorded items may contain personal data that reveal religious beliefs or political opinions, special categories under the General Data Protection Regulation (GDPR). Their processing for research needs the exception of Article 9(2)(j), which requires a basis in Union or Member State law, and the safeguards of Article 89(1), including data minimisation and pseudonymisation where the research allows it (European Parliament and Council of the European Union, 2016); the data-handling rules of Appendix D.9 and Section 9.4 are designed to meet them. Copying items for classification and coding is a reproduction of works, which Article 3 of the Directive on copyright in the Digital Single Market permits to research organisations for text and data mining for scientific research on works to which they have lawful access, with copies retained securely (European Parliament and Council of the European Union, 2019).

Order of preference and notice. These considerations fix the order of access routes in Appendix D.1: vetted access or a provider's research interface first, publicly accessible data second, and a sock-puppet audit only when neither can identify the estimand. In every route, the audit does not circumvent any technical access control, does not bypass challenges, blocks or rate limits, and does not misrepresent the researchers to the provider when asked; a puppet that the service restricts is stopped, not replaced by evasion. Telling a provider in advance that an audit is under way may change what it shows the audit accounts; under Article 40 the provider knows of the research through the procedure, and for a sock-puppet audit the registration and the ethics application record the choice and its justification. In either case the provider is informed before publication (Section 9.5).

9.2 Minimal footprint

An audit must not become a source of the harm it measures. Puppets never post, comment, reply, message, share, react, follow or report, never interact with a real user and never click advertisements. No persona is trained on antisemitic content, and flagged items are recorded but never opened, so the audit adds no engagement to them beyond the impression the service itself generates. False negatives can still be opened; their estimated count is reported, and a pilot share that is not small leads to a more conservative opening threshold. The ethics approval fixes caps on accounts, coding and the puppets' share of engagement (Section 8.3). Only a pilot is run unconditionally, and we do not run an audit too small to answer its question, because its footprint would buy no knowledge. Sequential stopping ends data collection as soon as the evidence is clear, and sessions are paced at human rates.

9.3 Annotator wellbeing

Validation requires people to read antisemitic content, some of it graphic or threatening, and repeated exposure is a documented burden (Steiger et al., 2021). Participation is voluntary, with informed consent, the right to skip items or withdraw, and fair pay. Daily time on hateful content is capped, batches interleave

flagged items with others, images and video are blurred unless needed, and confidential psychological support and debriefing are available to annotators and the research team; no one is required to code a given category of content. The coding volume is a registered cap set from these limits, not the other way round.

9.4 Data security

All data are held in an access-controlled, encrypted research environment, with logged access limited to named team members. Credentials are held in a secrets manager and destroyed at the end of the audit, identifiers of posting accounts are pseudonymised at collection (Appendix D.9), and raw content never leaves the environment. We will not publish lists of antisemitic items or accounts, which would serve as a directory for people seeking the content; examples are paraphrased.

9.5 Responsible disclosure

The provider receives the findings, the protocol and the code at least 30 days before public release, may correct factual errors and respond, and has no veto. Content encountered during the audit that may be illegal is reported through the provider's notice mechanism under Article 16 of the DSA (European Parliament and Council of the European Union, 2022) after the wave in which it was recorded ends, because removal during a wave would change the supply being measured. The ethics application states this delay and its justification for IRB approval, and the IRB may shorten it. Content that indicates an imminent threat to life or safety is reported immediately to the provider and to the competent authorities, whatever the effect on the audit.

9.6 Neutrality and the limits of a single audit

The protocol is platform-neutral, and the choice of service at registration is governed by the criteria of Appendix D.1, not by expectations about the result. An audit's conclusions hold for the service, surface, jurisdiction, window, personas, choice rule and classifier that it used. A finding that $\mathrm{AR}(d) > 1$ for one persona on one surface is not evidence about other surfaces or services, about the same service at another time, about the intentions of its designers, the legality of its conduct, or the exposure of real users; a finding that $\mathrm{AR}(d) \leq 1$ is equally limited. We will report every persona and every registered estimate, including null and unfavourable results, report scope variants under different definitions of antisemitism side by side, and disclose funding and any relationship with the provider or with organisations that campaign on platform regulation.

10 Discussion

What a well-designed audit could establish. Debate about recommender systems and hateful content tends to ask whether "the algorithm" amplifies it. The framework replaces that question with several that have precise answers under stated assumptions. For a stated service, recommendation surface, jurisdiction and audit window, and for each confirmatory persona, an audit run under the protocol of Section 8 would report the amplification ratio and the additive amplification relative to a named reference policy (Definition 3.1), corrected for classifier error in each arm (Theorem 6.1), with a confidence sequence that is approximately valid at whatever point the audit stopped (Algorithm 1 and Theorem 6.2), and a conclusion fixed in advance: material amplification, material de-amplification, no material difference at the registered margin, a difference of undetermined materiality, or an inconclusive result (Table 7). Only the four-persona materiality audit, rung R1 of the feasibility gate, would also report the persona-contrast decomposition

(Section 3.5). The Markov summaries of each arm (Proposition 4.1 and Theorem 4.2) would be reported as exploratory and model-based. Their value is that they reach beyond the audited session length, where short sessions can understate the effect of a change many times over (Section 4.3), but the protocol's slate states are observed through a classifier: in simulation the uncorrected chain can reverse the comparison of the arms, and only the corrected two-state chain was reliable (Section 7.5).

What the audit would in fact establish is smaller, and the paper's power calculations say so. Under plausible inputs, the four-persona materiality audit needs about 4,900 accounts and 170,000 coded exposures, and under illustrative capacity caps the feasibility gate selects rung R4: the detection of amplification, or de-amplification, for the Jewish-interest persona alone, with "difference detected; materiality undetermined" as the conclusion it is powered to reach. That is a modest result, but a defined one, and the alternative, an audit too small to answer its question, would spend its footprint on no knowledge (Section 8.3).

Such an audit could not establish everything that is asked of it, and the framework says exactly what it leaves out. A fixed-demand audit says nothing about how the algorithm shapes what people want, and without outside information on how demand forms, the total effect of the algorithm on real users is not identified (Proposition 3.4). A finding that $\mathrm{AR}(d) > 1$ for one persona is a statement about supply at that persona's demand. It is not a measurement of how much antisemitic content real users see, nor evidence about other surfaces, other services or other periods, nor about the intentions of the people who built the system or the legality of its conduct (Section 9.6). A finding that $\mathrm{AR}(d) \leq 1$ is limited in the same ways. Within those limits, an estimate of this kind is what a systemic-risk assessment under Article 34 of the DSA could be checked against: a provider's statement that its recommender does not raise exposure to antisemitic content is a claim about supply at given demand, and it is the claim an independent audit can test (Section 2.3).

Limitations. We stress six.

The moving target. An audit estimates amplification by the policies that ran during its window, and the estimate expires with the system it describes. Blocking keeps drift common to both arms out of the contrast, and sentinels and a test across waves make drift visible (Section 5.4), but nothing makes a result from one window a result about the next, as the period qualifier of Hosseinmardi et al. (2024) illustrates. A service that learns to recognise audit accounts breaks transport from puppets to people (Assumption A7), and the audit cannot test for it.

From puppets to people. A persona is a script; real users have histories, social graphs and preferences that respond to what they are shown. The step to real users rests on Assumption A7, which audit data cannot test. The choice rule adds a gap by design: puppets never open a flagged item, so $\mathrm{AR}(d)$ describes the supply to a demand profile that does not engage with antisemitic content, the right estimand for users who do not seek such content, not for those who do. Population statements need weights over demand profiles, which an audit does not supply.

Dependence on the classifier and the codebook. The true label is a coding decision that depends on the codebook and the definition of antisemitism behind it, which is why scope variants are reported side by side. The correction removes classifier bias only if the error rates hold within the strata where they are estimated, which emerging coded language (Zannettou et al., 2020), quotation and counter-speech (Röttger et al., 2021) and images and video can each break. Classifier error is modelled here, not eliminated.

Law and access. Terms of service, contract law and computer-crime statutes constrain sock-puppet audits (Section 9.1), and the practical reach of vetted access under Article 40(4) is still being established, so the services that are audited are not a random sample of services.

Heterogeneity across services. Surfaces and services differ in their reference policies: by Lemma 3.2(c),

changing the reference multiplies the ratio by a factor that does not involve the algorithm, so ratios are comparable across services only on a common reference or a stated ladder of references, with the same classifier, codebook and personas. Interference differs too, and because ξ is not identified, each conclusion carries its breakdown value rather than a corrected estimate (Proposition 5.2).

Feasibility. The audit the questions call for is not the audit a small lab can run. A materiality test across four personas needs thousands of accounts and coded exposures (Table 8), and the protocol therefore commits only to a pilot and a registered gate, which under illustrative caps selects detection for one persona. The decomposition across personas, materiality conclusions and evidence of no material difference are then out of reach. Platform-side access under Article 40(4), where it can deliver exposure under a fixed demand profile, removes the need for accounts and with it most of this constraint; until it does, single-persona detection audits are what independent researchers can responsibly run. The approximations add to this: the sequential guarantee is asymptotic, and at the registered sizes its level is checked by simulation and by a pilot check on ten puppets per cell (Section 8.2). In simulation the sequence on $\widetilde{E}_u$ was conservative at the registered design despite heavy tails, and the pilot check failed whenever the sequence did not hold (Section 7.6). This support holds only at the synthetic world's inputs, which are more exposed and less variable than the illustrative ones; rung R1 was not simulated, and tails rarer than ten pilot puppets can show would escape the pilot check.

Extensions. Three extensions seem most valuable. For *generative answer engines*, the policy is the system that generates the answer, a persona is a script of prompts, exposure is the share of answers that a validated classifier labels antisemitic, and a natural reference is the same system without generation; generation is stochastic at fixed inputs and has no fixed eligible pool, so the distinction between content and algorithmic supply must be redrawn. In *search*, demand is explicit in the query, a persona becomes a distribution over queries, and the same query without a profile is a natural reference (Robertson et al., 2023). *Cross-platform audits* with harmonised personas, one frozen classifier and a shared validation sample would make estimates comparable across services, and adaptive personas would let them study demand shaping for synthetic users (Section 3.5).

Why define amplification. A count of antisemitic items seen online says how much there is. It does not say where it came from. Content that users seek out calls for one kind of response, content that a recommender supplies beyond what users seek calls for another, and content that a recommender has taught users to seek calls for a third. These are different claims about responsibility, and they carry different duties under the law and different obligations for the people who design these systems. An audit can test the second claim, for a stated system and a stated alternative, and it can say plainly that it cannot test the third. Stating the comparison before measuring it, correcting for the errors of the instrument, stopping as soon as the evidence allows and fixing the rules before the data are seen are how an audit can take seriously both the people who are targeted by antisemitic content and the evidence about where it comes from.

Declarations

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AI assistance. This paper was drafted with substantial assistance from AI language models (Anthropic's Claude), directed and reviewed by the Kornstein Lab; all derivations were checked numerically and by independent automated review, and all simulation numbers are produced by the released code.

Code and data availability. The code for the simulation study (the `ampsim` Python package, the experiment driver `run_experiments.py` with its deterministic seeds, and the script that produces every figure and table of Section 7) is in the paper's repository, together with the simulation output (Appendix C.5). All data in this paper are synthetic and are regenerated by that code. No real platform was audited, and no real account, item or user data were used.

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Appendix A Proofs and further results for Sections 3–4

Throughout, vectors are columns, $\mathbf{1}$ is the vector of ones, $\boldsymbol{e}_x$ is the xth unit vector, and for a matrix B we write $B_{x\cdot}$ for its xth row, as a row vector. Two elementary lemmas are used repeatedly.

Lemma A.1 (Zero-sum vectors). *Let $\boldsymbol{v} \in \mathbb{R}^{\mathcal{X}}$ with $\boldsymbol{v}^\top \mathbf{1} = 0$.*

(a) For every $\boldsymbol{h} \in \mathbb{R}^m$, $|\boldsymbol{v}^\top \boldsymbol{h}| \le \frac{1}{2}\|\boldsymbol{v}\|_1 \operatorname{osc}(\boldsymbol{h})$.

(b) For every real matrix B with rows indexed by $\mathcal{X}$, $\|\boldsymbol{v}^\top B\|_1 \le \operatorname{erg}(B)\,\|\boldsymbol{v}\|_1$, where $\operatorname{erg}(B) = \frac{1}{2}\max_{x,y}\|B_{x\cdot} - B_{y\cdot}\|_1$.

(c) If every row of B has the same sum, then $\operatorname{osc}(B\boldsymbol{h}) \le \operatorname{erg}(B)\operatorname{osc}(\boldsymbol{h})$.

Proof. Write $\boldsymbol{v} = \boldsymbol{v}^+ - \boldsymbol{v}^-$ with $\boldsymbol{v}^\pm \ge \mathbf{0}$ the positive and negative parts. Since $\boldsymbol{v}^\top\mathbf{1} = 0$, both parts have total mass $s = \frac{1}{2}\|\boldsymbol{v}\|_1$. (a) $\boldsymbol{v}^\top\boldsymbol{h} = (\boldsymbol{v}^+)^\top\boldsymbol{h} - (\boldsymbol{v}^-)^\top\boldsymbol{h} \le s\max\boldsymbol{h} - s\min\boldsymbol{h}$, and the same bound holds for $-\boldsymbol{v}$. (b) If $s = 0$ there is nothing to prove. Otherwise $\boldsymbol{p} = \boldsymbol{v}^+/s$ and $\boldsymbol{q} = \boldsymbol{v}^-/s$ are probability vectors and $\boldsymbol{v}^\top B = s\sum_{x,y} p_x q_y (B_{x\cdot} - B_{y\cdot})$. By the triangle inequality, $\|\boldsymbol{v}^\top B\|_1 \le s\max_{x,y}\|B_{x\cdot} - B_{y\cdot}\|_1 = 2s\operatorname{erg}(B)$. (c) For rows x, y, the vector $(B_{x\cdot} - B_{y\cdot})^\top$ sums to zero, so (a) gives $(B\boldsymbol{h})_x - (B\boldsymbol{h})_y \le \frac{1}{2}\|B_{x\cdot} - B_{y\cdot}\|_1 \operatorname{osc}(\boldsymbol{h}) \le \operatorname{erg}(B)\operatorname{osc}(\boldsymbol{h})$. □

Lemma A.2 (Integrals against a difference of distributions). *Let F and G be probability measures on a measurable space $\mathcal{D}$, let S be a measurable set with $F(S) = G(S) = 1$, and let f be bounded and measurable. Then $\left|\int f\,d(F-G)\right| \le \operatorname{osc}_S(f)\,d_{\mathrm{TV}}(F,G)$, where $d_{\mathrm{TV}}(F,G) = \sup_B |F(B) - G(B)|$. If $\mathcal{D}$ is a metric space, F and G have finite first moments and f is L-Lipschitz, then $\left|\int f\,d(F-G)\right| \le L\,d_{\mathrm{W}}(F,G)$.*

Proof. Write the Jordan decomposition of the signed measure $F - G$ as $(F-G)^+ - (F-G)^-$. Since $(F-G)(\mathcal{D}) = 0$, $(F-G)^+(\mathcal{D}) = (F-G)^-(\mathcal{D}) = \sup_B (F-G)(B) = d_{\mathrm{TV}}(F,G)$. Both parts vanish outside S, because $|F-G|(\mathcal{D}\setminus S) \le F(\mathcal{D}\setminus S) + G(\mathcal{D}\setminus S) = 0$. Hence $\int f\,d(F-G) = \int_S f\,d(F-G)^+ - \int_S f\,d(F-G)^- \le d_{\mathrm{TV}}(F,G)(\sup_S f - \inf_S f)$, and the same holds for $-f$. The second bound is the Kantorovich–Rubinstein duality, $d_{\mathrm{W}}(F,G) = \sup\{\int h\,d(F-G) : h \text{ 1-Lipschitz}\}$ (Villani, 2009, Ch. 6), applied to $h = f/L$. □

A.1 Proof of Theorem 3.3

(a) By the definitions (3.4), $\tau_{\sup} + \tau_{\mathrm{dem}} + \iota = \bar{\mu}(\mathrm{alg};\mathrm{alg}) - \bar{\mu}(\mathrm{ref};\mathrm{ref}) = \tau$, which is the first identity. The two paths are the telescoping sums

$$\begin{aligned}\tau &= \underbrace{\{\bar{\mu}(\mathrm{alg};\mathrm{ref}) - \bar{\mu}(\mathrm{ref};\mathrm{ref})\}}_{\tau_{\sup}} + \underbrace{\{\bar{\mu}(\mathrm{alg};\mathrm{alg}) - \bar{\mu}(\mathrm{alg};\mathrm{ref})\}}_{\tau_{\mathrm{dem}}+\iota} \\ &= \underbrace{\{\bar{\mu}(\mathrm{alg};\mathrm{alg}) - \bar{\mu}(\mathrm{ref};\mathrm{alg})\}}_{\tau_{\sup}+\iota} + \underbrace{\{\bar{\mu}(\mathrm{ref};\mathrm{alg}) - \bar{\mu}(\mathrm{ref};\mathrm{ref})\}}_{\tau_{\mathrm{dem}}},\end{aligned}$$

where the labels under the braces follow from the definition of ι.

(b) The game has $v(\emptyset) = 0$, $v(\{\sup\}) = \tau_{\sup}$, $v(\{\mathrm{dem}\}) = \tau_{\mathrm{dem}}$ and $v(\{\sup,\mathrm{dem}\}) = \tau$. With two players, the Shapley value averages each player's marginal contribution over the two orders (Shapley, 1953): $\phi_{\sup} = \frac{1}{2}\{v(\{\sup\}) - v(\emptyset)\} + \frac{1}{2}\{v(\{\sup,\mathrm{dem}\}) - v(\{\mathrm{dem}\})\} = \frac{1}{2}\tau_{\sup} + \frac{1}{2}(\tau_{\sup} + \iota)$. This is the average of the two path attributions of (a), and it equals $\tau_{\sup} + \iota/2$. The expression for ϕ_{dem} follows by symmetry, and $\phi_{\sup} + \phi_{\mathrm{dem}} = \tau_{\sup} + \tau_{\mathrm{dem}} + \iota = \tau$. For uniqueness, write $v = v_1 + v_2 + v_{12}$ with $v_1(J) = \tau_{\sup}\mathbf{1}\{\sup \in J\}$,

$v_2(J) = \tau_{\text{dem}}\mathbf{1}\{\text{dem} \in J\}$ and $v_{12}(J) = \iota\,\mathbf{1}\{J = \{\text{sup}, \text{dem}\}\}$. Let ψ be an allocation that is efficient, symmetric, additive and gives nothing to null players. In v_1 demand is a null player, so $\psi_{\text{dem}}(v_1) = 0$ and, by efficiency, $\psi_{\text{sup}}(v_1) = \tau_{\text{sup}}$. Likewise $\psi(v_2) = (0, \tau_{\text{dem}})$. In v_{12} the players are symmetric, so each receives $\iota/2$. Additivity gives $\psi(v) = (\tau_{\text{sup}} + \iota/2, \tau_{\text{dem}} + \iota/2) = \phi$. For the ratio scale, apply the same computation to $v^{\log}(J) = \log \bar{\mu}(a_J; b_J) - \log \bar{\mu}(\text{ref}; \text{ref})$. Then

$$\phi^{\log}_{\text{sup}} = \tfrac{1}{2} \log \frac{\bar{\mu}(\text{alg}; \text{ref})}{\bar{\mu}(\text{ref}; \text{ref})} + \tfrac{1}{2} \log \frac{\bar{\mu}(\text{alg}; \text{alg})}{\bar{\mu}(\text{ref}; \text{alg})} = \tfrac{1}{2} \log\{\text{AR}(F_{\text{ref}})\, \text{AR}(F_{\text{alg}})\},$$

and similarly $\phi^{\log}_{\text{dem}} = \frac{1}{2} \log\{\text{DR}(\text{ref})\, \text{DR}(\text{alg})\}$. Efficiency gives $\phi^{\log}_{\text{sup}} + \phi^{\log}_{\text{dem}} = \log \text{TR}$, and exponentiating gives $\text{TR} = \Phi_{\text{sup}}\Phi_{\text{dem}}$. The log interaction is

$$v^{\log}(\{\text{sup}, \text{dem}\}) - v^{\log}(\{\text{sup}\}) - v^{\log}(\{\text{dem}\}) = \log \frac{\bar{\mu}(\text{alg}; \text{alg})\, \bar{\mu}(\text{ref}; \text{ref})}{\bar{\mu}(\text{alg}; \text{ref})\, \bar{\mu}(\text{ref}; \text{alg})},$$

whose exponential can be grouped either as $\text{AR}(F_{\text{alg}})/\text{AR}(F_{\text{ref}})$ or as $\text{DR}(\text{alg})/\text{DR}(\text{ref})$.

(c) By Assumption A1 and iterated expectations, $\bar{\mu}(a; b) = \mathbb{E}[\mathbb{E}\{E_U(a; b) \mid D_U(b)\}] = \mathbb{E}\{\mu_a(D_U(b))\} = \int \mu_a\, dF_b$; the integrals exist because $0 \le \mu_a \le 1$. This is (3.3). Substituting it into (3.4) gives $\tau_{\text{sup}} = \int \Delta\mu\, dF_{\text{ref}} = \text{AA}_{\text{ref}}(F_{\text{ref}})$, $\tau_{\text{dem}} = \int \mu_{\text{ref}}\, d(F_{\text{alg}} - F_{\text{ref}})$ and

$$\iota = \int \mu_{\text{alg}}\, dF_{\text{alg}} - \int \mu_{\text{alg}}\, dF_{\text{ref}} - \int \mu_{\text{ref}}\, dF_{\text{alg}} + \int \mu_{\text{ref}}\, dF_{\text{ref}} = \int \Delta\mu\, d(F_{\text{alg}} - F_{\text{ref}}).$$

Hence $\tau_{\text{sup}} + \iota = \int \Delta\mu\, dF_{\text{alg}} = \text{AA}_{\text{ref}}(F_{\text{alg}})$. Finally, $\phi_{\text{sup}} = \tau_{\text{sup}} + \frac{1}{2}\iota = \frac{1}{2}\int \Delta\mu\, dF_{\text{ref}} + \frac{1}{2}\int \Delta\mu\, dF_{\text{alg}} = \int \Delta\mu\, dF_{\text{mid}}$, and $\phi_{\text{dem}} = \tau_{\text{dem}} + \frac{1}{2}\iota = \int \{\mu_{\text{ref}} + \frac{1}{2}(\mu_{\text{alg}} - \mu_{\text{ref}})\}\, d(F_{\text{alg}} - F_{\text{ref}}) = \int \mu_{\text{mid}}\, d(F_{\text{alg}} - F_{\text{ref}})$.

(d) By (c), $\iota = \text{AA}_{\text{ref}}(F_{\text{alg}}) - \text{AA}_{\text{ref}}(F_{\text{ref}})$, which vanishes exactly when the two are equal. By (b) and (3.3), $\iota^{\times} = \text{AR}_{\text{ref}}(F_{\text{alg}})/\text{AR}_{\text{ref}}(F_{\text{ref}})$, which equals one exactly when the two ratios are equal. In case (i), every integral against $F_{\text{alg}} - F_{\text{ref}}$ vanishes, so $\iota = \tau_{\text{dem}} = 0$, the two path attributions in (a) both equal $\tau_{\text{sup}} = \tau$, and $\bar{\mu}(\text{alg}; \text{alg}) = \mu_{\text{alg}}(F) = \mu_{\text{ref}}(F) + \text{AA}_{\text{ref}}(F)$. In case (ii), $\Delta\mu = c$ on S, and F_{alg} and F_{ref} are both concentrated on S, so $\iota = c\{F_{\text{alg}}(S) - F_{\text{ref}}(S)\} = 0$. The bounds follow from Lemma A.2, applied to $f = \Delta\mu$ for ι and to $f = \mu_{\text{ref}}$ for τ_{dem}. □

A.2 Proof of Proposition 3.4

(a) $\text{AR}_\rho(\Lambda)$ and $\text{AA}_\rho(\Lambda)$ are functionals of Λ and of μ_{alg} and μ_ρ restricted to a set of Λ-probability one, here $\mathcal{D}_A$. The three special cases are Theorem 3.3(c).

(b) The audit observes only sessions of puppets whose demand is fixed at personas in $\mathcal{D}_A$. Under the conditions of Section 5, the joint law of these sessions is determined by the policies and the personas. It does not involve the demand profiles $D_U(b)$ of real users, so it is the same for every specification of the formation laws $(F_{\text{alg}}, F_{\text{ref}})$. Grant oracle knowledge: fix μ_{alg} and μ_{ref} on all of $\mathcal{D}$, and fix F_{alg}. For an arbitrary probability measure G on $\mathcal{D}$, a population model with $D_U(\text{ref}) \sim G$ and $\mathbb{E}\{E_U(a; \text{ref}) \mid D_U(\text{ref}) = d\} = \mu_a(d)$ satisfies Assumption A1. It leaves the law of the audit data unchanged and has $F_{\text{ref}} = G$. By (3.3), $\tau = \bar{\mu}(\text{alg}; \text{alg}) - \mu_{\text{ref}}(G)$, where $\bar{\mu}(\text{alg}; \text{alg}) = \mu_{\text{alg}}(F_{\text{alg}})$ is fixed. The identified set for τ is therefore $\{\bar{\mu}(\text{alg}; \text{alg}) - \mu_{\text{ref}}(G)\}$, as G ranges over all probability measures on $\mathcal{D}$. It remains to describe $\{\mu_{\text{ref}}(G)\}$. Let $s = \sup_{\mathcal{D}} \mu_{\text{ref}}$ and $i = \inf_{\mathcal{D}} \mu_{\text{ref}}$. Every $\mu_{\text{ref}}(G)$ lies in $[i, s]$. Point masses give every value $\mu_{\text{ref}}(d)$, and two-point mixtures give every convex combination of two such values, so the set contains the open interval (i, s). The endpoint s belongs to the set if and only if it is attained: if $\mu_{\text{ref}} < s$ everywhere, then $\mu_{\text{ref}}(G) < s$ for every G. The same holds for i. The identified set for τ is thus an interval with endpoints $\bar{\mu}(\text{alg}; \text{alg}) - s$

and $\bar{\mu}(\text{alg};\text{alg}) - i$, of length $s - i = \operatorname{osc}_{\mathcal{D}}(\mu_{\text{ref}})$. Since $\tau_{\text{dem}} = \mu_{\text{ref}}(F_{\text{alg}}) - \mu_{\text{ref}}(G)$, the same argument applies to τ_{dem}. In particular, τ and τ_{dem} are identified if and only if this interval is a point, that is, if and only if μ_{ref} is constant on $\mathcal{D}$. Likewise, as G ranges over the point masses, $\iota = \mathrm{AA}_{\text{ref}}(F_{\text{alg}}) - \int \Delta\mu \, dG$ and $\phi_{\text{dem}} = \int \mu_{\text{mid}} \, d(F_{\text{alg}} - G)$ take every value of $\mathrm{AA}_{\text{ref}}(F_{\text{alg}}) - \Delta\mu(d)$ and $\mu_{\text{mid}}(F_{\text{alg}}) - \mu_{\text{mid}}(d)$, so they are identified if and only if $\Delta\mu$, respectively μ_{mid}, is constant on $\mathcal{D}$. Without oracle knowledge, the values of μ_{alg} and μ_{ref} off $\mathcal{D}_A$ and the law F_{alg} are further unknowns, so the identified sets can only be larger.

(c) By (3.4), $\tau - \mathrm{AA}_{\text{ref}}(F_{\text{alg}}) = \tau - (\tau_{\text{sup}} + \iota) = \tau_{\text{dem}}$ and $\phi_{\text{sup}} - (\tau_{\text{sup}} + \iota) = -\iota/2$. The bounds follow from Theorem 3.3(d). □

A.3 Proof of Proposition 4.1

We use three standard facts about a unichain P with closed class C (Kemeny and Snell, 1960, Chs. 3–5); see also Seneta (2006). (U1) P has a unique stationary distribution ϖ, and $\varpi_x > 0$ if and only if $x \in C$. (U2) $T^{-1}\sum_{t=0}^{T-1} P^t \to \mathbf{1}\varpi^\top$. (U3) Every solution of $P\boldsymbol{v} = \boldsymbol{v}$ is constant, because $\boldsymbol{v} = T^{-1}\sum_{t<T} P^t\boldsymbol{v} \to \mathbf{1}\varpi^\top\boldsymbol{v}$.

(a) Algebra. If $(I - P + \mathbf{1}\varpi^\top)\boldsymbol{v} = \mathbf{0}$, multiplying on the left by $\varpi^\top$ gives $\varpi^\top\boldsymbol{v} = 0$. Then $(I-P)\boldsymbol{v} = \mathbf{0}$, so $\boldsymbol{v}$ is constant by (U3), and $\boldsymbol{v} = \mathbf{0}$. Thus Z exists. Since $(I - P + \mathbf{1}\varpi^\top)\mathbf{1} = \mathbf{1}$ and $\varpi^\top(I - P + \mathbf{1}\varpi^\top) = \varpi^\top$, we have $Z\mathbf{1} = \mathbf{1}$ and $\varpi^\top Z = \varpi^\top$. Hence $(I-P)Z = (Z^{-1} - \mathbf{1}\varpi^\top)Z = I - \mathbf{1}\varpi^\top$, and likewise $Z(I-P) = I - \mathbf{1}\varpi^\top$. For $H = Z - \mathbf{1}\varpi^\top$ this gives $H\mathbf{1} = \mathbf{0}$, $\varpi^\top H = \mathbf{0}^\top$ and $(I - P)H = H(I - P) = I - \mathbf{1}\varpi^\top$. Therefore $H(I - P)H = H - \mathbf{1}\varpi^\top H = H$ and $(I - P)H(I - P) = (I - \mathbf{1}\varpi^\top)(I - P) = I - P$. These are the defining equations of the group inverse (Meyer, 1975). For the formula for ϖ: $\varpi^\top(I - P + \mathbf{1}\mathbf{1}^\top) = \mathbf{1}^\top$, and $I - P + \mathbf{1}\mathbf{1}^\top$ is invertible by the argument used for Z. If $(I - P + \mathbf{1}\mathbf{1}^\top)\boldsymbol{v} = \mathbf{0}$, then multiplying by $\varpi^\top$ gives $\mathbf{1}^\top\boldsymbol{v} = 0$, so $(I - P)\boldsymbol{v} = \mathbf{0}$, $\boldsymbol{v} = c\mathbf{1}$ and $c|\mathcal{X}| = 0$.

(a) Limit theorems. Let $\tilde{\boldsymbol{f}} = \mathbf{1}_{\mathcal{A}} - \varpi(\mathcal{A})\mathbf{1}$ and $\boldsymbol{g} = \boldsymbol{g}_{\mathcal{A}} = H\mathbf{1}_{\mathcal{A}} = H\tilde{\boldsymbol{f}}$. Then $(I-P)\boldsymbol{g} = (I - \mathbf{1}\varpi^\top)\mathbf{1}_{\mathcal{A}} = \tilde{\boldsymbol{f}}$, so $\tilde{f}(X_t) = g(X_t) - (P\boldsymbol{g})(X_t)$ and

$$\sum_{t=1}^{T} \tilde{f}(X_t) = \sum_{t=1}^{T-1} d_{t+1} + g(X_1) - (P\boldsymbol{g})(X_T), \qquad d_{t+1} = g(X_{t+1}) - (P\boldsymbol{g})(X_t). \tag{A.1}$$

The d_{t+1} are bounded martingale differences for the natural filtration, with conditional variances $\sigma_g^2(X_t)$, where $\sigma_g^2(x) = \mathrm{Var}_{y\sim P(x,\cdot)}\{g(y)\}$. By the strong law for martingales with bounded increments (Hall and Heyde, 1980, Ch. 2), $T^{-1}\sum_t d_{t+1} \to 0$ almost surely, and the boundary terms in (A.1) are bounded. Hence $E - \varpi(\mathcal{A}) = T^{-1}\sum_t \tilde{f}(X_t) \to 0$ almost surely. The same argument applied to σ_g^2 instead of $\mathbf{1}_{\mathcal{A}}$ gives $T^{-1}\sum_t \sigma_g^2(X_t) \to \sum_x \varpi_x\sigma_g^2(x)$ almost surely. The martingale central limit theorem (Hall and Heyde, 1980, Ch. 3) then yields $T^{-1/2}\sum_t d_{t+1} \Rightarrow \mathrm{N}(0, \sum_x \varpi_x\sigma_g^2(x))$, and the boundary terms are negligible after division by $\sqrt{T}$. To identify the variance, use $\varpi^\top P = \varpi^\top$ and $P\boldsymbol{g} = \boldsymbol{g} - \tilde{\boldsymbol{f}}$:

$$\begin{aligned}\sum_x \varpi_x\sigma_g^2(x) &= \varpi^\top P(\boldsymbol{g}\circ\boldsymbol{g}) - \varpi^\top\{(P\boldsymbol{g})\circ(P\boldsymbol{g})\} = \varpi^\top(\boldsymbol{g}\circ\boldsymbol{g}) - \varpi^\top\{(\boldsymbol{g}-\tilde{\boldsymbol{f}})\circ(\boldsymbol{g}-\tilde{\boldsymbol{f}})\}\\ &= 2\varpi^\top(\tilde{\boldsymbol{f}}\circ\boldsymbol{g}) - \varpi^\top(\tilde{\boldsymbol{f}}\circ\tilde{\boldsymbol{f}}),\end{aligned}$$

where $\circ$ is the entrywise product. Since $\boldsymbol{g} = H\tilde{\boldsymbol{f}}$, this is the first expression for $\sigma^2_{\mathcal{A}}$.

(b) Multiplying the telescoping identity $\sum_{t=0}^{T-1} P^t(I - P) = I - P^T$ on the right by H, and using $(I - P)H = I - \mathbf{1}\varpi^\top$ and $P^t\mathbf{1} = \mathbf{1}$, gives

$$\sum_{t=0}^{T-1} P^t = T\,\mathbf{1}\varpi^\top + (I - P^T)H. \tag{A.2}$$

Substituting into (4.1) gives $e_T = \varpi(\mathcal{A}) + T^{-1}\nu^\top(I - P^T)\boldsymbol{g}_{\mathcal{A}}$. Since $(I-P^T)\mathbf{1} = \mathbf{0}$, $(I-P^T)H = (I-P^T)Z$, which gives the second form. Write $\nu^\top(I - P^T)\boldsymbol{g}_{\mathcal{A}} = b(\nu) + r_T$ with $b(\nu) = \nu^\top \boldsymbol{g}_{\mathcal{A}}$ and $r_T = -\nu^\top P^T \boldsymbol{g}_{\mathcal{A}} = -(\nu^\top P^T - \varpi^\top)\boldsymbol{g}_{\mathcal{A}}$, using $\varpi^\top \boldsymbol{g}_{\mathcal{A}} = 0$. Also $b(\nu) = (\nu - \varpi)^\top \boldsymbol{g}_{\mathcal{A}} = (\nu - \varpi)^\top Z \mathbf{1}_{\mathcal{A}}$, because $(\nu - \varpi)^\top \mathbf{1} = 0$. The vector $\nu^\top P^T - \varpi^\top$ sums to zero, so Lemma A.1(a) bounds $|r_T|$ as stated. If P is aperiodic, then $\nu^\top P^T \to \varpi^\top$.

(c) The event that a session has no antisemitic item is $\{X_1, \dots, X_T \in \bar{\mathcal{A}}\}$, whose probability is $\sum \nu_{x_1} P_{x_1 x_2} \cdots P_{x_{T-1} x_T}$ over $x_1, \dots, x_T \in \bar{\mathcal{A}}$, that is, $\nu_{\bar{\mathcal{A}}}^\top P_{\bar{\mathcal{A}}\bar{\mathcal{A}}}^{T-1} \mathbf{1}$. If $\mathcal{A}$ is accessible from every state, then making the states of $\mathcal{A}$ absorbing gives an absorbing chain, so the spectral radius of $P_{\bar{\mathcal{A}}\bar{\mathcal{A}}}$ is less than one (Kemeny and Snell, 1960, Ch. 3), and $I - P_{\bar{\mathcal{A}}\bar{\mathcal{A}}}$ is invertible. For $x \in \bar{\mathcal{A}}$, $\mathbb{P}_x(T_{\mathcal{A}} > n) = \mathbb{P}_x(X_1, \dots, X_{1+n} \in \bar{\mathcal{A}}) = (P^n_{\bar{\mathcal{A}}\bar{\mathcal{A}}}\mathbf{1})_x$, so $M_{x\mathcal{A}} = \sum_{n\geq 0}(P^n_{\bar{\mathcal{A}}\bar{\mathcal{A}}}\mathbf{1})_x = \{(I - P_{\bar{\mathcal{A}}\bar{\mathcal{A}}})^{-1}\mathbf{1}\}_x$. Now let P be irreducible and fix y, so that $\varpi_y > 0$. Let $u_x = (H_{yy} - H_{xy})/\varpi_y$. Then $u_y = 0$. The yth column of $(I - P)H = I - \mathbf{1}\varpi^\top$ reads $(I - P)H\boldsymbol{e}_y = \boldsymbol{e}_y - \varpi_y \mathbf{1}$, so $(I - P)\boldsymbol{u} = -(I - P)H\boldsymbol{e}_y/\varpi_y = \mathbf{1} - \boldsymbol{e}_y/\varpi_y$. For $x \neq y$ this says $u_x - \sum_{z\neq y} P_{xz} u_z = 1$, using $u_y = 0$. With $\mathcal{X}_{-y} = \mathcal{X} \setminus \{y\}$, this is the system $(I - P_{\mathcal{X}_{-y}\mathcal{X}_{-y}})\boldsymbol{u}_{\mathcal{X}_{-y}} = \mathbf{1}$, whose unique solution is $(M_{xy})_{x\neq y}$. Hence $M_{xy} = u_x$ for all x. Summing, $\sum_y \varpi_y M_{xy} = \sum_y (H_{yy} - H_{xy}) = \operatorname{tr} H - (H\mathbf{1})_x = \operatorname{tr} H$. For the spectrum, let $\mathcal{R} = \{\boldsymbol{v} : \varpi^\top \boldsymbol{v} = 0\}$. The space $\mathbb{R}^{\mathcal{X}}$ is the direct sum of $\operatorname{span}(\mathbf{1})$ and $\mathcal{R}$, and both are invariant under P and under H, because $P\mathbf{1} = \mathbf{1}$, $H\mathbf{1} = \mathbf{0}$, $\varpi^\top P = \varpi^\top$ and $\varpi^\top H = \mathbf{0}^\top$. On $\mathcal{R}$, $(I - P)H$ and $H(I - P)$ act as the identity, because $I - \mathbf{1}\varpi^\top$ does. So H restricted to $\mathcal{R}$ is the inverse of $I - P$ restricted to $\mathcal{R}$, and H is zero on $\operatorname{span}(\mathbf{1})$. The eigenvalues of P restricted to $\mathcal{R}$ are the eigenvalues of P with one copy of the eigenvalue 1 removed, namely $\lambda_2, \dots, \lambda_{|\mathcal{X}|}$, none of which equals one because 1 is a simple eigenvalue of an irreducible P. Hence $\operatorname{tr} H = \sum_{k=2}^{|\mathcal{X}|}(1 - \lambda_k)^{-1}$.

(d) From every transient state some closed class is accessible, so, as in (c), the spectral radius of $P_{\mathcal{T}\mathcal{T}}$ is less than one and $Z_{\mathcal{T}} = \sum_{n\geq 0} P^n_{\mathcal{T}\mathcal{T}}$ exists. The probability that a session started at $x \in \mathcal{T}$ stays in $\mathcal{T}$ for items $1, \dots, n+1$ and enters C_k at item $n+2$ is $(P^n_{\mathcal{T}\mathcal{T}} P_{\mathcal{T}C_k}\mathbf{1})_x$, so $B_{xk} = \sum_{n\geq 0}(P^n_{\mathcal{T}\mathcal{T}} P_{\mathcal{T}C_k}\mathbf{1})_x = (Z_{\mathcal{T}} P_{\mathcal{T}C_k}\mathbf{1})_x$, and the expected number of items before entering a closed class is $\sum_n \mathbb{P}_x(X_1, \dots, X_{1+n} \in \mathcal{T}) = (Z_{\mathcal{T}}\mathbf{1})_x$. For the long-run exposure, let T_C be the first time the chain is in a closed class. It is almost surely finite. On the event that the chain enters C_k, the strong Markov property and part (a), applied to the irreducible chain P restricted to C_k, give $T^{-1}\sum_{t=T_C}^{T} \mathbf{1}\{X_t \in \mathcal{A}\} \to \varpi^{(k)}(\mathcal{A})$ almost surely, while $T^{-1}\sum_{t<T_C} \mathbf{1}\{X_t \in \mathcal{A}\} \leq T_C/T \to 0$. Bounded convergence gives $\lim_T e_T(P, \nu) = \sum_k \mathbb{P}_\nu(\text{enter } C_k)\, \varpi^{(k)}(\mathcal{A})$, and $\mathbb{P}_\nu(\text{enter } C_k) = \nu(C_k) + \nu_{\mathcal{T}}^\top Z_{\mathcal{T}} P_{\mathcal{T}C_k}\mathbf{1}$. □

A.4 Proof of Theorem 4.2

Write $P = P_{\text{ref}}$, $\tilde{P} = P_{\text{alg}}$, $\tilde{\varpi} = \varpi_{\text{alg}}$ and $\Delta P = \tilde{P} - P$, so that $\Delta P\mathbf{1} = \mathbf{0}$, and $\boldsymbol{g} = \boldsymbol{g}_{\mathcal{A}}$.

(a) Since $\tilde{\varpi}^\top \tilde{P} = \tilde{\varpi}^\top$, $\tilde{\varpi}^\top(I - P) = \tilde{\varpi}^\top(I - \tilde{P}) + \tilde{\varpi}^\top \Delta P = \tilde{\varpi}^\top \Delta P$. Multiplying on the right by H and using $(I - P)H = I - \mathbf{1}\varpi^\top$ and $\tilde{\varpi}^\top \mathbf{1} = 1$ gives $\tilde{\varpi}^\top - \varpi^\top = \tilde{\varpi}^\top \Delta P H$, so $\tilde{\varpi}^\top(I - \Delta P H) = \varpi^\top$. To see that $I - \Delta P H$ is invertible, suppose $\boldsymbol{v}^\top(I - \Delta P H) = \mathbf{0}^\top$. Then $\boldsymbol{v}^\top \mathbf{1} = \boldsymbol{v}^\top \Delta P H \mathbf{1} = 0$, and $\boldsymbol{v}^\top(I - P) = \boldsymbol{v}^\top \Delta P H(I - P) = \boldsymbol{v}^\top \Delta P(I - \mathbf{1}\varpi^\top) = \boldsymbol{v}^\top \Delta P$. Hence $\boldsymbol{v}^\top(I - \tilde{P}) = \mathbf{0}^\top$. Because $\tilde{P}$ is unichain, $\boldsymbol{v} = c\tilde{\varpi}$, and $c = \boldsymbol{v}^\top \mathbf{1} = 0$. Thus $\tilde{\varpi}^\top = \varpi^\top(I - \Delta P H)^{-1}$. Along the path, apply this with $\Delta P = \varepsilon\Delta$: $\varpi(\varepsilon)^\top = \varpi^\top(I - \varepsilon\Delta H)^{-1}$, which is rational in ε by Cramer's rule and continuous wherever $P(\varepsilon)$ is unichain. The identity only requires the base matrix to be unichain, so it can be applied at any point ε of the path with increment $h\Delta$: $\varpi(\varepsilon + h)^\top - \varpi(\varepsilon)^\top = h\,\varpi(\varepsilon + h)^\top \Delta H(\varepsilon)$. Dividing by h and letting $h \to 0$ gives $d\varpi(\varepsilon)^\top/d\varepsilon = \varpi(\varepsilon)^\top \Delta H(\varepsilon)$. Since $Z(\varepsilon) = H(\varepsilon) + \mathbf{1}\varpi(\varepsilon)^\top$ and $\Delta\mathbf{1} = \mathbf{0}$, $\Delta Z(\varepsilon) = \Delta H(\varepsilon)$. Multiplying by $\mathbf{1}_{\mathcal{A}}$ at $\varepsilon = 0$ gives the derivative of $\varpi_{\mathcal{A}}$.

(b) By (a), $\tilde{\varpi}(\mathcal{A}) - \varpi(\mathcal{A}) = \boldsymbol{v}^\top \boldsymbol{g}$ and $\tilde{\varpi}^\top - \varpi^\top = \boldsymbol{v}^\top H$, with $\boldsymbol{v}^\top = \tilde{\varpi}^\top \Delta P$. Then $\boldsymbol{v}^\top \mathbf{1} = 0$

and $\|\boldsymbol{v}\|_1 \le \sum_x \tilde{\varpi}_x \|\Delta P_{x\cdot}\|_1 \le \|\Delta P\|_\infty$. Lemma A.1(a) gives $|\boldsymbol{v}^\top \boldsymbol{g}| \le \frac{1}{2}\|\Delta P\|_\infty \operatorname{osc}(\boldsymbol{g}) = \kappa_{\mathcal{A}} \|\Delta P\|_\infty$, and Lemma A.1(b) gives $\|\boldsymbol{v}^\top H\|_1 \le \operatorname{erg}(H)\|\Delta P\|_\infty$. By Proposition 4.1(c), $H_{xy} = H_{yy} - \varpi_y M_{xy}$, so $g_x = \sum_{y\in\mathcal{A}} H_{yy} - \sum_{y\in\mathcal{A}} \varpi_y M_{xy}$ and $\operatorname{osc}(\boldsymbol{g}) = \operatorname{osc}_x \sum_{y\in\mathcal{A}} \varpi_y M_{xy}$. Each sum lies in $[0, \sum_y \varpi_y M_{xy}] = [0, K(P)]$, so $\operatorname{osc}(\boldsymbol{g}) \le K(P)$. For any x, z, the vector $(H_{x\cdot} - H_{z\cdot})^\top$ sums to zero, so by Lemma A.1(a), $g_x - g_z = (H_{x\cdot} - H_{z\cdot})\mathbf{1}_{\mathcal{A}} \le \frac{1}{2}\|H_{x\cdot} - H_{z\cdot}\|_1 \le \operatorname{erg}(H)$. This proves $\kappa_{\mathcal{A}} \le \frac{1}{2}\min\{\operatorname{erg}(H), K(P)\}$. For a single state y, $\varpi_y M_{xy}$ is minimised at $x = y$, where it is zero, so $\kappa_y = \frac{1}{2}\varpi_y \max_x M_{xy}$. For sharpness, let $\Delta\mathbf{1} = \mathbf{0}$ and $\|\Delta\|_\infty \le 1$. By Lemma A.1(a) applied row by row, $|\varpi^\top \Delta \boldsymbol{g}| \le \sum_x \varpi_x \frac{1}{2}\|\Delta_{x\cdot}\|_1 \operatorname{osc}(\boldsymbol{g}) \le \kappa_{\mathcal{A}}$. Let x^+ maximise and x^- minimise $\boldsymbol{g}$, and let every row of Δ be $\frac{1}{2}(\boldsymbol{e}_{x^+} - \boldsymbol{e}_{x^-})^\top$. Then $\|\Delta\|_\infty = 1$, $\Delta_{x\cdot}\boldsymbol{g} = \frac{1}{2}\operatorname{osc}(\boldsymbol{g})$ for every x, and $\varpi^\top \Delta \boldsymbol{g} = \kappa_{\mathcal{A}}$. The matrix $P + \varepsilon\Delta$ is stochastic for small $\varepsilon > 0$ if and only if $P_{xx^-} > 0$ for every x. In general, a direction keeps $P + \varepsilon\Delta$ stochastic for small $\varepsilon > 0$ if and only if the negative entries of each row $\Delta_{x\cdot}$ lie in the support $S_x = \{y : P_{xy} > 0\}$. Then, by the argument of Lemma A.1(a), $\Delta_{x\cdot}\boldsymbol{g} \le \frac{1}{2}\|\Delta_{x\cdot}\|_1\{\max_y g_y - \min_{y\in S_x} g_y\}$, with equality for the row $\frac{1}{2}(\boldsymbol{e}_{x^+} - \boldsymbol{e}_{y_x})^\top$, where y_x minimises $\boldsymbol{g}$ over S_x. Weighting by ϖ_x and summing gives the largest first-order increase stated in Remark A.3.

(c) The matrix $P(\varepsilon)$ is stochastic and, for $\varepsilon > 0$, unichain, because every state leads to j in one step. Stationarity, $\varpi(\varepsilon)^\top\{(1-\varepsilon)P + \varepsilon\mathbf{1}\boldsymbol{e}_j^\top\} = \varpi(\varepsilon)^\top$, together with $\varpi(\varepsilon)^\top\mathbf{1} = 1$, reads $\varpi(\varepsilon)^\top\{I - (1-\varepsilon)P\} = \varepsilon\boldsymbol{e}_j^\top$. The matrix in braces is invertible because the spectral radius of $(1-\varepsilon)P$ is $1 - \varepsilon < 1$. The path has direction $\Delta = \mathbf{1}\boldsymbol{e}_j^\top - P$, and $\varpi^\top\Delta = \boldsymbol{e}_j^\top - \varpi^\top$. By (a), $d\varpi_{\mathcal{A}}/d\varepsilon|_0 = (\boldsymbol{e}_j - \varpi)^\top H\mathbf{1}_{\mathcal{A}} = g_{\mathcal{A}}(j)$, since $\varpi^\top H = \mathbf{0}^\top$. For every y, multiplying $H_{xy} = H_{yy} - \varpi_y M_{xy}$ by ϖ_x and summing over x gives $0 = \varpi^\top H\boldsymbol{e}_y = H_{yy} - \varpi_y \sum_x \varpi_x M_{xy}$, so $H_{yy} = \varpi_y \mathbb{E}_\varpi[T_y]$. Hence $g_{\mathcal{A}}(j) = \sum_{y\in\mathcal{A}} H_{jy} = \sum_{y\in\mathcal{A}}(H_{yy} - \varpi_y M_{jy}) = \sum_{y\in\mathcal{A}} \varpi_y(\mathbb{E}_\varpi[T_y] - M_{jy})$. Dividing by $\varpi(\mathcal{A}) > 0$ gives the logarithmic derivative. If $\mathcal{A} = \{j\}$, then $M_{jj} = 0$ and the expression reduces to $\mathbb{E}_\varpi[T_j]$. For $P = \mathbf{1}\varpi^\top$, $Z = (I - \mathbf{1}\varpi^\top + \mathbf{1}\varpi^\top)^{-1} = I$ and $H = I - \mathbf{1}\varpi^\top$, so $g_{\mathcal{A}}(j) = 1 - \varpi(\mathcal{A})$ for $j \in \mathcal{A}$.

(d) Because P is irreducible and $S \ne \mathcal{X}$, $q(S) > 0$. Let $x \in S$. To reach $j \notin S$, the chain must first leave S. From each state of S it stays in S for one more step with probability at least $1 - q(S)$, so by induction $\mathbb{P}_x(T_{\mathcal{X}\setminus S} > n) \ge \{1 - q(S)\}^n$. Hence $M_{xj} \ge \mathbb{E}_x T_{\mathcal{X}\setminus S} = \sum_{n\ge 0} \mathbb{P}_x(T_{\mathcal{X}\setminus S} > n) \ge 1/q(S)$. By (b), $\kappa_j = \frac{1}{2}\varpi_j \max_x M_{xj} \ge \varpi_j/\{2q(S)\}$, and $\mathbb{E}_\varpi[T_j] \ge \sum_{x\in S} \varpi_x M_{xj} \ge \varpi(S)/q(S)$.

(e) The telescoping identity $\tilde{P}^t - P^t = \sum_{s=0}^{t-1} \tilde{P}^s \Delta P\, P^{t-1-s}$ and (4.1) give

$$e_T(\tilde{P}, \nu) - e_T(P, \nu) = \frac{1}{T}\sum_{t=1}^{T-1}\sum_{s=0}^{t-1} (\nu^\top \tilde{P}^s \Delta P)\,(P^{t-1-s}\mathbf{1}_{\mathcal{A}}).$$

Each row vector $\boldsymbol{v}^\top = \nu^\top \tilde{P}^s \Delta P$ sums to zero and has $\|\boldsymbol{v}\|_1 \le \|\Delta P\|_\infty$, because $\nu^\top \tilde{P}^s$ is a probability vector. By Lemma A.1(a) and (c), $|\boldsymbol{v}^\top P^r \mathbf{1}_{\mathcal{A}}| \le \frac{1}{2}\|\Delta P\|_\infty \operatorname{osc}(P^r\mathbf{1}_{\mathcal{A}}) \le \frac{1}{2}\|\Delta P\|_\infty \operatorname{erg}(P^r)$. Summing over the pairs (t, s), and noting that $r = t - 1 - s$ takes the value r for $T - 1 - r$ pairs, gives the first bound. The second follows from $\operatorname{erg}(P^r) \le 1$ and $\sum_{r=0}^{T-2}(T - 1 - r) = T(T-1)/2$. For the derivative, put $\Delta P = \varepsilon\Delta$, divide by ε and let $\varepsilon \to 0$, so that $\tilde{P}^s \to P^s$. □

Remark A.3 (Sharpness of Theorem 4.2(b)). Over formal directions, with $\Delta\mathbf{1} = \mathbf{0}$ and $\|\Delta\|_\infty \le 1$, the supremum of $|d\varpi_{\mathcal{A}}/d\varepsilon|_{\varepsilon=0}$ equals $\kappa_{\mathcal{A}}$. It is attained by moving mass $\frac{1}{2}$ in every row from a minimiser to a maximiser of $\boldsymbol{g}_{\mathcal{A}}$. Over directions that keep $P_{\mathrm{ref}} + \varepsilon\Delta$ stochastic for small $\varepsilon > 0$, the largest first-order increase is $\frac{1}{2}\sum_x \varpi_x\{\max_y g_{\mathcal{A}}(y) - \min_{y:P_{\mathrm{ref}}(x,y)>0} g_{\mathcal{A}}(y)\}$. This equals $\kappa_{\mathcal{A}}$ when every row of P_{ref} charges a minimiser of $\boldsymbol{g}_{\mathcal{A}}$, for instance when $P_{\mathrm{ref}} > 0$, and is smaller otherwise. The proof is the last paragraph of the proof of part (b) above.

A.5 Estimation from audit sessions

Section 4.4 summarises this result. Within one arm of an audit, fix the policy and the persona and observe n independent sessions $(X_{s1}, \dots, X_{sT})$, $s = 1, \dots, n$. Let n_{0x} be the number of sessions that start in x, let $n_{xy} = \sum_s \sum_{t<T} \mathbf{1}\{X_{st} = x, X_{s,t+1} = y\}$ count transitions, and let $n_{x+} = \sum_y n_{xy}$. Let $\bar{n}_x = \sum_{t=1}^{T-1} (\nu^\top P^{t-1})_x$ be the expected number of departures from x per session, and $\Sigma_x = \operatorname{diag}(P_{x\cdot}) - P_{x\cdot}^\top P_{x\cdot}$, where $P_{x\cdot}$ is the xth row of P as a row vector. For a function $\Psi(\nu, P)$, let $\boldsymbol{c} = \partial\Psi/\partial\nu$ and let $G_{x\cdot} = \partial\Psi/\partial P_{x\cdot}$, a gradient determined up to a constant in each row.

Proposition A.4 (Estimation). *Let Assumption A2 hold within the arm, and let $\bar{n}_x > 0$ for every x.*

(a) The maximum-likelihood estimators are $\hat{\nu}_x = n_{0x}/n$ and $\hat{P}_{xy} = n_{xy}/n_{x+}$.

(b) As $n \to \infty$ with T fixed, $\sqrt{n}(\hat{\nu} - \nu)$ and the rows $\sqrt{n}(\hat{P}_{x\cdot} - P_{x\cdot})$ are asymptotically independent and normal, with covariance matrices $\operatorname{diag}(\nu) - \nu\nu^\top$ and $\Sigma_x/\bar{n}_x$.

(c) If Ψ is differentiable at (ν, P), then $\sqrt{n}\{\Psi(\hat{\nu}, \hat{P}) - \Psi(\nu, P)\}$ converges in distribution to $\mathrm{N}(0, \sigma_\Psi^2)$, with

$$\sigma_\Psi^2 = \operatorname{Var}_{x\sim\nu}(c_x) + \sum_x \frac{1}{\bar{n}_x} \operatorname{Var}_{y\sim P(x,\cdot)}(G_{xy}).$$

(d) For long-run exposure, $\Psi = \varpi(\mathcal{A})$, one may take $\boldsymbol{c} = \mathbf{0}$ and $G_{xy} = \varpi_x\, g_{\mathcal{A}}(y)$, so that $\sigma_\Psi^2 = \sum_x (\varpi_x^2/\bar{n}_x) \operatorname{Var}_{y\sim P(x,\cdot)}\{g_{\mathcal{A}}(y)\}$. For session exposure, $\Psi = e_T$, one may take $c_x = T^{-1} \sum_{t=0}^{T-1} (P^t \mathbf{1}_{\mathcal{A}})_x$ and $G_{xy} = T^{-1} \sum_{t=1}^{T-1} \sum_{s=0}^{t-1} (\nu^\top P^s)_x (P^{t-1-s} \mathbf{1}_{\mathcal{A}})_y$.

Parts (a) and (b) are the classical theory of inference for Markov chains. Anderson and Goodman (1957) treat exactly this design, many independent sequences of fixed length, and Billingsley (1961) surveys the case of one long chain. The likelihood factorises into one multinomial kernel per row, and each row behaves asymptotically like a multinomial sample of size $n\bar{n}_x$. Part (c) is the delta method (van der Vaart, 1998, Ch. 3), and part (d) supplies the gradients, the first from Theorem 4.2(a). The two arms of an audit are independent, so the log amplification ratio $\log\{\hat{\varpi}_{\mathrm{alg}}(\mathcal{A})/\hat{\varpi}_{\mathrm{ref}}(\mathcal{A})\}$ has asymptotic variance $\sigma_{\mathrm{alg}}^2/\{n_{\mathrm{alg}}\varpi_{\mathrm{alg}}(\mathcal{A})^2\} + \sigma_{\mathrm{ref}}^2/\{n_{\mathrm{ref}}\varpi_{\mathrm{ref}}(\mathcal{A})^2\}$, with the σ^2 of part (d) for each arm.

A.6 A sticky reference

Table 9 isolates stickiness. Besides the reference P_{ref} of Section 4.5, consider the sticky reference

$$P_{\mathrm{stk}} = \begin{pmatrix} .994 & .005 & .001 \\ .05 & .89 & .06 \\ .05 & .20 & .75 \end{pmatrix},$$

with rows and columns ordered (N, B, A), in which neutral content rarely leads elsewhere and the borderline region is hard to leave. The two references have nearly the same long-run exposure, about 2.3%, and the same memoryless benchmark. Redirecting one recommendation in a hundred to A raises long-run exposure to 3.7% under P_{ref}, close to the memoryless 3.3%. Under the sticky reference it raises it to 8.0%, an increase of 5.6 percentage points from a nudge of one percentage point. This is Theorem 4.2(c) at work: the mean time to reach A from equilibrium is 262 items under P_{stk} against 62 under P_{ref}. By Theorem 4.2(d), with $S = \{\mathsf{N}\}$ and $q(S) = 0.006$, it is at least 149. The first-order prediction overshoots slightly, 8.5%, because a nudge of this size is no longer small for so sensitive a chain. Finite sessions show the same ordering. Under the sticky reference, 20-item sessions see five times as much antisemitic content after the nudge as before.

Table 9. A nudge of size $\varepsilon = 0.01$ toward A under two references with nearly the same long-run exposure (Theorem 4.2(c)). **Illustrative, invented matrices; not data.** Sessions start in N; $\mathcal{A} = \{\mathsf{A}\}$. All values are exact computations from the matrices in the text.

	P_{ref}	P_{stk}
Kemeny's constant K	3.13	21.1
Sensitivity coefficient $\kappa_{\mathcal{A}}$	0.742	3.31
Mean items to A from equilibrium, $\mathbb{E}_{\varpi}[T_{\mathsf{A}}]$	61.9	261.8
Memoryless benchmark $(1-\varpi_{\mathsf{A}})/\varpi_{\mathsf{A}}$	42.1	41.3
Long-run exposure $\varpi(\mathcal{A})$: before nudge	0.0232	0.0236
after nudge, exact	0.0375	0.0796
after nudge, first order $\varpi_{\mathsf{A}}(1+\varepsilon\,\mathbb{E}_{\varpi}[T_{\mathsf{A}}])$	0.0376	0.0855
after nudge, memoryless $\varpi_{\mathsf{A}}+\varepsilon(1-\varpi_{\mathsf{A}})$	0.0330	0.0334
Session exposure e_{20}: before nudge	0.0207	0.0086
after nudge	0.0339	0.0439
Session exposure e_{100}: before nudge	0.0227	0.0192
after nudge	0.0368	0.0702

A.7 Proof of Proposition A.4

(a) The likelihood of n sessions is $\prod_s \nu_{X_{s1}} \prod_{t<T} P_{X_{st}X_{s,t+1}} = \prod_x \nu_x^{n_{0x}} \prod_x \prod_y P_{xy}^{n_{xy}}$. It is a product of multinomial kernels, one for ν and one for each row of P, and each is maximised by the empirical frequencies.

(b) For session s, let $\boldsymbol{Y}_s^{\nu} = \boldsymbol{e}_{X_{s1}} - \nu$ and, for each x, $\boldsymbol{Y}_{s,x} = \sum_{t=1}^{T-1} \mathbf{1}\{X_{st} = x\}\,(\boldsymbol{e}_{X_{s,t+1}} - P_{x\cdot}^{\top})$. These vectors are independent and identically distributed across sessions. Each summand of $\boldsymbol{Y}_{s,x}$ has conditional mean zero given $X_{s1}, \dots, X_{st}$, so all the vectors have mean zero, and cross-products of summands at different times have mean zero. At equal times, $\mathbf{1}\{X_{st} = x\}\,\mathbf{1}\{X_{st} = x'\} = 0$ for $x \neq x'$, and for $x = x'$ the conditional covariance of $\boldsymbol{e}_{X_{s,t+1}}$ given $X_{st} = x$ is Σ_x. Hence $\operatorname{Cov}(\boldsymbol{Y}_{s,x}, \boldsymbol{Y}_{s,x'}) = \mathbf{1}\{x = x'\}\,\bar{n}_x \Sigma_x$. Conditioning on $X_{s1}, \dots, X_{st}$ also shows that $\operatorname{Cov}(\boldsymbol{Y}_s^{\nu}, \boldsymbol{Y}_{s,x}) = 0$. The multivariate central limit theorem gives joint asymptotic normality of $n^{-1/2} \sum_s (\boldsymbol{Y}_s^{\nu}, (\boldsymbol{Y}_{s,x})_x)$, with block-diagonal covariance $\operatorname{diag}\{\operatorname{diag}(\nu) - \nu\nu^{\top}, (\bar{n}_x \Sigma_x)_x\}$. Finally, $\sqrt{n}(\hat{\nu} - \nu) = n^{-1/2} \sum_s \boldsymbol{Y}_s^{\nu}$, and $\sqrt{n}(\hat{P}_{x\cdot} - P_{x\cdot})^{\top} = (n/n_{x+})\, n^{-1/2} \sum_s \boldsymbol{Y}_{s,x}$, where $n_{x+}/n \to \bar{n}_x > 0$ almost surely by the strong law of large numbers. Slutsky's lemma completes the proof.

(c) By the delta method (van der Vaart, 1998, Thm. 3.1), $\Psi(\hat{\nu}, \hat{P}) - \Psi(\nu, P) = \boldsymbol{c}^{\top}(\hat{\nu} - \nu) + \sum_x G_{x\cdot}(\hat{P}_{x\cdot} - P_{x\cdot})^{\top} + o_p(n^{-1/2})$. Its asymptotic variance is $\boldsymbol{c}^{\top}\{\operatorname{diag}(\nu) - \nu\nu^{\top}\}\boldsymbol{c} + \sum_x \bar{n}_x^{-1} G_{x\cdot} \Sigma_x G_{x\cdot}^{\top}$. Each quadratic form is a variance: $\boldsymbol{c}^{\top}\{\operatorname{diag}(\nu) - \nu\nu^{\top}\}\boldsymbol{c} = \operatorname{Var}_{x\sim\nu}(c_x)$ and $G_{x\cdot}\Sigma_x G_{x\cdot}^{\top} = \operatorname{Var}_{y\sim P(x,\cdot)}(G_{xy})$. Both are unchanged when a constant is added to $\boldsymbol{c}$ or to a row of G, which is why the gradients need only be determined up to such constants: estimation errors in $\hat{\nu}$ and in each row of $\hat{P}$ sum to zero.

(d) $\varpi(\mathcal{A})$ does not depend on ν, so $\boldsymbol{c} = \mathbf{0}$. By Theorem 4.2(a), the derivative of $\varpi(\mathcal{A})$ along a direction Δ with $\Delta\mathbf{1} = \mathbf{0}$ is $\sum_x \varpi_x \sum_y \Delta_{xy} g_{\mathcal{A}}(y) = \sum_{x,y} G_{xy}\Delta_{xy}$ with $G_{xy} = \varpi_x g_{\mathcal{A}}(y)$. For e_T, the derivative along a change $\delta\nu$ of the initial distribution is $T^{-1} \sum_{t=0}^{T-1} \delta\nu^{\top} P^t \mathbf{1}_{\mathcal{A}}$. The derivative along Δ is the expression in Theorem 4.2(e), $T^{-1} \sum_{t=1}^{T-1} \sum_{s=0}^{t-1} \nu^{\top} P^s \Delta P^{t-1-s} \mathbf{1}_{\mathcal{A}} = \sum_{x,y} G_{xy}\Delta_{xy}$, with G as stated. □

Appendix B Proofs and further results for Sections 5–6

Throughout, kTS is the number of recorded positions of one puppet, and for a puppet u we write TP_u, FP_u, FN_u and TN_u for the numbers of its recorded positions that are true positives, false positives, false

negatives and true negatives, and $\mathrm{PP}_u = \mathrm{TP}_u + \mathrm{FP}_u$ and $\mathrm{PN}_u = \mathrm{FN}_u + \mathrm{TN}_u$ for the numbers of its flagged (predicted positive) and unflagged positions. Thus $kTS\,\bar{E}_u = \mathrm{TP}_u + \mathrm{FN}_u$ and $kTS\,\bar{Q}_u = \mathrm{PP}_u$. All statements are conditional on the environment of the window (Section 5.1). We use a superpopulation formulation: the vectors of potential records $\{\mathcal{R}_u(z) : z \in \mathcal{D}_A \times \{\mathrm{alg}, \mathrm{ref}\}\}$ of distinct puppets are independent, and the assignment $((D_u, \Pi_u))_u$ is independent of them.

B.1 Two counting lemmas

Lemma B.1 (Counting identity). *For every cell, $q_\pi(d) = \mathrm{Se}_\pi(d)\mu_\pi(d) + \{1 - \mathrm{Sp}_\pi(d)\}\{1 - \mu_\pi(d)\} = \{1 - \mathrm{Sp}_\pi(d)\} + J_\pi(d)\mu_\pi(d)$, and $q_\pi(d)\mathrm{PPV}_\pi(d) = \mathrm{Se}_\pi(d)\mu_\pi(d)$, where $\mathrm{PPV}_\pi(d) = \mathbb{E}\mathrm{TP}_u/\mathbb{E}\mathrm{PP}_u$ is the exposure-weighted positive predictive value.*

Proof. Take expectations in $kTS\,\bar{Q}_u = \mathrm{TP}_u + \mathrm{FP}_u$. By definition (5.1), $\mathbb{E}\mathrm{TP}_u = \mathrm{Se}_\pi(d)\,\mathbb{E}(\mathrm{TP}_u + \mathrm{FN}_u) = \mathrm{Se}_\pi(d)\,kTS\,\mu_\pi(d)$, and $\mathbb{E}\mathrm{FP}_u = \mathbb{E}(\mathrm{FP}_u + \mathrm{TN}_u) - \mathbb{E}\mathrm{TN}_u = \{1 - \mathrm{Sp}_\pi(d)\}\,kTS\,\{1 - \mu_\pi(d)\}$. Dividing by kTS gives the first identity, and rearranging gives the second. Finally, $q_\pi(d)\mathrm{PPV}_\pi(d) = \mathbb{E}\mathrm{TP}_u/(kTS) = \mathrm{Se}_\pi(d)\mu_\pi(d)$. (If $\mu_\pi(d) = 0$ or 1, the undefined rate can be given any value and the identities still hold.) □

Lemma B.2 (Confirmed-flagged exposure). *Let $Y_u = \bar{Q}_u\widehat{P}_u$ as in Section 6.2, with $Y_u = 0$ if the puppet has no flagged position. Then $\mathbb{E}(Y_u \mid \mathcal{R}_u) = \mathrm{TP}_u/(kTS)$, and $\mathbb{E}Y_u = \mathrm{Se}_\pi(d)\mu_\pi(d)$ for a puppet of cell (d, π).*

Proof. If $\mathrm{PP}_u = 0$ then $Y_u = 0 = \mathrm{TP}_u/(kTS)$. If $0 < \mathrm{PP}_u \le m$, all flagged positions are coded, so $\widehat{P}_u = \mathrm{TP}_u/\mathrm{PP}_u$. If $\mathrm{PP}_u > m$, the number of confirmed positives among m positions drawn without replacement is hypergeometric with mean $m\mathrm{TP}_u/\mathrm{PP}_u$. In every case $\mathbb{E}(\widehat{P}_u \mid \mathcal{R}_u) = \mathrm{TP}_u/\mathrm{PP}_u$, and since $\bar{Q}_u = \mathrm{PP}_u/(kTS)$ is a function of the record, $\mathbb{E}(Y_u \mid \mathcal{R}_u) = \mathrm{TP}_u/(kTS)$. Taking expectations and using (5.1) gives $\mathbb{E}Y_u = \mathrm{Se}_\pi(d)\mu_\pi(d)$. □

B.2 Proof of Theorem 5.1

(a) Identification. By Assumption A3, the observed record of puppet u is $\mathcal{R}_u(D_u, \Pi_u)$. Because the assignment is independent of the potential records, $\mathbb{P}\{\mathcal{R}_u(D_u, \Pi_u) \in \cdot \mid (D_u, \Pi_u) = (d, \pi)\} = \mathbb{P}\{\mathcal{R}_u(d, \pi) \in \cdot\}$. By Assumption A4(a), this law $F_{d\pi}$ is the same for every puppet. Hence $q_\pi(d) = \mathbb{E}\{\bar{Q}_u \mid (D_u, \Pi_u) = (d, \pi)\}$ is the mean of $\bar{Q}$ under $F_{d\pi}$, a functional of the law of the observed data. By Assumption A6, $\bar{E}_u$ under $F_{d\pi}$ is the session average of draws of $E_u(\pi; d)$, so its mean is $\mu_\pi(d)$. Lemma B.1 with Assumption A5 gives $q_\pi(d) = \{1 - \mathrm{Sp}(d)\} + J(d)\mu_\pi(d)$, and since $J(d) > 0$, $\mu_\pi(d) = \{q_\pi(d) + \mathrm{Sp}(d) - 1\}/J(d)$. The expressions for $\mathrm{AA}(d)$ and $\mathrm{AR}(d)$ follow by subtraction and division within persona d, in which $\mathrm{Sp}(d) - 1$ cancels from the difference and $J(d)$ from the ratio. A linear combination $\sum c_{d\pi}\mu_\pi(d)$ equals $\sum c_{d\pi}\{q_\pi(d) + \mathrm{Sp}(d) - 1\}/J(d)$. The three terms of the persona-contrast decomposition are such combinations. For the demand contrast, $\mu_{\mathrm{ref}}(d) - \mu_{\mathrm{ref}}(d_0) = \{q_{\mathrm{ref}}(d) + \mathrm{Sp}(d) - 1\}/J(d) - \{q_{\mathrm{ref}}(d_0) + \mathrm{Sp}(d_0) - 1\}/J(d_0)$. Neither Sp nor J cancels unless both rates coincide across the two personas. Correcting with common rates when they differ gives the wrong contrast, by Theorem 6.1(c) applied to personas instead of arms.

(b) Asymptotic normality. Conditionally on the assignment, the observed records are independent, and those of the $n_{d\pi}$ puppets of cell (d, π) are identically distributed with law $F_{d\pi}$. Since $\bar{Q}_u \in [0, 1]$, the variance $\sigma^2_{d\pi}$ is finite, and the central limit theorem gives $\sqrt{n_{d\pi}}(\widehat{q}_{d\pi} - q_\pi(d)) \Rightarrow \mathrm{N}(0, \sigma^2_{d\pi})$. With $n_{d\pi}/n \to p_{d\pi}$, $\sqrt{n}(\widehat{q}_{d\pi} - q_\pi(d)) \Rightarrow \mathrm{N}(0, \sigma^2_{d\pi}/p_{d\pi})$. The cell means are independent given the assignment, and since the limit does not depend on the assignment, they are jointly asymptotically normal with a diagonal covariance matrix. The map $(q_a, q_r) \mapsto (q_a - q_r)/J$ is linear, which gives the law of

$\widehat{\mathrm{AA}}(d)$. The map $(q_a, q_r) \mapsto \log(q_a + \mathrm{Sp} - 1) - \log(q_r + \mathrm{Sp} - 1)$ is differentiable at the truth when $\mu_{\mathrm{alg}}(d), \mu_{\mathrm{ref}}(d) > 0$, with gradient $\{1/(J\mu_{\mathrm{alg}}), -1/(J\mu_{\mathrm{ref}})\}$, so the delta method (van der Vaart, 1998, Ch. 3) gives the law of $\log \widehat{\mathrm{AR}}(d)$. Finally, $\sigma^2_{d\pi} = \mathrm{Var}(S^{-1}\sum_s Q_{us}) = S^{-2}\sum_{s,s'} \mathrm{Cov}(Q_{us}, Q_{us'})$. If the sessions are exchangeable, with common mean q, variance σ^2_{s} and pairwise correlation γ, this equals $S^{-2}\{S\sigma^2_{\mathrm{s}} + S(S-1)\gamma\sigma^2_{\mathrm{s}}\} = \sigma^2_{\mathrm{s}}\{1 + (S-1)\gamma\}/S$, so deff $= 1 + (S-1)\gamma$.

(c) Variance estimation. In the saturated regression of Q_{us} on cell indicators, the coefficient of cell (d, π) is the mean of its $n_{d\pi}S$ sessions, which equals $\widehat{q}_{d\pi}$ because every puppet has S sessions. With S sessions for every puppet, its bread is $(n_{d\pi}S)^{-1}$ and its residuals are $e_{us} = Q_{us} - \widehat{q}_{d\pi}$. The cluster-robust meat is $\sum_u(\sum_s e_{us})^2 = S^2\sum_u(\bar{Q}_u - \widehat{q}_{d\pi})^2$. The sandwich is therefore $n^{-2}_{d\pi}\sum_u(\bar{Q}_u - \widehat{q}_{d\pi})^2$, and multiplying by $n_{d\pi}/(n_{d\pi} - 1)$ gives $s^2_{d\pi}/n_{d\pi}$. By the strong law of large numbers, $s^2_{d\pi} \to \sigma^2_{d\pi}$ almost surely, and Slutsky's lemma gives asymptotic coverage $1 - \alpha$ for the Wald intervals. The estimator that treats sessions as independent is $\widehat{\sigma}^2_{\mathrm{s},d\pi}/(n_{d\pi}S)$, where $\widehat{\sigma}^2_{\mathrm{s},d\pi}$ is the sample variance of the $n_{d\pi}S$ values Q_{us}. By the strong law applied to the puppet-level sums $\sum_s(Q_{us} - q)^2$, $\widehat{\sigma}^2_{\mathrm{s},d\pi} \to S^{-1}\sum_s \mathbb{E}(Q_{us} - q)^2 = \sigma^2_{\mathrm{s},d\pi}$. Hence $n_{d\pi}$ times this estimator converges to $\sigma^2_{\mathrm{s},d\pi}/S = \sigma^2_{d\pi}/\mathrm{deff}_{d\pi}$, which differs from $\sigma^2_{d\pi}$ unless $\mathrm{deff}_{d\pi} = 1$ and is smaller when $\mathrm{deff}_{d\pi} > 1$.

(d) Blocked design. Under Assumption A4(b), randomisation within waves makes the observed records of the puppets of wave w and cell (d, π) independent draws from $F_{wd\pi}$, and records of different waves independent. Part (a), applied within wave w with the wave's error rates (equal across arms), gives $\mu_{w\pi}(d) = \{q_{w\pi}(d) + \mathrm{Sp}_w - 1\}/J_w$. Hence $\widehat{\mu}_{wd\pi} = (\widehat{q}_{wd\pi} + \mathrm{Sp}_w - 1)/J_w$ is unbiased for $\mu_{w\pi}(d)$, and $\sum_w \omega_w\widehat{\mu}_{wd\pi}$ is unbiased for $\bar{\mu}_\pi(d)$. With W fixed and $n_{wd\pi}/n \to p_{wd\pi} > 0$, the $2W$ wave-cell means are independent and asymptotically normal as in (b), so $\sqrt{n}\{\sum_w \omega_w\widehat{\mu}_{wd\pi} - \bar{\mu}_\pi(d)\} \Rightarrow \mathrm{N}\{0, \sum_w \omega^2_w\sigma^2_{wd\pi}/(J^2_w p_{wd\pi})\}$, independently for the two arms. The difference and the delta method for the log ratio, with gradient $(1/\bar{\mu}_{\mathrm{alg}}, -1/\bar{\mu}_{\mathrm{ref}})$, give the stated laws. Consistency of the stratified variance estimator follows wave by wave as in (c). □

B.3 Interference: proofs and further results

Proof of Proposition 5.2(a). Under Assumption A8, waves are independent, and within wave w, given the load ℓ_w, the puppets of cell (d, π) are independent draws from a law with mean exposure $m_{d\pi}(\ell_w)$ and arm-independent error rates. This is Assumption A4(b) with $F_{wd\pi}$ the law at load ℓ_w. By Theorem 5.1(d), each corrected wave-cell mean is unbiased for $m_{d\pi}(\ell_w)$, so $\mathbb{E}\widehat{\mathrm{AA}}(d) = \bar{m}_{\mathrm{alg}}(d) - \bar{m}_{\mathrm{ref}}(d)$ with $\bar{m}_\pi(d) = \sum_w \omega_w m_{d\pi}(\ell_w)$. Writing $\mathrm{AA}(\ell) = m_{d,\mathrm{alg}}(\ell) - m_{d,\mathrm{ref}}(\ell)$, so that $\mathrm{AA}(0) = \mathrm{AA}^0(d)$,

$$\left|\mathbb{E}\widehat{\mathrm{AA}}(d) - \mathrm{AA}^0(d)\right| = \left|\sum_w \omega_w\{\mathrm{AA}(\ell_w) - \mathrm{AA}(0)\}\right| \le \sum_w \omega_w\xi_\Delta\ell_w = \xi_\Delta\bar{\ell}.$$

By the triangle inequality, $|\mathrm{AA}(\ell) - \mathrm{AA}(0)| \le |m_{d,\mathrm{alg}}(\ell) - m_{d,\mathrm{alg}}(0)| + |m_{d,\mathrm{ref}}(\ell) - m_{d,\mathrm{ref}}(0)| \le (\xi_{\mathrm{alg}} + \xi_{\mathrm{ref}})\ell$, so the smallest admissible ξ_Δ is no larger than $\xi_{\mathrm{alg}} + \xi_{\mathrm{ref}}$ (an arbitrary admissible ξ_Δ need not be). For the ratio, $|\bar{m}_\pi(d) - \mu_\pi(d)| \le \xi_\pi\bar{\ell}$ in the same way. If $\bar{m}_\pi(d) > \xi_\pi\bar{\ell}$, then $\min\{\bar{m}_\pi(d), \mu_\pi(d)\} \ge \bar{m}_\pi(d) - \xi_\pi\bar{\ell} > 0$, and the mean value theorem gives $|\log\bar{m}_\pi(d) - \log\mu_\pi(d)| \le |\bar{m}_\pi(d) - \mu_\pi(d)|/\min\{\bar{m}_\pi(d), \mu_\pi(d)\} \le \xi_\pi\bar{\ell}/\{\bar{m}_\pi(d) - \xi_\pi\bar{\ell}\}$. Adding the two arms bounds $|\log\{\bar{m}_{\mathrm{alg}}(d)/\bar{m}_{\mathrm{ref}}(d)\} - \log\mathrm{AR}^0(d)|$. For attainment, take $m_{d,\mathrm{alg}}(\ell) = \mu_{\mathrm{alg}}(d) + \xi_\Delta\ell$ and $m_{d,\mathrm{ref}}(\ell) = \mu_{\mathrm{ref}}(d)$, which satisfies Assumption A8 with $\xi_{\mathrm{alg}} = \xi_\Delta$ and $\xi_{\mathrm{ref}} = 0$ and has bias exactly $\xi_\Delta\bar{\ell}$. Taking instead $m_{d,\mathrm{alg}}(\ell) = \mu_{\mathrm{alg}}(d) + \xi_{\mathrm{alg}}\ell$ and $m_{d,\mathrm{ref}}(\ell) = \mu_{\mathrm{ref}}(d) - \xi_{\mathrm{ref}}\ell$, with $\xi_{\mathrm{ref}}\ell_{\max} \le \mu_{\mathrm{ref}}(d)$, gives $\xi_\Delta = \xi_{\mathrm{alg}} + \xi_{\mathrm{ref}}$ and bias $(\xi_{\mathrm{alg}} + \xi_{\mathrm{ref}})\bar{\ell}$.

Proof of Proposition 5.2(d). By (a), $\mathrm{AA}^0(d) \in [\mathbb{E}\widehat{\mathrm{AA}} - \xi_\Delta\bar{\ell}, \mathbb{E}\widehat{\mathrm{AA}} + \xi_\Delta\bar{\ell}]$. On the event $|\widehat{\mathrm{AA}} - \mathbb{E}\widehat{\mathrm{AA}}| \le z_{1-\alpha/2}\widehat{\mathrm{se}}$, whose probability is $1 - \alpha$ to the accuracy of the normal approximation of Theorem 5.1, the

interval $\widehat{\mathrm{AA}} \pm (\xi_\Delta \bar{\ell} + z_{1-\alpha/2}\widehat{\mathrm{se}})$ contains $\mathrm{AA}^0(d)$. It excludes zero if and only if $|\widehat{\mathrm{AA}}| > \xi_\Delta \bar{\ell} + z_{1-\alpha/2}\widehat{\mathrm{se}}$, that is, if and only if $\xi_\Delta < \xi^*$. Parts (b) and (c) are Propositions B.3 and B.4 below. □

Proposition B.3 (Direction at a fixed state). *Suppose a policy fills each recorded position independently, choosing item c with probability proportional to $f_c\, p_c^{\theta}$, where $f_c \geq 0$ is a relevance factor fixed by the puppet's current state, p_c is the item's popularity and $\theta > 0$. Suppose the wave's puppets shift popularity to $p = (1-\ell)p^0 + \ell p^{\mathrm{P}}$, where p^{P} is the popularity generated by the puppets' openings. Let $\chi(\ell)$ be the probability that a position shows an item with $L_c = 1$.*

(i) If p^{P} puts no mass on items with $L_c = 1$, then $\chi(\ell)$ is non-increasing in ℓ.

(ii) If $\theta = 1$, then for every p^{P}, $\chi(\ell) = (1-\epsilon_\ell)\,\chi(0) + \epsilon_\ell\,\chi^{\mathrm{P}}$, where $\chi^{\mathrm{P}} = \sum_{c:L_c=1} f_c p_c^{\mathrm{P}} / \sum_c f_c p_c^{\mathrm{P}}$ and $\epsilon_\ell \in [0,1)$ increases from $\epsilon_0 = 0$. Exposure then rises with load if and only if $\chi^{\mathrm{P}} > \chi(0)$, which a small share of puppet engagement on false negatives with large f_c can produce.

Proof. Fix the puppet's state, and so the factors f_c. Let $\mathcal{A}_1 = \{c : L_c = 1\}$, and for a popularity vector p let $s_1(p) = \sum_{c\in\mathcal{A}_1} f_c p_c^{\theta}$ and $s(p) = \sum_c f_c p_c^{\theta}$, assumed positive, so that the probability that a position shows an item with $L_c = 1$ is $s_1(p)/s(p)$. Let $0 \leq \ell < \ell' < 1$. Then $p(\ell') = (1-\epsilon)p(\ell) + \epsilon p^{\mathrm{P}}$ with $\epsilon = (\ell'-\ell)/(1-\ell) \in (0,1]$, as one checks by expanding both sides. For $c \in \mathcal{A}_1$, $p_c^{\mathrm{P}} = 0$, so $p_c(\ell') = (1-\epsilon)p_c(\ell)$ and $s_1\{p(\ell')\} = (1-\epsilon)^{\theta} s_1\{p(\ell)\}$. For $c \notin \mathcal{A}_1$, $p_c(\ell') \geq (1-\epsilon)p_c(\ell)$, and $x \mapsto x^{\theta}$ is increasing, so $s\{p(\ell')\} - s_1\{p(\ell')\} \geq (1-\epsilon)^{\theta}[s\{p(\ell)\} - s_1\{p(\ell)\}]$. Hence $s\{p(\ell')\} \geq (1-\epsilon)^{\theta} s\{p(\ell)\}$ and $s_1\{p(\ell')\}/s\{p(\ell')\} \leq s_1\{p(\ell)\}/s\{p(\ell)\}$, which is (i). For (ii), with $\theta = 1$, both sums are linear in p, so $s_1\{p(\ell)\} = (1-\ell)s_1(p^0) + \ell s_1(p^{\mathrm{P}})$ and $s\{p(\ell)\} = (1-\ell)s(p^0) + \ell s(p^{\mathrm{P}})$. Hence

$$\chi(\ell) = (1-\epsilon_\ell)\frac{s_1(p^0)}{s(p^0)} + \epsilon_\ell \frac{s_1(p^{\mathrm{P}})}{s(p^{\mathrm{P}})}, \qquad \epsilon_\ell = \frac{\ell\, s(p^{\mathrm{P}})}{(1-\ell)s(p^0) + \ell\, s(p^{\mathrm{P}})},$$

with $s_1(p^0)/s(p^0) = \chi(0)$ and $s_1(p^{\mathrm{P}})/s(p^{\mathrm{P}}) = \chi^{\mathrm{P}}$. The weight ϵ_ℓ is zero at $\ell = 0$ and increases in ℓ, so $\chi(\ell) - \chi(0) = \epsilon_\ell\{\chi^{\mathrm{P}} - \chi(0)\}$, whose sign is that of $\chi^{\mathrm{P}} - \chi(0)$. Both statements fix the factors f_c, that is, the puppet's state. □

Proposition B.4 (Randomised saturation). *Under the conditions of Proposition 5.2, suppose waves are randomised to two loads $\ell_1 < \ell_2$, with weights $\omega^{(1)}$ and $\omega^{(2)}$ summing to one, and let $\mathrm{AA}(\ell) = m_{d,\mathrm{alg}}(\ell) - m_{d,\mathrm{ref}}(\ell)$ be twice differentiable on $[0,\ell_2]$ with $|\mathrm{AA}''| \leq \zeta$. The extrapolated estimator $\widehat{\mathrm{AA}}^{\mathrm{ex}} = \{\ell_2\widehat{\mathrm{AA}}(\ell_1) - \ell_1\widehat{\mathrm{AA}}(\ell_2)\}/(\ell_2 - \ell_1)$, built from the blocked estimators at each load, satisfies*

$$\left|\mathbb{E}\widehat{\mathrm{AA}}^{\mathrm{ex}} - \mathrm{AA}^0(d)\right| \leq \tfrac{1}{2}\,\zeta\,\ell_1\ell_2,$$

and its variance is $\{\ell_2^2 v_1 + \ell_1^2 v_2\}/(\ell_2-\ell_1)^2$, where v_j is the variance of $\widehat{\mathrm{AA}}(\ell_j)$. The slope $\{\widehat{\mathrm{AA}}(\ell_2) - \widehat{\mathrm{AA}}(\ell_1)\}/(\ell_2-\ell_1)$ estimates the average derivative of AA on $[\ell_1,\ell_2]$, and a test that it is zero is a test of differential interference.

Proof. Let $f = \mathrm{AA}$ on $[0,\ell_2]$ and let g be the linear function with $g(\ell_j) = f(\ell_j)$, $j = 1,2$. Then $g(0) = \{\ell_2 f(\ell_1) - \ell_1 f(\ell_2)\}/(\ell_2-\ell_1)$. The interpolation remainder for twice-differentiable f is standard: for every x there is ℓ_x^* in the smallest interval containing x, ℓ_1, ℓ_2 with $f(x) - g(x) = \frac{1}{2}f''(\ell_x^*)(x-\ell_1)(x-\ell_2)$. (Proof: for $x \notin \{\ell_1,\ell_2\}$, the function $y \mapsto f(y) - g(y) - \{f(x)-g(x)\}(y-\ell_1)(y-\ell_2)/\{(x-\ell_1)(x-\ell_2)\}$ vanishes at ℓ_1, ℓ_2 and x, so by Rolle's theorem its second derivative vanishes at some ℓ_x^*.) At $x = 0$ this gives $|f(0) - g(0)| \leq \frac{1}{2}\zeta\ell_1\ell_2$. By Theorem 5.1(d) applied to the waves at each load, $\mathbb{E}\widehat{\mathrm{AA}}(\ell_j) = f(\ell_j)$, and the extrapolated estimator is linear, so $\mathbb{E}\widehat{\mathrm{AA}}^{\mathrm{ex}} = g(0)$. The two sets of waves are independent, so

the variance of the linear combination is $\{\ell_2^2 v_1 + \ell_1^2 v_2\}/(\ell_2 - \ell_1)^2$. The slope estimator converges to $\{f(\ell_2) - f(\ell_1)\}/(\ell_2 - \ell_1) = (\ell_2 - \ell_1)^{-1} \int_{\ell_1}^{\ell_2} f'$, which is zero when f is constant, that is, without differential interference on $[\ell_1, \ell_2]$. □

B.4 Rotation when cells cannot share a wave

If capacity does not allow every cell in every wave, cells can be rotated across waves in a balanced incomplete block design, in which every pair of cells shares a wave equally often. Within-wave contrasts then identify every pairwise contrast, and the window average is estimated by combining them. The contrast between alg and ref for the same persona should always share a wave, because it is the contrast that drift would otherwise confound.

B.5 Proof of Theorem 6.1, and the variances of corrected estimators

Proposition B.5 (Variance of corrected estimators). *Under Assumption A5, let* $(\widehat{\mathrm{Se}}, \widehat{\mathrm{Sp}})$ *come from an external design, and let the numbers of puppets* n_a, n_r *and of coded items* n_1, n_0 *grow in fixed proportions. Then* $\widehat{\mathrm{AA}}$ *and* $\log \widehat{\mathrm{AR}}$ *of Theorem 6.1(b) have asymptotic variances*

$$\mathrm{Var}(\widehat{\mathrm{AA}}) \simeq \frac{1}{J^2}\Big\{v_a + v_r + \mathrm{AA}^2\,(v_{\mathrm{Se}} + v_{\mathrm{Sp}})\Big\}, \tag{B.1}$$

$$\mathrm{Var}(\log \widehat{\mathrm{AR}}) \simeq \frac{1}{J^2}\left\{\frac{v_a}{\mu_a^2} + \frac{v_r}{\mu_r^2} + \left(\frac{1}{\mu_a} - \frac{1}{\mu_r}\right)^2 v_{\mathrm{Sp}}\right\}, \tag{B.2}$$

where $v_\pi = \sigma_\pi^2/n_\pi$ *is the cluster-robust variance of* $\widehat{q}_\pi$ *(Theorem 5.1),* $v_{\mathrm{Se}} = \mathrm{Se}(1-\mathrm{Se})/n_1$ *and* $v_{\mathrm{Sp}} = \mathrm{Sp}(1-\mathrm{Sp})/n_0$. *Replacing every unknown by its estimate gives consistent variance estimators. The sensitivity enters* (B.1) *through* J *and through* v_{Se}*; in* (B.2) *its estimation error does not appear, and it enters only through the common factor* J^{-2}. *With validation stratified by arm and external designs per arm,* $\mathrm{Var}(\widehat{\mu}_\pi^{\mathrm{st}}) \simeq J_\pi^{-2}\{v_\pi + \mu_\pi^2 v_{\mathrm{Se},\pi} + (1-\mu_\pi)^2 v_{\mathrm{Sp},\pi}\}$*, and the two arms are independent.*

Proof of Theorem 6.1(a). By Lemma B.1 and Assumption A5, $q_\pi = (1-\mathrm{Sp}) + J\mu_\pi$. Since $\mu_a = \mathrm{AR}\,\mu_r$,

$$\frac{q_a}{q_r} = \frac{1-\mathrm{Sp}}{q_r} + \frac{J\mu_r}{q_r}\,\mathrm{AR} = w_0 + (1-w_0)\,\mathrm{AR},$$

because $(1-\mathrm{Sp}) + J\mu_r = q_r$. Here $w_0 \geq 0$, and $w_0 < 1$ because $J\mu_r > 0$. Also $w_0 = 0$ if and only if $\mathrm{Sp} = 1$. The difference is $q_a - q_r = J(\mu_a - \mu_r)$.

Proof of Theorem 6.1(b) and Proposition B.5. Let the numbers of puppets and of coded items grow with n in fixed proportions. By Theorem 5.1(b), $\widehat{q}_a$ and $\widehat{q}_r$ are independent and asymptotically normal with variances v_a and v_r. In the external design, $\widehat{\mathrm{Se}}$ and $\widehat{\mathrm{Sp}}$ are proportions in two independent binomial samples, hence asymptotically normal with variances v_{Se} and v_{Sp}, and independent of each other and of the audit. If the numbers n_1 and n_0 are themselves random, as when a random sample of exposures is coded, the same holds conditionally on them, and the limit does not depend on them. The vector $(\widehat{q}_a, \widehat{q}_r, \widehat{\mathrm{Se}}, \widehat{\mathrm{Sp}})$ is therefore jointly asymptotically normal with a diagonal covariance matrix. Let $\mathrm{RG}_\pi(q_\pi, \mathrm{Se}, \mathrm{Sp}) = (q_\pi + \mathrm{Sp} - 1)/(\mathrm{Se} + \mathrm{Sp} - 1)$ be the Rogan–Gladen map. Its partial derivatives at the truth are

$$\frac{\partial \mathrm{RG}_\pi}{\partial q_\pi} = \frac{1}{J}, \qquad \frac{\partial \mathrm{RG}_\pi}{\partial \mathrm{Se}} = -\frac{q_\pi + \mathrm{Sp} - 1}{J^2} = -\frac{\mu_\pi}{J}, \qquad \frac{\partial \mathrm{RG}_\pi}{\partial \mathrm{Sp}} = \frac{J - (q_\pi + \mathrm{Sp} - 1)}{J^2} = \frac{1-\mu_\pi}{J}.$$

For $\mathrm{AA} = \mathrm{RG}_a - \mathrm{RG}_r$, the gradient with respect to $(q_a, q_r, \mathrm{Se}, \mathrm{Sp})$ is $J^{-1}(1, -1, -\mathrm{AA}, -\mathrm{AA})$, which gives (B.1). For $\log \mathrm{AR} = \log(q_a + \mathrm{Sp} - 1) - \log(q_r + \mathrm{Sp} - 1)$, in which J has cancelled, the gradient

is $\{1/(J\mu_a), -1/(J\mu_r), 0, 1/(J\mu_a) - 1/(J\mu_r)\}$, which gives (B.2). The delta method gives asymptotic normality, and the continuous mapping theorem gives consistency of the plug-in variance estimators. The stratified variance is the same computation for each arm with its own rates, the arms being independent.

Lemma B.6 (Range of the naive ratio). *In the setting of Theorem 6.1(c), if the error rates are only known to lie in* $\mathrm{Se}_\pi \in [s_-, s_+]$ *and* $\mathrm{Sp}_\pi \in [p_-, p_+]$*, with* $s_- + p_- > 1$*, the set of values of* $\mathrm{AR}^{\mathrm{nv}}$ *compatible with* (μ_a, μ_r) *is the interval*

$$\mathrm{AR} \cdot \left[\frac{r_-(\mu_a)}{r_+(\mu_r)}, \frac{r_+(\mu_a)}{r_-(\mu_r)}\right], \qquad r_\mp(\mu) = s_\mp + (1 - p_\pm)\frac{1-\mu}{\mu},$$

and it contains values on both sides of one whenever $r_-(\mu_a)/r_+(\mu_r) < 1/\mathrm{AR} < r_+(\mu_a)/r_-(\mu_r)$.

Proof of Theorem 6.1(c) and Lemma B.6. By Lemma B.1, $q_\pi/\mu_\pi = \mathrm{Se}_\pi + (1 - \mathrm{Sp}_\pi)(1 - \mu_\pi)/\mu_\pi = r_\pi$, so $q_a/q_r = (r_a\mu_a)/(r_r\mu_r) = \mathrm{AR}\, r_a/r_r$. With pooled rates, the corrected arm means converge to $(q_\pi + \overline{\mathrm{Sp}} - 1)/\overline{J}$, and $q_\pi + \overline{\mathrm{Sp}} - 1 = J_\pi\mu_\pi + \overline{\mathrm{Sp}} - \mathrm{Sp}_\pi$ by Lemma B.1. The factor $\overline{J}$ cancels from the ratio, which gives (6.2). For the range, fix $\mu \in (0, 1)$. The function $(\mathrm{Se}, \mathrm{Sp}) \mapsto \mathrm{Se} + (1 - \mathrm{Sp})(1 - \mu)/\mu$ is continuous, increasing in Se and decreasing in Sp, so on the rectangle $[s_-, s_+] \times [p_-, p_+]$ it takes every value between its minimum $r_-(\mu)$ at (s_-, p_+) and its maximum $r_+(\mu)$ at (s_+, p_-). On this rectangle $J \geq s_- + p_- - 1 > 0$. The two arms' rates vary independently, so r_a/r_r ranges over the image of a product of two connected sets under a continuous map, which is the interval with endpoints $r_-(\mu_a)/r_+(\mu_r)$ and $r_+(\mu_a)/r_-(\mu_r)$. Multiplying by AR gives the stated set, which contains values below and above one exactly when $r_-(\mu_a)/r_+(\mu_r) < 1/\mathrm{AR} < r_+(\mu_a)/r_-(\mu_r)$. For the mixture model, the expected number of recorded positions of type k is $kTS\, e_{\pi k}$, and each is flagged with probability q_k. Hence $\mathbb{E}\sum \mathbf{1}\left\{L_c = 1, \widehat{L}_c = 1\right\} = kTS\sum_{k:L_k=1} e_{\pi k}q_k$ and $\mathbb{E}\sum \mathbf{1}\{L_c = 1\} = kTS\,\mu_\pi$, which gives Se_π, and $1 - \mathrm{Sp}_\pi$ follows in the same way. Thus $\mathrm{Se}_a = \mathrm{Se}_r$ if and only if $\sum_{k:L_k=1}(e_{ak}/\mu_a - e_{rk}/\mu_r)q_k = 0$, and similarly for the specificities.

Proof of Theorem 6.1(d). If $(\widehat{\mathrm{Se}}_\pi, \widehat{\mathrm{Sp}}_\pi) \to (\mathrm{Se}_\pi, \mathrm{Sp}_\pi)$ in probability and $\widehat{q}_\pi \to q_\pi$, then, since RG_π is continuous where $J_\pi > 0$, $\widehat{\mu}^{\mathrm{st}}_\pi \to (q_\pi + \mathrm{Sp}_\pi - 1)/J_\pi = \mu_\pi$ by Lemma B.1. This uses no relation between the arms. For the identity (6.3), let arm π have PP flagged and PN unflagged recorded positions, so that $\widehat{q}_\pi = \mathrm{PP}/(\mathrm{PP} + \mathrm{PN})$. Let x_1 of the m_1 coded flagged positions and x_0 of the m_0 coded unflagged positions be antisemitic, and define the weighted counts $\mathrm{TP} = \mathrm{PP}\, x_1/m_1$, $\mathrm{FP} = \mathrm{PP}\,(m_1 - x_1)/m_1$, $\mathrm{FN} = \mathrm{PN}\, x_0/m_0$ and $\mathrm{TN} = \mathrm{PN}\,(m_0 - x_0)/m_0$. Then $\mathrm{TP} + \mathrm{FP} = \mathrm{PP}$ and $\mathrm{FN} + \mathrm{TN} = \mathrm{PN}$. Let $\mathrm{AP} = \mathrm{TP} + \mathrm{FN}$ and $\mathrm{TOT} = \mathrm{PP} + \mathrm{PN}$, so that $\widehat{\mathrm{Se}}_\pi = \mathrm{TP}/\mathrm{AP}$ and $\widehat{\mathrm{Sp}}_\pi = \mathrm{TN}/(\mathrm{TOT} - \mathrm{AP})$. Then

$$\widehat{q}_\pi + \widehat{\mathrm{Sp}}_\pi - 1 = \frac{\mathrm{PP}}{\mathrm{TOT}} - \frac{\mathrm{FP}}{\mathrm{TOT} - \mathrm{AP}} = \frac{\mathrm{TP}(\mathrm{TOT} - \mathrm{AP}) - \mathrm{FP}\,\mathrm{AP}}{\mathrm{TOT}(\mathrm{TOT} - \mathrm{AP})},$$
$$\widehat{\mathrm{Se}}_\pi + \widehat{\mathrm{Sp}}_\pi - 1 = \frac{\mathrm{TP}(\mathrm{TOT} - \mathrm{AP}) - \mathrm{FP}\,\mathrm{AP}}{\mathrm{AP}(\mathrm{TOT} - \mathrm{AP})},$$

using $\mathrm{PP} = \mathrm{TP} + \mathrm{FP}$ in the first numerator. Whenever $\widehat{J}_\pi \neq 0$ the ratio is $\mathrm{AP}/\mathrm{TOT} = \widehat{q}_\pi(x_1/m_1) + (1 - \widehat{q}_\pi)(x_0/m_0)$, which is (6.3). Applied to the records of one puppet, with coded positions drawn without replacement within its two strata, the hypergeometric means give $\mathbb{E}(x_1/m_1 \mid \mathcal{R}_u) = \mathrm{TP}_u/\mathrm{PP}_u$ and $\mathbb{E}(x_0/m_0 \mid \mathcal{R}_u) = \mathrm{FN}_u/\mathrm{PN}_u$, so the conditional mean of the right side is $(\mathrm{TP}_u + \mathrm{FN}_u)/(kTS) = \bar{E}_u$. An empty stratum contributes zero, exactly. With the unflagged stratum split into score bands, the same argument applies band by band.

Proof of Theorem 6.1(e). This is the last identity of Lemma B.1, divided between the arms. □

B.6 Proof of Theorem 6.2

Let $a > 0$ and $f(n) = (n^*)^{1/2} h(n/n^*)$, so that the radius of Algorithm 1 is $h_n = \widehat{\sigma}_n f(n)/n$. Since $\log((1+as)^{1/2}/\alpha) \geq \log(1/\alpha)$, also $f(n)^2 \geq 2\log(1/\alpha)\, n$ for every n.

Lemma B.7 (Time-uniform crossing with a plug-in scale). *Let $\psi_1, \psi_2, \ldots$ be independent and identically distributed, bounded, with mean zero and variance $\sigma_\psi^2 \in (0, \infty)$. Let R_n and $\widehat{\sigma}_n$ be random variables on the same space, with $R_n = O(\log\log n)$ and $\widehat{\sigma}_n \to \sigma_\psi$ almost surely. Fix $c > 0$ and let $n_{\mathrm{b}} = \lceil cn^* \rceil$. Then*

$$\limsup_{n^* \to \infty} \mathbb{P}\Big(\exists\, n \geq n_{\mathrm{b}} : \Big|\sum_{i \leq n} \psi_i + R_n\Big| > \widehat{\sigma}_n f(n)\Big) \leq \bar{\alpha}(c) = \mathbb{P}\big(\exists\, s \geq c : |\mathbb{B}_s| \geq h(s)\big) \leq \alpha,$$

where $\mathbb{B}$ is a standard Brownian motion. Moreover $\bar{\alpha}(c) \to \alpha$ as $c \downarrow 0$.

Proof. Step 1: Brownian motion. For $x \in \mathbb{R}$ let $\mathcal{E}_s(x) = \exp(x\mathbb{B}_s - x^2 s/2)$. Mixing over $x \sim \mathrm{N}(0, a)$ gives

$$\mathcal{M}_s = \int \mathcal{E}_s(x)\, \frac{e^{-x^2/(2a)}}{(2\pi a)^{1/2}}\, dx = (1+as)^{-1/2} \exp\Big\{\frac{a\mathbb{B}_s^2}{2(1+as)}\Big\},$$

a continuous nonnegative martingale with $\mathcal{M}_0 = 1$ (Robbins, 1970). A direct computation shows that $\mathcal{M}_s \geq 1/\alpha$ if and only if $|\mathbb{B}_s| \geq h(s)$. By Ville's inequality (Ville, 1939), $\mathbb{P}(\sup_s \mathcal{M}_s \geq 1/\alpha) \leq \alpha$, so $\bar{\alpha}(c) \leq \bar{\alpha}(0) \leq \alpha$. For the limit $c \downarrow 0$: by the law of the iterated logarithm $\mathcal{M}_s \to 0$ almost surely. Let $s^* = \inf\{s : \mathcal{M}_s \geq 1/\alpha\}$. The stopped martingale $\mathcal{M}_{s \wedge s^*}$ is bounded by $1/\alpha$ and has mean one, and it converges to $\alpha^{-1}\mathbf{1}\{s^* < \infty\}$, so bounded convergence gives $\mathbb{P}(s^* < \infty) = \alpha$, that is, $\bar{\alpha}(0) = \alpha$. Since $|\mathbb{B}_0| = 0 < h(0)$ and the paths are continuous, $\{s^* < \infty\} = \bigcup_{c>0}\{\exists\, s \geq c : |\mathbb{B}_s| \geq h(s)\}$, an increasing union as $c \downarrow 0$, so $\bar{\alpha}(c) \uparrow \alpha$. Finally, let $G_c(y) = \mathbb{P}\{\sup_{s \geq c} |\mathbb{B}_s|/h(s) \geq y\}$. The supremum over $[c, \infty)$ is attained almost surely, because the ratio is continuous and tends to zero, so $G_c(1) = \bar{\alpha}(c)$, and $G_c(1-\epsilon) \downarrow G_c(1)$ as $\epsilon \downarrow 0$ by continuity from above.

Step 2: coupling. Because the ψ_i are bounded, they have a finite moment generating function, and the strong approximation of Komlós et al. (1975, 1976) gives, after enlarging the probability space if necessary, a standard Brownian motion $\mathbb{W}$ with $\sum_{i \leq n} \psi_i - \sigma_\psi \mathbb{W}_n = O(\log n)$ almost surely. The enlargement can carry the whole data sequence, by drawing each observation from its conditional law given ψ_i, so R_n and $\widehat{\sigma}_n$ live on the same space. Fix $\epsilon \in (0, 1/3)$. Since $f(n) \geq \{2\log(1/\alpha)n\}^{1/2}$, the ratio $\{|\sum_{i \leq n} \psi_i - \sigma_\psi \mathbb{W}_n| + |R_n|\}/f(n)$ tends to zero almost surely, and so does $\sup_{n \geq n'} |\widehat{\sigma}_n - \sigma_\psi|$ as $n' \to \infty$. Hence the event E_{n^*} that, for all $n \geq n_{\mathrm{b}}$, $|\sum_{i \leq n} \psi_i - \sigma_\psi \mathbb{W}_n| + |R_n| \leq \epsilon\sigma_\psi f(n)$ and $\widehat{\sigma}_n \geq (1-\epsilon)\sigma_\psi$, has probability tending to one as $n^* \to \infty$. On E_{n^*}, for $n \geq n_{\mathrm{b}}$, the event $|\sum_{i \leq n} \psi_i + R_n| > \widehat{\sigma}_n f(n)$ implies $|\mathbb{W}_n| > (1-2\epsilon) f(n)$. By Brownian scaling, $\mathbb{B}_s = (n^*)^{-1/2} \mathbb{W}_{sn^*}$ is a standard Brownian motion, and $|\mathbb{W}_n| > (1-2\epsilon) f(n)$ is the event $|\mathbb{B}_{n/n^*}| > (1-2\epsilon) h(n/n^*)$ with $n/n^* \geq c$. Hence the probability in the lemma is at most $\mathbb{P}(E_{n^*}^c) + G_c(1-2\epsilon)$. Letting $n^* \to \infty$ and then $\epsilon \downarrow 0$ gives the bound $G_c(1) = \bar{\alpha}(c)$. □

Tying the mixture to n^* is essential. With a mixture that does not depend on n^*, $f(n)$ grows like $(n \log n)^{1/2}$, faster than the law of the iterated logarithm allows a random walk to grow, and the probability of a crossing after n' pairs tends to zero as $n' \to \infty$ for every α. A limit statement of that kind is true but carries no information about the level.

Proof of Theorem 6.2(a). Let $\vartheta = \mathbb{E}O_a/\mathbb{E}O_r = \lambda\mathrm{AR}$. Suppose $\log \mathrm{AR} \notin C_n$. If $\log \mathrm{AR} < \log \widehat{\vartheta}_n - \log \lambda_+ - h_n$, then $\log \widehat{\vartheta}_n - \log \vartheta > h_n + \log(\lambda_+/\lambda) \geq h_n$. If $\log \mathrm{AR} > \log \widehat{\vartheta}_n - \log \lambda_- + h_n$, then $\log \widehat{\vartheta}_n - \log \vartheta < -h_n - \log(\lambda/\lambda_-) \leq -h_n$. In both cases $n|\log \widehat{\vartheta}_n - \log \vartheta| > \widehat{\sigma}_n f(n)$. Now $O_u \in [0, 1]$, so $\psi_i = O_{a,i}/\mathbb{E}O_a - O_{r,i}/\mathbb{E}O_r$ is bounded. A second-order Taylor expansion of the logarithm gives

$n(\log\widehat{\vartheta}_n - \log\vartheta) = \sum_{i\le n}\psi_i + R_n$ with $|R_n| \le Cn\{(\bar{O}_a - \mathbb{E}O_a)^2 + (\bar{O}_r - \mathbb{E}O_r)^2\}$ for large n, and by the law of the iterated logarithm $R_n = O(\log\log n)$ almost surely. Moreover, $\widehat{\sigma}_n^2 \to \mathbb{E}\psi_1^2 = \sigma_\psi^2$ almost surely, by the strong law of large numbers and the continuity of $(x, y) \mapsto (O_a/x - O_r/y)^2$ at $(\mathbb{E}O_a, \mathbb{E}O_r)$, where both are positive. Lemma B.7 gives $\limsup_{n^*\to\infty}\mathbb{P}(\exists\, n \ge n_\mathrm{b} : \log\mathrm{AR} \notin C_n) \le \bar{\alpha}(c) \le \alpha$, with $\bar{\alpha}(c) \to \alpha$ as $c \to 0$. If $N \ge n_\mathrm{b}$, then $\{\log\mathrm{AR} \notin C_N\} \subseteq \{\exists\, n \ge n_\mathrm{b} : \log\mathrm{AR} \notin C_n\}$. Finally, rule S1 states that $\log\mathrm{AR} > \log\delta$ when $\mathrm{LB} > \log\delta$. If the truth is $\log\mathrm{AR} \le \log\delta$, then $\log\mathrm{AR} < \mathrm{LB}$, so some C_n after the burn-in excludes $\log\mathrm{AR}$, because LB and UB are formed only from sets after the burn-in. The same argument applies to S2.

Proof of Theorem 6.2(b). Let $\vartheta^* = \mathbb{E}O_a/\mathbb{E}O_r$ and $X_i = O_{a,i} - \vartheta^* O_{r,i} \in [-\vartheta^*, 1]$. The X_i are independent with mean zero. Let $b_i^+ \in [0, c/\vartheta^*]$ and $b_i^- \in [0, c]$, with $c < 1$, be predictable bets, that is, functions of $X_1, \dots, X_{i-1}$. Then $1 + b_i^+ X_i \ge 1 - c > 0$ and $1 - b_i^- X_i \ge 1 - c > 0$, and $\mathbb{E}(1 \pm b_i^\pm X_i \mid X_1, \dots, X_{i-1}) = 1$. Hence $\mathrm{Cap}_n^\pm = \prod_{i\le n}(1 \pm b_i^\pm X_i)$ are nonnegative martingales with initial value one, and so is $\mathrm{Cap}_n = \frac{1}{2}\mathrm{Cap}_n^+ + \frac{1}{2}\mathrm{Cap}_n^-$, the hedged capital process of Waudby-Smith and Ramdas (2024). By Ville's inequality, $\mathbb{P}(\exists\, n : \mathrm{Cap}_n(\vartheta^*) \ge 1/\alpha) \le \alpha$. On the complementary event, ϑ^* is never rejected, and since $\lambda \in [\lambda_-, \lambda_+]$, $\log\mathrm{AR} = \log\vartheta^* - \log\lambda \in C_n^{\mathrm{bet}}$ for every n. □

B.7 Proof of Proposition 6.3

Variances. Assume, as in (6.5), that given the true labels $\mathcal{L}_u$ of a puppet's recorded positions the flags are independent, with probability Se for a position with $L_c = 1$ and $1 - \mathrm{Sp}$ for one with $L_c = 0$. Then $\mathbb{E}(\bar{Q}_u \mid \mathcal{L}_u) = (1 - \mathrm{Sp}) + J\bar{E}_u$ and $\mathrm{Var}(\bar{Q}_u \mid \mathcal{L}_u) = (kTS)^{-1}\{\bar{E}_u\mathrm{Se}(1 - \mathrm{Se}) + (1 - \bar{E}_u)\mathrm{Sp}(1 - \mathrm{Sp})\}$. The law of total variance gives $\sigma_\pi^2 = J^2\,\mathrm{Var}(\bar{E}_u) + \sigma_{\mathrm{c},\pi}^2$, and $\mathrm{Var}(\bar{E}_u) = \sigma_{E,\pi}^2$ by the computation in the proof of Theorem 5.1(b), applied to the true session exposures.

(a) By Proposition B.5, with n puppets per arm, $\log\widehat{\mathrm{AR}}$ is approximately normal with mean $\log r_1$ and variance $J^{-2}\sum_\pi \sigma_\pi^2/(n\mu_\pi^2) + V_{\mathrm{val}} = V/n + V_{\mathrm{val}}$. For the pooled design, $V_{\mathrm{val}} = V_{\mathrm{val}}^{\mathrm{pool}}$ is the last term of (B.2). For the design stratified by arm, the gradient of $\log\mu_\pi = \log(q_\pi + \mathrm{Sp}_\pi - 1) - \log J_\pi$ with respect to $(q_\pi, \mathrm{Se}_\pi, \mathrm{Sp}_\pi)$ is $J_\pi^{-1}\{1/\mu_\pi, -1, (1 - \mu_\pi)/\mu_\pi\}$, and independence of the validation samples across arms and labels gives $V_{\mathrm{val}}^{\mathrm{st}}$. For double sampling, $\log\mu_\pi = \log\{q_\pi\mathrm{PPV}_\pi + (1 - q_\pi)\mathrm{FOR}_\pi\}$ has gradient $\mu_\pi^{-1}(\mathrm{PPV}_\pi - \mathrm{FOR}_\pi, q_\pi, 1 - q_\pi)$ with respect to $(q_\pi, \mathrm{PPV}_\pi, \mathrm{FOR}_\pi)$, and independent coding within the two strata gives $V_{\mathrm{val}}^{\mathrm{ds}}$ and the factor $(\mathrm{PPV}_\pi - \mathrm{FOR}_\pi)^2$ in place of J_π^{-2}. Take $r_1 > r_0$; the other case is symmetric. The interval excludes $\log r_0$ from above with probability $\Phi\{\Delta(V/n + V_{\mathrm{val}})^{-1/2} - z_{1-\alpha/2}\}$, and the probability of excluding it from below is negligible. This is at least $1 - \beta$ if and only if $V/n + V_{\mathrm{val}} \le \Delta^2/z^2$. If $\Delta^2 \le z^2 V_{\mathrm{val}}$ this is impossible, and otherwise it is equivalent to $n \ge z^2 V/(\Delta^2 - z^2 V_{\mathrm{val}})$. Letting the other validation stratum grow without bound and setting the remaining sizes equal across arms, the condition $\Delta^2 > z^2 V_{\mathrm{val}}$ becomes $n_0 > n_0^{\min}$, $n_{0\pi} > n_{0,\mathrm{st}}^{\min}$ or $m_{0\pi} > m_0^{\min}$ for the three designs. With a perfect classifier, $J = 1$ and $\sigma_{\mathrm{c},\pi}^2 = 0$, so $V = V_0$.

(b) Expanding (6.5), $\sigma_{E,\pi}^2 + \sigma_{\mathrm{c},\pi}^2/J^2 = \sigma_{\mathrm{s},\pi}^2\gamma_\pi + \{\sigma_{\mathrm{s},\pi}^2(1 - \gamma_\pi) + \varsigma_\pi^2/(kTJ^2)\}/S$, so $V = v_\mathrm{p} + v_\mathrm{s}/S$. With $V_{\mathrm{val}} = 0$, the required n is proportional to V, and the cost is proportional to $g(S) = (v_\mathrm{p} + v_\mathrm{s}/S)(c_p + Sc_s) = v_\mathrm{p}c_p + v_\mathrm{s}c_s + v_\mathrm{p}c_sS + v_\mathrm{s}c_p/S$. This is convex on $(0, \infty)$, and $g'(S) = v_\mathrm{p}c_s - v_\mathrm{s}c_p/S^2$ vanishes at S^*.

(c) The radius per unit standard deviation at n^* is $f(n^*)/n^* = h(1)(n^*)^{-1/2} = c_\alpha(n^*)^{-1/2}$. With known nuisance, the standard deviation per pair of the linearised log ratio is $V^{1/2}$, so the radius at n^* is $c_\alpha(V/n^*)^{1/2}$. Under the same normal approximation as in (a), the lower end of C_n exceeds $\log r_0$ with probability $\Phi\{\Delta(n/V)^{1/2} - c_\alpha\}$, which is at least $1 - \beta$ when $n \ge n_{\mathrm{seq}}$. The running lower end LB of Algorithm 1 is at least the lower end of C_n, so the power by the horizon is at least the power at the horizon.

(d) By Lemma B.2, $\mathrm{Var}(Y_u) = \mathrm{Var}\{\mathrm{TP}_u/(kTS)\} + \mathbb{E}\,\mathrm{Var}(Y_u \mid \mathcal{R}_u)$. Under the flag model above,

$\mathbb{E}\{\mathrm{TP}_u/(kTS) \mid \mathcal{L}_u\} = \mathrm{Se}\bar{E}_u$ and $\mathrm{Var}\{\mathrm{TP}_u/(kTS) \mid \mathcal{L}_u\} = \bar{E}_u\mathrm{Se}(1-\mathrm{Se})/(kTS)$, so $\mathrm{Var}\{\mathrm{TP}_u/(kTS)\} = \mathrm{Se}^2\sigma^2_{E,\pi} + \mu_\pi\mathrm{Se}(1-\mathrm{Se})/(kTS)$. Given the record, Y_u is $\mathrm{PP}_u/(kTS)$ times a hypergeometric proportion, with variance $\{\mathrm{PP}_u/(kTS)\}^2\mathrm{PPV}_u(1-\mathrm{PPV}_u)m^{-1}(\mathrm{PP}_u - m)/(\mathrm{PP}_u - 1)$ when $\mathrm{PP}_u > m$, where $\mathrm{PPV}_u = \mathrm{TP}_u/\mathrm{PP}_u$. Replacing $\mathrm{PP}_u/(kTS)$ by q_π and PPV_u by PPV_π, and dropping the finite-population factor, gives $q^2_\pi\mathrm{PPV}_\pi(1-\mathrm{PPV}_\pi)/m$. Dividing by $(\mathbb{E}Y_u)^2 = (\mathrm{Se}\mu_\pi)^2$ and summing over the arms gives V_Y. Finally, the lower end of C_n is $\log\widehat{\vartheta}_n - \log\lambda_+ - h_n$, and $\log\vartheta = \log r_1 + \log\lambda_S \geq \log r_1 + \log\lambda_-$. In the worst case $\lambda_S = \lambda_-$, the lower end is centred at $\log r_1 - \log(\lambda_+/\lambda_-)$, and the argument of (c) with Δ replaced by $\Delta - \log(\lambda_+/\lambda_-)$ gives the horizon. □

Remark B.8 (A model-based session variance). If sessions follow the chain of Assumption A2 with $k = 1$ and start near stationarity, Proposition 4.1(a) gives $\sigma^2_{\mathcal{A}}/T$ for the within-session part of $\sigma^2_{\mathrm{s},\pi}$. By the law of total variance, this is approximately a lower bound for $\sigma^2_{\mathrm{s},\pi}$, since it ignores differences between puppets and between sessions. It is a check on pilot estimates, not a substitute for them.

Appendix C Simulation details

This appendix documents the simulation study of Section 7. All data are synthetic. The catalogue, the classifier, the personas and the users are generated from random numbers and invented parameters, and no real platform, item or user is represented. The source of truth is `results/sim_results.json`, which records the configurations (`meta.configs`), the seeds, the software versions and the runtimes of every run, as well as the results. `results/SUMMARY.md` prints the same values, formatted by `code/write_summary.py`. The code is the package `code/ampsim` (version 1.0.0), with modules for the catalogue and classifier (`world`), the recommender and the accounts (`engine`), validation and correction (`measurement`), the double-sampled exposure of the protocol (`doublesample`), Markov chains (`markov`) and confidence sequences (`sequential`), and the scripts `run_experiments.py`, `make_figures.py` and `write_summary.py`.

C.1 Configuration

Table 10 lists the parameters of the synthetic world, and Table 11 the four synthetic personas. The codes P1–P4 are used in every figure, table and paragraph of Section 7, so that no panel can be read as a finding about a real group of users. Three mechanisms need more detail than the tables give.

Catalogue. An item of topic t and type j has the embedding $x_c = (e_t + a_j e_A + 0.35\,\epsilon_c)/\|e_t + a_j e_A + 0.35\,\epsilon_c\|$, where e_t is the direction of topic t, e_A the antisemitism direction, a_j the loading of the type and ϵ_c standard normal in eight dimensions. The numbers of borderline and antisemitic items in each topic are proportional to the topic's size and to its relative propensity, scaled so that 4% of all items are antisemitic and twice as many are borderline. Half of the antisemitic items are coded. Each item also has an engagement prior ("bait"), a background popularity w_c (log-normal, with a loading on bait) and a posting rate (log-normal, with a multiplier by type). The persona seed pool is the set of the 200 neutral items most similar to the persona's taste vector. No seed is borderline or antisemitic.

Popularity and load. Global popularity is $p_c = Vw_c$ when it is frozen, where V is the background volume of openings per round. In E2 it is updated after every round as $p \leftarrow \delta p + (1-\delta)(Vw + o)$, where o counts the puppets' openings in the round and $\delta = 0.5^{1/100}$, a half-life of one day of 100 rounds. Between sessions the remaining rounds of the day pass with $o = 0$. For a target traffic share ρ, $V = n(1-\rho)/\rho$, where n is the number of puppets and sentinels. The realised traffic share is the puppets' openings divided by those

openings plus V times the number of active rounds. The load is the puppets' share of the engagement with the items they opened, $\sum_c o_c / \sum_{c:o_c>0}(o_c + V w_c\, r)$ over the r active rounds of a wave, averaged over waves.

Evolving demand. After each formation session in E3, a user's taste moves toward the mean embedding of the items opened, $\theta \leftarrow (1 - 0.05)\theta + 0.05\,\bar{x}$, renormalised. For $x \in \{\mathsf{B}, \mathsf{A}\}$, the appetite becomes $\alpha_x + 0.4 f_x - 0.1\{\alpha_x - \alpha_x(0)\}$, capped at 2, where f_x is the share of the session's openings in state x. The share of the eligible pool drawn from followed borderline and antisemitic items becomes $s + 0.02(f_{\mathsf{B}} + f_{\mathsf{A}}) - 0.1\{s - s(0)\}$, clipped to $[0, 0.5]$; it starts at zero. Real users start from their persona's taste and appetites plus normal noise (s.d. 0.15 on taste, then renormalised, and 0.3 on the appetites for B and A), with new accounts ($v_u = 0$) and no training.

C.2 Experiments

E1. For each of the four regimes, one pool of 1,000 protocol puppets per persona and policy (algorithmic, reverse-chronological, random, popularity-only), simulated in chunks of 1,000 accounts with popularity frozen. The pools are saved in `results/raw/pool_<regime>.npz` as per-puppet impression counts by true type and score band. An audit draws $U \in \{5, 10, 20, 40, 80\}$ puppets per cell with replacement, $R = 1{,}000$ times. Its validation sample draws 100 impressions per score band and arm without replacement (all of a band if it has fewer), for 500 per arm; 250 and 1,000 are also run with $U = 20$. Pooled error rates are estimated from both arms' samples, weighted by inverse inclusion probabilities. The variance of the corrected estimators is computed by the delta method from the between-puppet covariance of the band shares and binomial variances, with the finite-population correction, for the validated band prevalences. The true AR and AA are those of the pools, with Monte Carlo standard errors.

E1b. For each variant, 1,000 protocol puppets per persona and arm (algorithmic and reverse-chronological), recording the true and flagged shares of each of their 10 sessions; in the bucket variant every account has a shift drawn from $\mathrm{N}(0, 1)$ that is added to the algorithm's weight on bait. Audits draw $U \in \{10, 20, 40\}$ puppets per cell, $R = 2{,}000$ times. Known error rates are the realised exposure-weighted rates of the pool, pooled over the arms or by arm. External validation draws $\widehat{\mathrm{Se}}$ from $\mathrm{Bin}(300, \mathrm{Se})/300$ and $\widehat{\mathrm{Sp}}$ from $\mathrm{Bin}(1{,}500, \mathrm{Sp})/1{,}500$, pooled or separately per arm, and uses the variances (B.1)–(B.2) or those of Theorem 6.1(d).

E2. Three designs, each with 15 protocol puppets per persona and arm (algorithmic, reverse-chronological and popularity-only) and 8 untrained sentinels with P1's follow profile on the algorithmic feed: simultaneous (seed pool of 40 items, one wave); low co-viewing (seed pool of 200, one wave); staggered (seed pool of 40, five waves with gaps of five days). The simultaneous design is run at traffic shares 0, 0.02, 0.05, 0.1, 0.2 and 0.3, the other two at 0, 0.1 and 0.3, each with $R = 24$ replications. Baseline sentinels are measured before each audit without feeding popularity. The randomised-saturation estimates pair replication i at the lower share with replication $i + 12$ (modulo 24) at the higher share, so that the two audits are independent. The slopes ξ are the largest ratios of |shift| to load over the grid. The sensitivity interval of Proposition 5.2(d) uses ξ_Δ at this simulated value, an oracle that no audit observes (`E2.sensitivity_xi_source` in the JSON).

E3. 3,000 real users and two formed copies, as in Section 7.4, with checkpoints at $H \in \{3, 10, 20, 40\}$ sessions. At each checkpoint, each copy is evaluated for three sessions under each policy (four cells), and a fixed-demand counterpart of each cell (the user's current demand held fixed, a new account, three burn-in and three evaluation sessions) checks Assumption A1. Standard errors are delta-method standard errors over users. The persona-contrast decomposition and the protocol-puppet audit use the ND pool of E1.

Table 10. Configuration of the synthetic world (`meta.configs` of `results/sim_results.json`). All values are invented. Topic order: news, sport, entertainment, culture and community, regional news, conspiracy; these are labels of invented clusters of items. Type order: neutral, borderline, coded antisemitic, explicit antisemitic. The personas are in Table 11.

Parameter	Value
Catalogue	
Items; embedding dimension	5,000; 8
Topic shares of items	0.30, 0.18, 0.18, 0.08, 0.11, 0.15
Propensity of borderline and antisemitic items by topic	0.4, 0.1, 0.1, 1.0, 1.5, 2.0
Antisemitic share; borderline-to-antisemitic ratio; coded share	0.04; 2; 0.5
Loading a_j on e_A by type; embedding noise s.d.	0, 0.5, 0.7, 1.4; 0.35
Bait: mean by type; s.d.	0, 0.5, 0.4, 0.3; 0.3
Background popularity: s.d. of log; loading on bait	1.0; 1.0
Posting rate: multiplier by type; s.d. of log	1, 1.2, 1.3, 1.5; 0.5
Item neighbours; persona seed pool	15; 200 neutral items
Recommender	
Eligible pool; trending candidates; slate k	50; 15; 4
Algorithmic score: weights on $\langle v_u, x_c \rangle$, log popularity, bait; Gumbel scale	2.0, 0.3, 1.0; 0.5
Embedding update $v_u \leftarrow (1-\lambda)v_u + \lambda x_c$	$\lambda = 0.1$
Popularity-only feed: Gumbel scale	0.05
Choice and sessions	
Protocol puppets: s_{safe}; similarity threshold	−0.5; 0.25
Naturalistic: weights on taste, bait, position, current item	3.0, 0.8, −0.3, 2.0
Naturalistic: utility of opening nothing	1.0 (E3); −50, so never (E4)
Training: days × seeds per day; seeds drawn per puppet	3 × 10; 30
Measurement: sessions × steps T; rounds per day	10 × 20 (4 of the 10 in E4b, E5b); 100
Demand evolution (E3)	
Taste drift; appetite gain, decay, cap	0.05; 0.4, 0.1, 2.0
Followed-fringe gain, decay, cap	0.02, 0.1, 0.5
Classifier regimes: flag rate by type	
ND, non-differential	0.02, 0.02, 0.85, 0.85
DU, borderline false positives	0.01, 0.40, 0.85, 0.85
DD, coded false negatives	0.02, 0.02, 0.35, 0.95
NDI, item-level noise	as ND in expectation; realised 0.016, 0.023, 0.86, 0.79

Table 11. The synthetic personas and their codes. Each is a demand profile in the invented catalogue: weights over its topics (Table 10), which set the eligible pool and the seed items, and appetites, utility shifts for the content states (N, B, A) used only by naturalistic accounts (E3, E4). The second column names the candidate persona of Appendix D.2 for which each profile stands in; no profile describes a real group of users. Antisemitic share: share of antisemitic items among the topics followed, weighted by the topic weights. μ_{ref}: impression exposure of protocol puppets under the reverse-chronological feed, regime ND (E1 pools).

Code	Stands in for	Synthetic demand profile: topic weights	Appetites	Antisemitic share	μ_{ref}	E3 share
P1 (d_0)	General interest (baseline)	three broad topics: news 1, sport 0.8, entertainment 0.8	$(0, -0.5, -2.5)$	0.012	0.018	0.6
P2	Jewish interest	culture and community 1, news 0.4, entertainment 0.25	$(0, -1.0, -4.0)$	0.041	0.054	0.1
P3	Middle East news	regional news 1, news 0.5, culture and community 0.2	$(0, 0, -2.0)$	0.064	0.082	0.15
P4	Conspiracy-adjacent	conspiracy 1, news 0.3, regional news 0.25	$(0, 0.5, -1.0)$	0.092	0.113	0.15

E4. Base runs: 250 untrained naturalistic accounts per persona and policy (algorithmic, reverse-chronological), three burn-in sessions, ten estimation sessions of 20 steps and one session of 200 steps. Perturbation runs use the algorithmic policy only. In the tilt runs the algorithm's score of every item with $L_c = 1$ is raised by $\varepsilon \in \{0, 0.1, 0.25, 0.5, 1\}$. In the replacement runs the opened item is replaced by a uniformly drawn antisemitic item with probability $\varepsilon \in \{0, 0.01, 0.02, 0.05\}$. Transition matrices are maximum-likelihood estimates over the states of consumed items; a row without departures takes the pooled next-state distribution. Bootstrap standard errors resample accounts (200 resamples).

Detailed pools. E4b, E5b and E5s use the E1 pools of regimes ND, DU and DD, re-simulated with the same seeds and chunks while recording, for the algorithmic and reverse-chronological puppets, per-session impression counts by type and score band, the type and band of every opened position, and slate states (`results/raw/pool_detail_<regime>.npz`). The recording draws no random numbers; the counts are identical to the E1 pools and the per-session counts add up to them (`E5b.pools` in the JSON). After its ten sessions, every puppet runs one further session of 200 steps.

E4b. For each regime, persona and arm, the true slate state of a step is the most severe true state among its $k = 4$ positions, and the observed state is A if a position is flagged, else B if a position lies in score band $[-1, 0)$, else N. Chains are estimated from the transitions and first states of the 10 sessions of all 1,000 puppets, as three-state chains and collapsed to {A, not A}. Predictions on the impression scale multiply slate functionals by $\bar{k}_{\mathcal{A}}/k$, with $\bar{k}_{\mathcal{A}}$ (1.02–1.26) from the same sessions. The correction inverts the slate confusion matrix C in $C^{\top}\,\mathrm{diag}(\boldsymbol{\nu})PC$ (Proposition A.4), sets negative entries to zero and renormalises the rows. Item-level confusion is the cell's known exposure-weighted rate of each observed class by true state. Under the protocol's approximation, a slate not in A is read as A with probability $1 - \mathrm{Sp}^k$ and a slate in A with probability $1 - (1 - \mathrm{Se})\mathrm{Sp}^{k-1}$; with the true composition, the rows average over the simulated joint distribution of the numbers of borderline and antisemitic positions per slate. The three-state correction is computed only where the condition number of C is below 100. Long-run targets are the shares in steps 100–199 of the 200-step session. Replications: 120 puppets per arm drawn with replacement, each with four of its sessions drawn without replacement, $R = 500$.

E5b. For each persona, regime (ND, DD) and case (true arms; null, both arms from the reverse-chronological pool), $R = 2{,}000$ audits of 120 pairs, each puppet drawn with replacement from the detailed pool with four

of its ten sessions drawn without replacement. Codes per puppet: $\min(10, \cdot)$ flagged positions; $\min(5, \cdot)$ opened positions (never flagged under the avoid-on-flag rule, as the code checks); 20 other unflagged positions over score bands 0–2, allocated by D'Hondt rounding of $20\, n_h s_h / \sum_j n_j s_j$ with at least one per non-empty band, where $s_h = \{f_h(1 - f_h)\}^{1/2}$ and f_h is the false-omission rate of band h among non-opened positions, pooled over the regime's personas and arms ($s_h = 0.014, 0.070, 0.285$ under ND and $0.070, 0.173, 0.354$ under DD). Within a stratum the coded positions are a simple random sample, so true positives are hypergeometric. Sequences: Algorithm 1 with $\lambda = 1$ and its version for pair differences, with $b \in \{0, 1\}$; betting sequences of Theorem 6.2(b) on a grid of 300 log-spaced ratios in $[0.1, 10]$ and of 801 values of AA in $[-0.1, 0.1]$. Coverage is evaluated at the ends of waves. The closed form is $V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}}$ of Appendix D.8 with the pool's μ, q, Se, Sp, PPV and FOR, $kTS = 320$, and the simulated variance of a puppet's true exposure. *Pilot check*: 200 pilots per configuration of 10 puppets per arm, each coded on all flagged positions, 10 openings and 40 band-stratified unflagged positions; $B = 1{,}000$ resampled audits of the registered design, in which each pair draws pilot puppets with replacement and the registered codes are re-drawn without replacement from the pilot's codes, with the ratio and difference of the pilot's mean double-sampled exposures as the truth. The check fails when the estimated miss rate exceeds $\alpha + \{\alpha(1 - \alpha)/B\}^{1/2}$.

E5s. The statistic and the pilot check of E5b, with new draws, under two stress designs: waves of 5 or of 2 pairs up to 120 pairs, no burn-in, and the mixture tuned to 40 pairs. They exist only to show whether the pilot check detects a failing sequence.

E5 (benchmark). For each persona and for regimes ND and DD, $R = 2{,}000$ audits of 400 pairs (one algorithmic and one reverse-chronological puppet), drawn with replacement from the E1 pools. For each puppet, $\min(10, \text{flagged positions})$ flagged positions are coded by a hypergeometric draw. Variants: $\alpha \in \{0.05, 0.0125\}$; burn-in $b \in \{0, 1\}$; range $\{1\}$, $[0.95, 1.05]$ (ND) or $[0.8, 1.2]$ (DD). The mixture is tuned to 40 pairs; $\delta = 1.25$; stopping rules are evaluated with $W_{\max} \in \{8, 40\}$. The exact betting sequence uses $R = 200$ audits and a grid of 300 log-spaced values of the ratio on $[0.1, 10]$. Power calculations use the target power 0.8.

C.3 Seeds and Monte Carlo error

Seeds. Every random stream is `numpy.random.SeedSequence([20261103, id, *keys])`, with experiment ids world 0, E1 1, E2 2, E3 3, E4 4, E5 5, E4b 6, E5b 7 and E5s 8; E1b and the detailed pools use the id of E1. The world is `SeedSequence([20261103, 0])`. The keys are as follows, with regimes indexed ND 0, DU 1, DD 2, NDI 3, policies alg 0, chron 1, random 2, popular 3, and cases true arms 0, null 1.

- E1: $(0, \text{regime})$ for the assignment of puppets to chunks, $(1, \text{regime}, \text{chunk})$ for their simulation, $(2, \text{regime}, \text{reference}, \text{persona}, U)$ for the audits, and $(3, \text{regime}, \text{reference}, \text{persona}, n_{\mathrm{val}})$ for the validation-size runs. The detailed pools use the first two.
- E1b: $(10, \text{regime}, v)$, $(11, \text{regime}, v, \text{chunk})$ and $(12, 100 \cdot \text{regime} + v, \text{persona}, U)$, with $v = 1$ for the bucket variant and 0 otherwise.
- E2: $(1, \text{design}, \text{replication})$, the same for every traffic share, which gives common random numbers across shares.

Table 12. Runs of the simulation study (`meta.runtime_s.runs`). Wall time of the run, and CPU time by part summed over the worker processes, in seconds.

Run	Experiments	Wall	CPU by part	Status
1	E1, E2, E3, E4	755.2	E1 pools 76.5, 76.1, 77.7, 76.1 (ND, DU, DD, NDI); E2 565.0; E3 384.8; E4 189.9	current for E1–E4
2	E5	329.6	E5 328.6	superseded
3	E1b, E5	311.7	E1b 36.9; E5 286.2	superseded
4	E5	11.4	E5 10.5	current for E5
5	E1b	27.4	E1b 39.1	current for E1b
6	E4b, E5b	521.7	detailed pools 39.7, 37.6, 36.7 (ND, DU, DD); E5b 855.9; E4b 19.7	current for E4b, E5b
7	E5s	455.2	E5s 905.4	current for E5s

- E3: (0) for the users, $(1, \text{policy})$ for the audit at F_0, $(2, \text{formation policy}, H)$ for formation, and $(3, H, \text{formation}, \text{evaluation})$ and $(4, H, \text{formation}, \text{evaluation})$ for the cells and their fixed-demand counterparts.
- E4: $(1, 0)$ for the base runs and $(1, 1)$ for every perturbation run, which gives common random numbers across ε; (9) for the bootstrap.
- E4b: (regime) for the replications. E5b: (regime, case, persona), shared by the audits and the pilot check. E5s: (regime, case, persona) for the audits and (regime, case, persona, 1 + design) for the pilot check.
- E5: $(2, \text{regime}, \text{persona})$.

Monte Carlo error. The MCSE of a coverage or rejection rate c is $\{c(1-c)/R\}^{1/2}$: 0.7 points near 95% with R = 1,000, 0.5 with R = 2,000 (0.25 near 98.75%), and 4.4 with R = 24. Means and biases are reported with their standard errors over replications. The true values of E1, E5 and E5b are means over the pools of 1,000 puppets per cell. Their MCSE is at most 0.020 for AR relative to the reverse-chronological reference, and at most 0.035 over all references. A pilot check's failure rate over 200 pilots has an MCSE of at most 3.5 points.

C.4 Software, runs and runtimes

The study was run with Python 3.13.18, NumPy 2.5.3 and SciPy 1.18.1, with two worker processes, single-threaded linear algebra and reduced priority (`nice`). Figures were drawn with Matplotlib 3.11.2. The 29 unit tests in `code/tests/` pass; those added for E4b and E5b check the conditional unbiasedness of $\widetilde{E}_u$ and of its re-draw from pilot codes, the allocation rule, the detailed recording, the slate confusion matrix and the corrected chain. The results come from seven runs (Table 12), with a total wall time of 2,412.1 seconds (`meta.runtime_s.total_wall`). Each run after the first updated only the experiments it names, loading the E1 pools from `results/raw/`. Runs 2 and 3 evaluated an earlier version of Algorithm 1 and were replaced by run 4; no number from them appears in this paper. Their history, and the run logs, are documented in `code/README.md`.

C.5 Reproduction

From `papers/amplification/code/`, the current runs, the summary, the figures and tables, and the tests are reproduced by

```
export OPENBLAS_NUM_THREADS=1
nice -n 10 python run_experiments.py --only E1,E2,E3,E4
nice -n 10 python run_experiments.py --only E5 --merge
nice -n 10 python run_experiments.py --only E1b --merge
nice -n 10 python run_experiments.py --only E4b,E5b --merge
nice -n 10 python run_experiments.py --only E5s --merge
python write_summary.py      # results/SUMMARY.md
python make_figures.py       # figures/fig-sim-*.pdf, tables/tab-sim-*.tex
python -m pytest tests/ -q   # 29 tests
```

The first command takes about 13 minutes with two workers, the fourth and fifth about 9 and 8, and the others under a minute each (Table 12). `run_experiments.py` without `-only` runs all experiments at once from the same seeds, and `-quick -out <dir>` makes a smoke run. Tables 5, 6, 13 and 14 are generated by `make_figures.py`. Every other number in Section 7 is read from `results/sim_results.json`, or computed from values in it (for instance a relative error, a ratio of two cells, or a range over personas).

C.6 Further results

Table 13 gives the results of E1b that Section 7.2 summarises, together with the checks of the bounds of Proposition 5.2 from E2. Table 14 summarises E2–E5, Figure 6 shows the interference runs of E2, Figure 7 the decomposition of E3, Figure 8 the perturbation runs of E4, and Figure 9 the benchmark E5. Nine further results support Section 7.

- *E1b, variance estimators.* With pools that record every session, sessions of a puppet are nearly uncorrelated (design effects 0.96–1.23) and both variance estimators of Theorem 5.1 cover. The form AR r_a/r_r of (6.2) reproduces every naive ratio, and $\mathrm{AR}^{\mathrm{pool}}$ matches the median pooled correction to within 0.006. With external validation, the delta-method s.d. of $\log \widehat{\mathrm{AR}}$ is within −5.3 to +3.9% of the Monte Carlo s.d. when $\mathrm{cv}_{\mathrm{Sp}} \le 0.102$, but understates by 15–23% for P1 ($\mathrm{cv}_{\mathrm{Sp}} = 0.246$) and by up to 33% under DD (0.312).

- *E1b, known error rates.* With 20 puppets per cell and no buckets, cluster-robust intervals for AA cover in 93.7–94.6% of audits, and session-level ones in 92.8–95.0%. With buckets, cluster-robust intervals for log AR cover in 93.2–94.2% and session-level ones in 69.3–85.0%. Under DD the pooled correction with external validation covers in 73.5, 78.5, 15.7 and 1.5% of audits and the correction by arm in 95.0–97.9%. The attenuation formula (6.1) gives 1.325, 1.231, 1.534 and 1.033 under ND against naive ratios of 1.316, 1.232, 1.532 and 1.027; it would be exact if the realised error rates of the two arms were identical. Under DD it fails (1.289 against 1.182 for P1), while the limit $\mathrm{AR}^{\mathrm{pool}}$ of (6.2) (1.494, 1.254, 1.487, 0.861) matches the median pooled correction with 20 puppets per cell (1.489, 1.250, 1.483, 0.861); for P4 both lie below one, although the truth is 1.053. In the delta-method comparison, the median s.d. for P1 under ND is 0.159 against a Monte Carlo s.d. of 0.188 with 20 puppets, and the mean of the delta-method variance overstates, because a few audits with a corrected reference exposure near zero dominate it.

- *E2, interference.* Even at traffic share 0.02 the bias of $\widehat{\mathrm{AA}}_{\mathrm{chron}}$ is −1.5 to −3.5% of the no-audit value for three personas and −19% for the fourth. For three personas ξ_Δ is 0.008–0.009 against the popularity-only feed and 0.025–0.037 against the reverse-chronological feed. Seeding from 200 rather than 40 items barely helps (Table 14). The breakdown value ξ^* is well above the simulated slope only for P3 (0.101 against 0.037).

- *E2, bounds.* The slopes ξ are estimated from the same grid on which the bound of Proposition 5.2(a) is checked, so that bound holds by construction for AA. The ratio bound holds with room: at share 0.3, |bias of log $\widehat{\mathrm{AR}}$| is 0.070–0.120 against bounds of 0.087–0.323. The extrapolated estimators of part (c) have biases within the bound $\zeta\ell_1\ell_2/2$ for every pair of loads; for loads closer than those of shares 0.1 and 0.3 they cost 4.4–5.9 times the variance of one audit.

- *E3.* The initial demand F_0 draws the personas P1–P4 with shares 0.6, 0.1, 0.15 and 0.15. After 40 formation sessions TR = 1.45 = 1.24 × 1.17. As formation lengthens, supply stays between 1.52 and 1.70 points, while demand shaping grows from 0.30 and the interaction falls from −0.04. Assumption A1 holds closely: no cell differs from fixed-demand accounts with the same demand by more than 0.11 points (SE 0.05). The fixed-demand audit at F_0 gives AR = 1.36; protocol puppets with the persona mix of the population give AA = 1.91 points and AR = 1.42, because the avoid-on-flag rule is part of the persona. In the persona contrasts relative to P1, the supply contrast is 1.40 points, the demand contrasts are 3.65, 6.45 and 9.54, the interactions 0.41, 4.08 and −0.92 (MCSE 0.05–0.07), and the totals 5.46, 11.93 and 10.02, for P2, P3 and P4.

- *E4, perturbations* (Figure 8). The chain reproduces e_{20} within 0.0002 in all eight cells, an in-sample check. When the algorithm's score of antisemitic items is tilted, the exact identity of Theorem 4.2(a), applied to the estimated chains, tracks the simulated change (at $\varepsilon = 1$, 14.26 against 14.77 percentage points for P3). The bound of part (b) holds but is loose, by factors of 1.3–3.3 for P3 and P4 and 3.2–120 for P1 and P2, because $\|P_1 - P_0\|_\infty$ includes the rarely visited, noisily estimated row of A. When the next item is replaced by an antisemitic one with probability $\varepsilon \le 0.05$, the closed form of Theorem 4.2(c), computed from the base chain alone, predicts long-run exposure within 9%. Its multiplier shows: for P1, with $\mathbb{E}_\varpi[T_\mathsf{A}] = 345$ items, replacing 1% of items raises long-run exposure from 0.31% to 1.30%, and for P3, which reaches antisemitic items quickly anyway, from 3.56% to 4.64%.

- *E4, hitting.* The probability of reaching A within 20 items is predicted less well than exposure. It is overpredicted by a factor of up to 1.9 where it is small (0.061 against 0.032 for P1 under the reverse-chronological feed), presumably because the pooled chain treats accounts as exchangeable when a few accounts carry most visits to A. The mean time to reach A from equilibrium is 302–363 items for P1 and P2 and 16–41 for P3 and P4.

- *E5b and E5s.* The pilot check for the sequence for AA fails in 0–1.5% of pilots at the registered design and $\alpha = 0.05$, and in 66.0–90.0% under the stress design with waves of five pairs, whose sequence for AA covers in 92.7–96.2%, so it fails less clearly than the one for AR. At $\alpha = 0.025$ the stress designs cover in 92.6–95.6% (waves of five) and 71.5–79.6% (waves of two) for AR, and the check fails in 99.0–100% and 100% of pilots. Every pilot had a positive mean exposure in both arms, so the check could always be computed. On the registered design the betting sequence for AA excludes zero only for P3 (1.6% of audits under ND and 47.2% under DD, at $\alpha = 0.05$).

- *E5, benchmark.* The confidence sequence for AA, on corrected pair differences with known error rates, covers in 94.5–96.7% of audits. With the range $[0.95, 1.05]$ under ND, P1 stops after 5.9 waves instead of 5.1 and concludes material amplification in 68.5% of audits. Under DD with the range $[0.8, 1.2]$ at $\alpha = 0.0125$, P2 and P4 end inconclusive in 97.8% and 100% of audits. For P4 (AR = 1.07, λ_SAR < 1) under DD with $\lambda_S = 1$ assumed, the audit detects de-amplification within eight waves in 48.5% of audits. With the range $[0.8, 1.2]$ the sequence excludes AR = 1 by wave 8

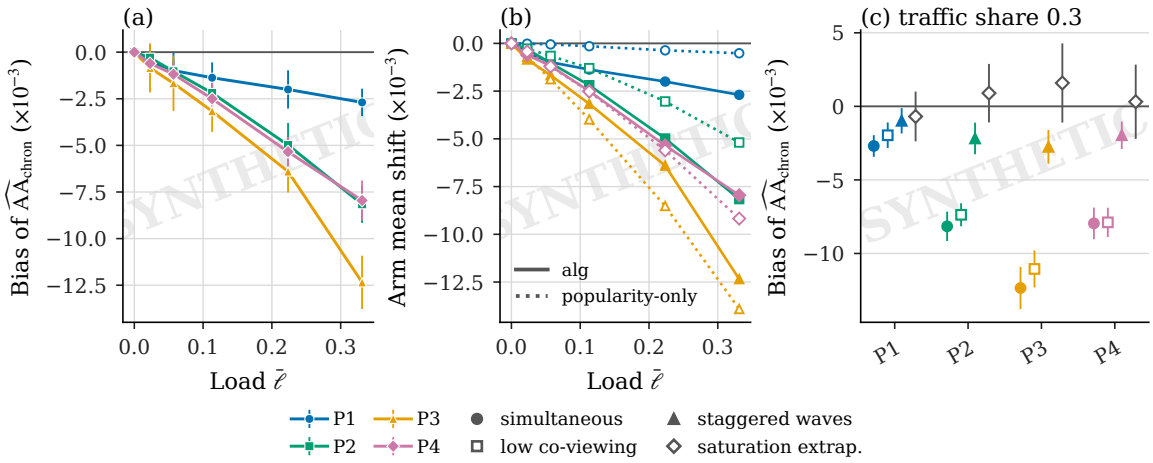


Figure 6. Synthetic data, E2: interference through shared popularity ($R = 24$ paired replications; bars ±2 SE). (a) Bias of the additive amplification relative to the reverse-chronological feed against the load $\bar{\ell}$, simultaneous design. (b) Shift in mean exposure of the algorithmic arm (solid) and the popularity-only arm (dotted); the reverse-chronological arm does not move. (c) Bias at traffic share 0.3: simultaneous (load 0.33), low co-viewing (0.33), staggered waves (0.12), and randomised-saturation extrapolation from shares 0.1 and 0.3.

in 49.6, 5.5, 92.8 and 0.0% of audits, against 96.1, 85.0, 100 and 54.8% with $\lambda_S = 1$. By 400 pairs the exact betting sequence excludes AR = 1 only for P3, and its median width at 40 pairs is 2.2–4.6, against 0.21–0.45 for Algorithm 1. The closed form V_Y of Proposition 6.3(d) overstates the simulated variance by 5–23%. The simulated power for detecting AR ≠ 1 is 0.82–0.93 at n_{fix} and 0.85–1.00 by n_{seq}; with $\lambda_S \in [0.95, 1.05]$ no horizon exists for P4, since $|\log \text{AR}| = 0.042 < \log(1.05/0.95)$.

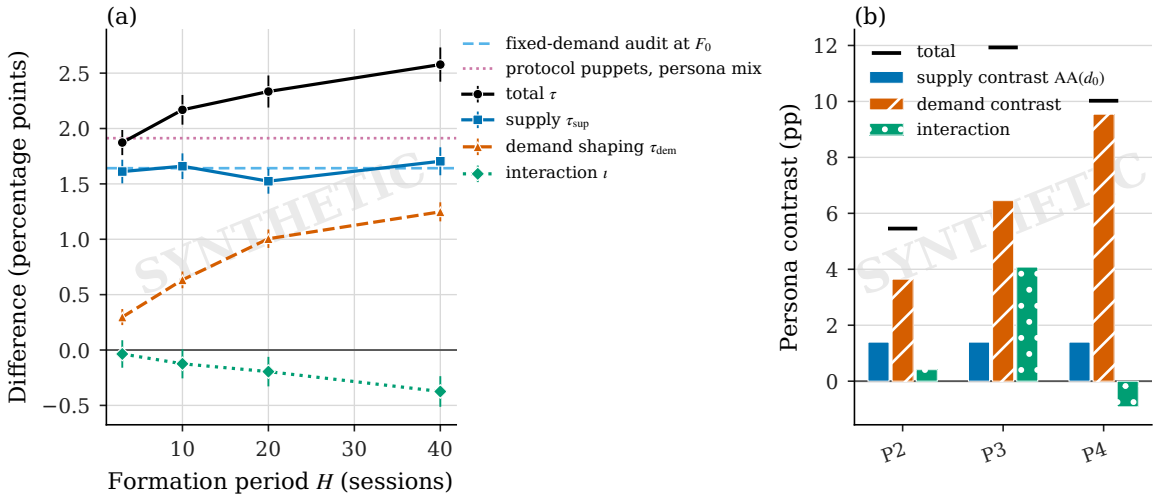


Figure 7. Synthetic data, E3: the decomposition of Theorem 3.3. (a) Components for 3,000 synthetic users whose demand evolves, against the length H of the formation period (reference: reverse-chronological; bars ±2 SE over users). Lines: the fixed-demand audit at the initial demand F_0, and protocol puppets with the persona mix of the population. (b) Persona-contrast decomposition relative to P1 (d_0; protocol puppets, regime ND); black marks: total $\mu_{\text{alg}}(d) - \mu_{\text{ref}}(d_0)$.

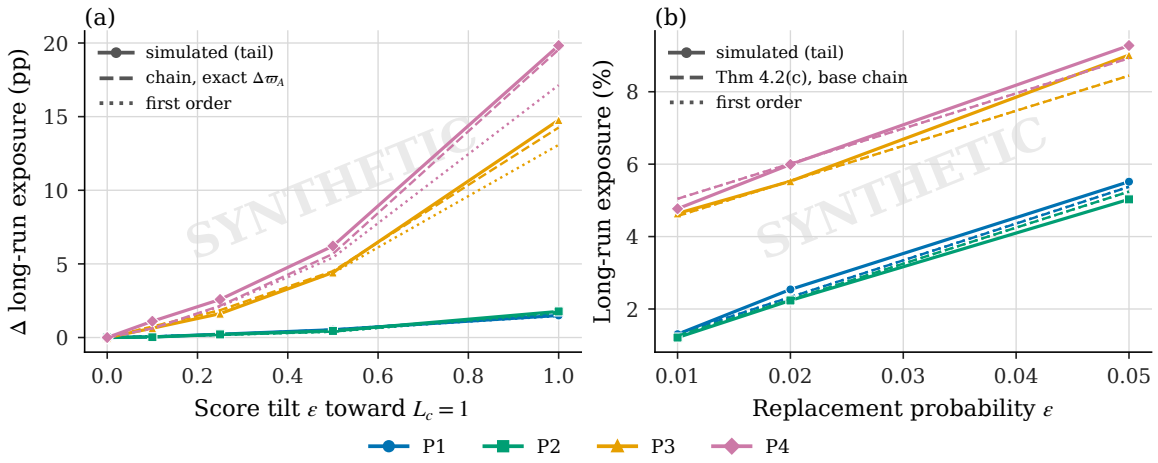


Figure 8. Synthetic data, E4: perturbations of the chains of consumed items (algorithmic policy, 250 naturalistic accounts per persona, common random numbers across ε). (a) Score tilt ε toward items with $L_c = 1$: simulated change in long-run exposure, the exact change between the estimated chains (Theorem 4.2(a)), and its first-order term. (b) Replacement of the next item by an antisemitic one with probability ε: simulation, the closed form of Theorem 4.2(c) from the base chain alone (dashed), and its first-order term $\varpi(\mathcal{A})(1 + \varepsilon\,\mathbb{E}_{\varpi}[T_{\mathsf{A}}])$ (dotted).

Table 13. Synthetic data, E1b and E2: checks of Theorems 5.1 and 6.1 and Proposition 5.2. E1b: $R = 2{,}000$ audits; error rates known unless stated; external validation with $n_1 = 300$ coded positives and $n_0 = 1{,}500$ coded negatives. Design effect: $S\sigma^2/\sigma_s^2$ of Theorem 5.1 for $S = 10$ sessions. Sessions iid: variance that treats the sessions of a puppet as independent. Experiment buckets: every account has a persistent random shift (s.d. 1) in the algorithm's weight on the engagement prior. s.d. of log $\widehat{\mathrm{AR}}$: Monte Carlo, and the median over audits of the delta-method value; $\mathrm{cv}_{\mathrm{Sp}}$ as in Section 6.1. Interference block: E2, simultaneous design ($R = 24$). There ξ_Δ is the largest $|\text{bias}|/\bar{\ell}$ over the grid of loads and ζ the largest second divided difference of the bias in the load, so the bound of part (a) holds by construction and is attained where the largest ratio occurs.

Quantity	P1	P2	P3	P4
Non-differential error, profiles as simulated (20 puppets per cell)				
Design effect, alg; ref	1.08; 0.96	1.07; 1.01	1.23; 1.04	1.10; 0.99
Coverage AA: cluster-robust; sessions iid (%)	94.2; 94.8	94.6; 95.0	93.7; 92.8	94.2; 94.2
Coverage log AR: cluster-robust; sessions iid (%)	94.5; 95.2	95.0; 95.2	93.8; 93.2	94.1; 94.3
Naive AR; $w_0 + (1 - w_0)$AR; true AR	1.316; 1.325; 1.767	1.232; 1.231; 1.334	1.532; 1.534; 1.690	1.027; 1.033; 1.040
External validation: s.d. of log $\widehat{\mathrm{AR}}$, MC; delta (median)	0.188; 0.159	0.062; 0.061	0.050; 0.050	0.039; 0.039
$\mathrm{cv}_{\mathrm{Sp}} = v_{\mathrm{Sp}}^{1/2}/(J\mu_r)$	0.246	0.081	0.053	0.039
coverage log AR; AA (%)	94.2; 94.2	94.8; 95.3	95.0; 94.0	94.0; 94.1
Non-differential error, accounts in experiment buckets (20 puppets per cell)				
Design effect, alg; ref	3.47; 1.04	3.61; 1.02	5.65; 1.02	2.75; 1.03
Coverage AA: cluster-robust; sessions iid (%)	93.0; 76.4	93.1; 75.6	93.0; 62.9	93.2; 82.5
Coverage log AR: cluster-robust; sessions iid (%)	94.2; 85.0	93.7; 78.5	93.3; 69.3	93.2; 82.2
Differential error (coded false negatives), external validation, 20 puppets per cell				
True AR; naive $\mathrm{AR}^{\mathrm{nv}}$; $\mathrm{AR}^{\mathrm{pool}}$ (eq. 6.2)	1.784; 1.182; 1.494	1.353; 1.162; 1.254	1.780; 1.349; 1.487	1.053; 0.891; 0.861
Corrected, pooled: median AR; coverage (%)	1.489; 73.5	1.250; 78.5	1.483; 15.7	0.861; 1.5
Corrected, stratified by arm: median AR; coverage (%)	1.765; 97.9	1.348; 95.4	1.785; 95.0	1.050; 95.3
Interference (reference chron): Proposition 5.2(a) and (c)				
$\lvert\text{bias AA}\rvert$; $\xi_\Delta\bar{\ell}$ at share 0.3 ($\times 10^{-3}$)	2.69; 7.20	8.16; 8.16	12.35; 12.35	7.95; 8.76
$\lvert\text{bias log AR}\rvert$; ratio bound at share 0.3	0.088; 0.323	0.120; 0.146	0.095; 0.110	0.070; 0.087
Extrapolation 0.05/0.1: $\lvert\text{bias}\rvert$; $\zeta\ell_1\ell_2/2$ ($\times 10^{-3}$)	0.61; 0.83	0.13; 0.94	0.19; 1.44	0.10; 1.08
Extrapolation 0.1/0.2: $\lvert\text{bias}\rvert$; $\zeta\ell_1\ell_2/2$ ($\times 10^{-3}$)	0.73; 3.27	0.66; 3.70	0.13; 5.67	0.37; 4.23
Extrapolation 0.1/0.3: $\lvert\text{bias}\rvert$; $\zeta\ell_1\ell_2/2$ ($\times 10^{-3}$)	0.69; 4.84	0.90; 5.49	1.59; 8.41	0.31; 6.27

Table 14. Synthetic data: summary of E2–E5. E2: bias of the additive amplification relative to the reverse-chronological feed at traffic share 0.3 ($R = 24$; SE 0.4–0.7 $\times 10^{-3}$), simultaneous design (load 0.33), low co-viewing (0.33), staggered waves (0.12). E3: the "real users" are synthetic accounts whose demand evolves. E4: consumed items of naturalistic accounts, and the impression exposure of the same accounts. E4b: slate states of protocol puppets; long-run impression exposure in the second half of a 200-step session, and its prediction by chains of true, observed and corrected slate states. E5 (benchmark): $R = 2{,}000$ audits (MCSE about 0.5 near 95%); coverage over all 80 waves; peeking: fixed-sample interval after each of waves 1–8; false detection: the sequence excludes AR = 1 when it is true; n_{fix}, n_{seq} from V_Y of Proposition 6.3(d) for detecting AR $\neq 1$; last row: regime DD.

Quantity	P1	P2	P3	P4
E2: interference (reference chron, traffic share 0.3, load 0.33)				
AA^0 ($\times 10^{-3}$)	14.3	18.2	54.9	3.2
Bias of $\widehat{AA}$ ($\times 10^{-3}$)	−2.7	−8.2	−12.3	−8.0
with low co-viewing	−2.0	−7.4	−11.1	−7.9
with staggered waves	−1.0	−2.2	−2.8	−2.0
E3: real users, formation $H = 40$ sessions (pp): $\tau = 2.58$, $\tau_{\text{sup}} = 1.70$, $\tau_{\text{dem}} = 1.25$, $\iota = -0.37$; audit at F_0: 1.64				
E4: Markov model of consumed items, naturalistic accounts, algorithmic policy (%)				
e_{20} predicted / simulated	0.28 / 0.28	0.26 / 0.26	3.16 / 3.16	3.62 / 3.60
e_{200} predicted / simulated	0.30 / 0.38	0.28 / 0.22	3.36 / 3.54	3.84 / 3.89
impression exposure of the same accounts	2.85	6.20	13.40	12.00
AR of consumed items; of impressions	0.87; 1.60	0.86; 1.14	1.40; 1.67	0.62; 1.06
$\kappa_{\mathcal{A}}$; Kemeny K	0.50; 2.07	0.51; 1.98	0.51; 2.04	0.52; 2.17
E4b: slate chains of protocol puppets, regime ND, long-run impression exposure, alg; ref (%)				
Simulated (second half of a 200-step session)	3.14; 1.76	7.38; 5.37	14.29; 8.15	12.23; 11.37
Chain of true slate states	3.17; 1.76	7.26; 5.40	13.85; 8.22	11.85; 11.30
Chain of observed slate states, uncorrected	4.57; 3.36	7.93; 6.36	13.67; 8.72	11.87; 11.36
Corrected, minimal composition	3.22; 1.77	7.38; 5.47	14.35; 8.36	12.19; 11.62
Corrected, true composition	3.19; 1.76	7.26; 5.40	13.87; 8.20	11.85; 11.31
E4b: slate chains of protocol puppets, regime DD, long-run impression exposure, alg; ref (%)				
Simulated (second half of a 200-step session)	3.24; 1.77	7.53; 5.36	15.19; 8.16	12.53; 11.44
Chain of true slate states	3.24; 1.77	7.34; 5.37	14.56; 8.09	12.19; 11.32
Chain of observed slate states, uncorrected	3.79; 3.09	6.57; 5.52	10.48; 7.24	8.96; 9.58
Corrected, minimal composition	3.31; 1.80	7.62; 5.53	16.22; 8.41	13.27; 11.97
Corrected, true composition	3.22; 1.78	7.31; 5.39	14.57; 8.06	12.16; 11.30
E5, benchmark design: Algorithm 1 on Y_u, reference chron, waves of 5 pairs, $\alpha = 0.05$ unless stated				
True AR	1.79	1.33	1.67	1.04
Time-uniform coverage, burn-in 1 wave; none (%)	95.0; 92.6	96.2; 93.5	95.7; 93.3	95.2; 93.2
peeking, waves 1–8 (%)	71.2	69.8	69.5	69.8
Null AR= 1: false detection by wave 8, Alg. 1; peeking (%)	2.9; 30.3	2.1; 29.2	2.5; 29.8	3.4; 33.0
P(excludes AR= 1) by wave 8 (%)	100.0	99.8	100.0	9.0
n_{fix}; n_{seq} from V_Y (pairs)	6; 12	10; 20	2; 4	267; 512
Power at n_{fix} (fixed); by horizon (sequential)	0.87; 0.92	0.85; 0.85	0.93; 1.00	0.82; –
Differential Se: λ_S; coverage with λ_S=1; with [0.8, 1.2] (%)	0.86; 2.8; 99.6	0.93; 19.8; 100.0	0.83; 0.0; 99.4	0.82; 0.0; 99.6

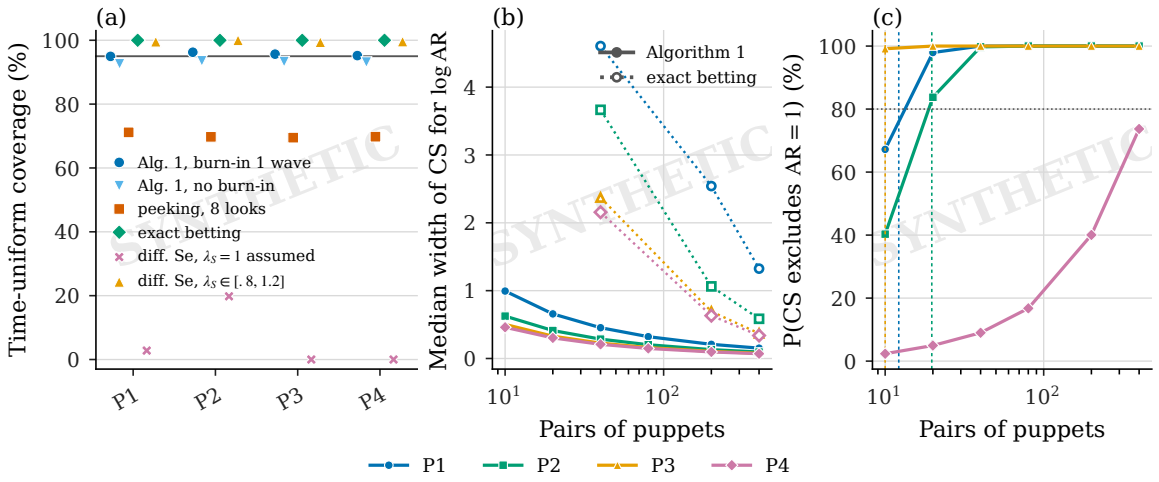


Figure 9. Synthetic data, E5 (benchmark: conventional small-lab design, Algorithm 1 on the confirmed-flagged statistic Y_u, waves of five pairs, ten sessions per puppet; R = 2,000; exact betting R = 200). (a) Time-uniform coverage at $\alpha = 0.05$ over 80 waves, with and without one burn-in wave; a fixed-sample interval recomputed after each of the first eight waves (peeking); exact betting; and, under DD, $\lambda_S = 1$ assumed or the range $[0.8, 1.2]$. (b) Median width for log AR: Algorithm 1 (solid), exact betting (dotted). (c) Probability of excluding AR = 1; dashed verticals: the horizon n_{seq} of Proposition 6.3(d), at least two waves (beyond the axis for P4).

Appendix D Registered protocol

This appendix is the full text of the protocol summarised in Section 8. The stopping and decision rules and the feasibility gate are stated in Sections 8.2 and 8.3, and the ethical and legal conditions in Section 9. Every number below is a design input, a cap, or an evaluation of a closed form at stated illustrative inputs; none is a result.

D.1 Service, surface and access route

The unit of audit is one recommendation *surface* (for example a home feed, or a list of items recommended next to the current one) of one service, in one jurisdiction, during one audit window. The service and surface are chosen at registration by three criteria, in this order:

(i) *A reference policy is available.* AR and AA require exposure under a reference policy $\pi = \text{ref}$ (Definition 3.1). On a live service, the candidates are the non-profiling option that very large online platforms must offer under Article 38 of the DSA, or a reverse-chronological option where offered (Section 2.3), and logged-out sessions (Section 2.2). The random-from-eligible-pool and popularity-only reference policies of Definition 3.1 are not generally available to an outside auditor. They are evaluated in Section 7, and on a live service only if data obtained under Article 40(4) permit it. If no reference policy is available for a surface, no amplification is estimated for it.

(ii) *A lawful access route exists.* In order of preference: data obtained as a vetted researcher under Article 40(4) of the DSA, or through a research interface the provider offers, where these can deliver exposure under a fixed demand profile; access to publicly accessible data under Article 40(12); and a sock-puppet audit only where neither route can identify the estimand, and only after the legal review and ethics approval of Section 9.

(iii) *The classifier is validated for the content.* The service's dominant language(s) and formats (text, images, video transcripts) must be covered by the frozen classifier and by the annotators of Appendix D.5.

The jurisdiction is an EU Member State, so that the DSA routes apply and the Article 38 option is in scope. All puppets present from that jurisdiction.

D.2 Personas, puppets and sessions

Personas. A persona d is a demand profile. It has three components: a pool of seed items used to train the account; a choice rule that selects one item from each slate; and an interaction budget. There are four candidate personas, listed in the registered order of priority that the feasibility gate of Section 8.3 uses. The gate selects the set $\mathcal{D}_A$ of confirmatory personas from them. Their seed pools are compiled and screened before registration, and two coders confirm that no seed item is antisemitic under the codebook of Appendix D.5.

(a) *Jewish interest*: Jewish culture, religion, history and community news. Users who seek such content may be targeted by antisemitic replies and recommendations, so this persona measures what the algorithm supplies to an audience that is often the target of the content.

(b) *Conspiracy-adjacent*: content on conspiracy theories about elites, finance or health that is not itself antisemitic under the codebook. This is the persona whose demand lies closest to antisemitic content without requiring any engagement with it.

(c) *Middle East news*: news and commentary on Israel, the Palestinian territories and the region, drawn from a seed pool balanced across outlets and viewpoints by two coders working independently.

(d) *General interest* (d_0, the baseline): mainstream news, sport, entertainment and how-to content.

The first two come first because they ask the questions that motivate the audit. The baseline persona is needed mainly for the persona-contrast decomposition (Appendix D.6), which is estimated only if all four personas are audited. No persona is trained on antisemitic content. A persona that sought it out would inflate the engagement metrics of hateful items, which the protocol forbids (Section 9.2). The audit therefore asks what each policy *supplies* to profiles whose demand is legitimate or adjacent, not what it delivers to users who seek antisemitic content.

Choice rule and the avoid-on-flag constraint. At each step t the puppet observes a slate. The first k positions are recorded and scored by the frozen classifier. The puppet then opens the highest-ranked recorded item whose score is below a registered threshold s_{safe} and whose similarity to the persona's seed pool exceeds a registered threshold. If no recorded item meets both conditions, it opens the highest-ranked recorded item below s_{safe}. If there is none, it opens the first item below s_{safe} further down the slate, which is recorded but not counted among the exposure positions. Items scoring above s_{safe} are recorded as exposures but never opened. The threshold s_{safe} lies below the classifier's decision threshold, so that no flagged item is opened. It is set conservatively enough that the pilot's estimate of the share of openings that are false negatives is small (Appendix D.4). The choice rule is part of the persona, so it is applied identically under both policies, and $\mathrm{AR}(d)$ is defined conditional on it (Assumption A6). Because the rule depends on the classifier, false negatives can still be opened. Their share is estimated from coded openings (Appendix D.5), and it enters the interference analysis (Appendix D.4) and the residual footprint (Section 9.2).

Exposure. Exposure counts *impressions*, which settles the choice that Section 3.1 leaves to the protocol. The exposure E_{us} of session s of puppet u is the share of its kT recorded positions, k at each of T steps, that show an item with $L_c = 1$, and the flagged share Q_{us} is the share with $\widehat{L}_c = 1$, exactly as in Section 5.1. An item recorded at several positions counts at each, and opening an item adds no further impression. Exposure is therefore not a property of the items a puppet opens. Under the avoid-on-flag rule, an opened item is antisemitic only if it is a false negative. The registered k is the number of items visible without scrolling on the surface's default layout, fixed during the pilot. With $k = 1$, exposure reduces to the top-ranked item. Appendix D.6 describes how the Markov model of Section 4 is fed with these impressions.

Session structure and waves. Each puppet passes through a *training phase* of S_{train} sessions, one per day, in which it opens a random subset of its persona's seed pool with a fixed dwell time per item. A *measurement phase* of S sessions follows, at most one per day at a randomised time, each of T steps from a registered entry point (the home surface, or a seed item drawn at random from the pool). Drawing a different random subset of seeds for each puppet limits the overlap in viewing between puppets of the same persona, which is a channel of interference (Appendix D.4). Puppets run in waves. A wave occupies $S_{\text{train}} + S$ days, and consecutive waves are separated by a washout in which no audit account is active (Appendix D.4). With the candidate values of Table 15, a wave cycle lasts three weeks. Table 15 lists the design parameters and how each is fixed.

D.3 Randomisation

All accounts are created in the same way, before any is assigned, so that features of account creation cannot differ systematically by arm. The audit runs in waves $w = 1, \ldots, W$, with $W \le W_{\max}$, that are separated

Table 15. Design parameters of the audit. Candidate values are illustrative and are fixed at registration, by the pilot or by the feasibility gate, as indicated; none is a result.

Parameter	Candidate	Fixed by
Confirmatory personas $\mathcal{D}_A$	1, 2 or 4 of the 4 candidates, in priority order	Feasibility gate (Section 8.3)
Policy arms	alg, ref	Reference policy available on the surface (Appendix D.1)
Pairs per persona n (trained puppets per arm)	From the gate; 122 on rung R4 under the inputs of Table 8	Proposition 6.3 with pilot inputs (Section 8.3)
Capacity caps: accounts $U_{\max}$ (all accounts created, pilot and sentinels included); coded positions (pilot included)	400; 15,000	Registration, with ethics approval and budget (Sections 9.2 and 9.3)
Wave budget $W_{\max}$; washout between waves	6; at least 14 days	Registration; washout checked in the pilot (Appendix D.4)
Load cap $\ell_{\max}$	0.01	Registration (Appendix D.4)
Burn-in b; tuning constant; planned pairs	1 wave; $a = -2\log\alpha + \log(1 - 2\log\alpha)$; $n^* = n$	Registration (Algorithm 1)
Training sessions S_{train}; seeds opened per session	3; 10	Pilot (stability of recommendations)
Measurement sessions S per puppet	4	Proposition 6.3(b) with pilot costs and variances
Steps per session T; positions recorded k	20; visible slate (5 in Table 8)	Pilot; surface layout
Codes per puppet: flagged m; unflagged m'	10; 25 (5 of them openings)	Algorithm 1; pilot (Appendix D.5)
Sentinel puppets per arm and wave (not coded)	2	Fixed
Pilot: trained puppets per cell; coding	10; all flagged and 50 unflagged positions per puppet	Fixed (Section 8.3)
Error level α; margin δ; power	0.05; 1.25; 0.8 at AR = δ^2	Registration (Section 8.2)
Range $[\lambda_-, \lambda_+]$ for λ_S (secondary analysis)	$[0.95, 1.05]$	Registration (Appendix D.6)

by washouts. Within each wave, puppets are assigned completely at random to the persona–policy cells of the confirmatory personas, in equal numbers per arm. Persona training is therefore randomised, and policy is randomised independently of it. Seed subsets, session start times and the order in which puppets run within a day are also randomised. All puppets are trained in the same way whatever their arm. The policy contrast is applied only at the start of the measurement phase: puppets in the reference arm switch to the non-profiling or chronological option, or run their measurement sessions logged out with the same choice rule. Training is thus identical in distribution across arms, and the only systematic difference at measurement is the policy. This is the design of Theorem 5.1.

Waves are blocks. Every wave contains both arms of every confirmatory persona, so a change in the service's policy during the window affects both arms of a wave equally. The pairs of Algorithm 1 are formed within waves, by pairing each puppet of the algorithmic arm at random with a distinct puppet of the reference arm of the same wave and persona. The wave weights ω_w of the blocked estimator of Theorem 5.1(d) are the shares of pairs, so that the blocked estimator and the confidence sequence target the same window average (Appendix D.7). If attrition leaves the arms of a wave unequal, the surplus puppets are excluded from the pairs at random, and attrition is reported by arm. On rungs R1 and R2, if at least four waves are run, wave sizes are randomised for the saturation design of Appendix D.4: consecutive waves are grouped in pairs, and one wave of each pair, chosen at random, has twice as many puppets per cell as the other. On rungs R3 and R4 all waves have equal size (Section 8.3). Changes announced by the service during the window are logged as they occur.

D.4 Interference controls

Puppets share the learning system they audit. Through it, one puppet's behaviour can change what another is shown, which violates Assumption A3. The protocol limits interference by design, keeps the load within a registered cap, and collects evidence about what remains, so that the bound and the breakdown value of Proposition 5.2 can be reported. It cannot estimate the constants ξ_π and ξ_Δ of Assumption A8, because no audit observes the no-audit state $\ell = 0$.

- *Isolation.* Puppets never follow, message, mention or otherwise interact with any account, including each other. Each runs in its own browser profile with separate storage, and no identifier is shared between puppets.
- *Limited co-viewing in training.* Each seed item may be opened by at most a registered number of puppets per wave, and seed subsets are drawn so that puppets of the same persona overlap as little as the pool allows.
- *Washout.* Consecutive waves are separated by at least 14 days in which no audit account, sentinels included, is active, so that waves are the non-interfering groups of Assumption A8. The length is fixed at registration from what the service publishes about its update cycles, and it is checked in the pilot, whose sentinels run before, during and after the wave. The washout is lengthened if their exposure has not returned to its pre-wave level. No washout resets engagement counters that the service accumulates without decay, so the load is computed cumulatively: the openings of earlier waves count against the items' cumulative engagement.
- *Load cap.* After each wave, an approximate upper bound on its load ℓ_w is computed from public engagement counts, as the puppets' openings of the items they opened divided by the increase in those items' public engagement counts, cumulated as above. The registered cap is $\ell_{\max} = 0.01$. Wave sizes are set at the gate so that the load projected from the pilot stays below the cap. If a confirmatory

wave exceeds it, later waves are made smaller and the deviation is reported. Within a wave, staggered start times lower the instantaneous load further.

- *Device and network hygiene.* Each puppet keeps one consistent device profile and one stable network location within the study jurisdiction for its whole life. Network locations are not rotated to evade a service's detection or rate limits, and no technical access control is circumvented (Section 9.1). A puppet that is challenged, restricted or suspended is stopped and not replaced within the wave. Its attrition and the reason are recorded, and attrition is compared across arms.
- *False-negative mass.* Because false negatives are opened, some audit engagement reaches antisemitic items. By Proposition 5.2(b), whether load then raises or lowers exposure depends on how concentrated this engagement is on antisemitic items, relative to the baseline exposure. We estimate the share of openings that are false negatives, χ_{open}, an unweighted analogue of χ^{P} in Proposition 5.2(b), from the coded openings (Appendix D.5). We report it with its confidence interval next to the estimated exposures $\mu_\pi(d)$. We make no claim about the sign of the interference bias, and we use only the two-sided bound of Proposition 5.2(a) and the breakdown value of part (d). If the pilot's upper Clopper–Pearson bound for χ_{open} is not small relative to the pilot exposures, s_{safe} is lowered before the first confirmatory wave.
- *Sentinels.* In each wave, two untrained puppets per arm run measurement sessions alongside the trained puppets. Across waves of different load, the slope of their exposure in the load is evidence about interference for an untrained profile. It is neither an estimate of ξ_Δ for the audited personas, whose sensitivity to load can differ, nor a bound on it. Sentinels also track drift that is common to both arms.
- *Randomised saturation.* When wave sizes are randomised, on rungs R1 and R2 (Appendix D.3), the two load levels satisfy $\ell_2 \approx 2\ell_1 \le \ell_{\max}$. Proposition 5.2(c) then gives a test of differential interference and the extrapolated estimator $\widehat{\text{AA}}^{\text{ex}}$. Its variance is several times that of the blocked estimator, so it is a robustness analysis, and even on rungs R1 and R2 the test has little power. Under Assumption A8, $|\text{AA}(\ell_2) - \text{AA}(\ell_1)| \le \xi_\Delta(\ell_1 + \ell_2)$ by the triangle inequality, so the saturation contrast also gives a data-based lower value for the interference constant, $|\widehat{\text{AA}}(\ell_2) - \widehat{\text{AA}}(\ell_1)|/(\ell_1 + \ell_2)$.

Reporting. For each confirmatory persona we report the bound on $\bar{\ell}$, the estimate of χ_{open}, the sentinel slopes, the saturation slope where available, and the breakdown value ξ^* of Proposition 5.2(d). Because the final analysis follows sequential monitoring, ξ^* is computed with the radius of the confidence sequence for $\text{AA}(d)$ (Appendix D.7) in place of $z_{1-\alpha/2}\widehat{\text{se}}$. The product $\xi^*\bar{\ell}$ is the largest shift of $\text{AA}(d)$ by differential interference that leaves the sign of the conclusion intact, and it is stated with every conclusion. A conclusion whose ξ^* lies below the lower value from the saturation contrast is reported as *fragile to interference*.

D.5 Classifier and validation sample

Frozen classifier. The classifier follows the lab's detection proposal (Kornstein Lab for Antisemitism Research, 2026d). Its version, decision threshold, s_{safe} and the score band that defines the observed borderline state of the Markov analysis are fixed and hashed at registration and are not changed during the audit. Every recorded item receives a score; $\widehat{L}_c$ is the thresholded score.

What the validation must deliver. Three requirements of Sections 5 and 6 fix the design. First, error rates must be estimated by arm, because they differ when the arms show different content (Theorem 6.1(c)), and by persona, because persona contrasts need the error rates of each persona (Theorem 5.1(a)). Second, they

must be exposure-weighted, as in (5.1), because the target is the classifier's performance on what each cell is shown, not on distinct items. Third, coding must be done within puppets. Algorithm 1 codes each puppet's flagged positions, and puppet-level double sampling makes each puppet's corrected exposure conditionally unbiased (Theorem 6.1(d)), so that the cluster-robust variance carries the validation uncertainty. The validation floors of Proposition 6.3(a) decide between the designs that meet these requirements. Under the inputs of Table 8, a Rogan–Gladen correction with error rates estimated by arm from coded negatives needs about 21,400 coded negatives per arm and persona on rung R1 before any number of puppets suffices. Internal double sampling needs about 2,600 coded unflagged exposures per arm and persona, and its coding grows with the number of puppets. A fixed validation sample of 2,000 items, as in a conventional small-lab design, would give about 125 per arm and persona. The protocol registers double sampling within puppets.

Sampling within puppets. For each puppet, after its last measurement session, recorded positions are drawn from its own record for coding, in three strata:

(i) $m = 10$ flagged positions, uniformly without replacement, or all of them if the puppet has fewer, as Algorithm 1 requires;

(ii) 5 positions of items that the puppet opened, which estimate the false-negative share of openings χ_{open};

(iii) 20 other unflagged positions, stratified by registered classifier-score bands, with the allocation across bands set by the rule of Neyman (1934) from pilot estimates of the false-omission rate in each band.

Strata (ii) and (iii) make up the $m' = 25$ unflagged codes per puppet. Within a stratum, every recorded position has the same probability of being drawn, so an item shown at several positions is proportionally more likely to be drawn, and each drawn position is weighted by its stratum's size divided by the number drawn from it. The sample therefore represents exposures, not distinct items. A distinct item drawn more than once, within a puppet or across puppets, is coded once, and its adjudicated label is attached to every draw. Coding effort is thus counted in distinct items, while estimation is per exposure. In the pilot, all flagged positions and 50 unflagged positions per puppet are coded, to estimate the inputs of Section 8.3 and the false-omission rate by score band.

Coding. Two trained annotators code each sampled item independently. They are blind to arm, persona, puppet, stratum and classifier score; items from all strata and cells are interleaved in mixed batches, and they work from a written codebook. The codebook has four primary categories: *antisemitic* (including coded forms); *quotes, reports or criticises antisemitism*; *borderline or ambiguous*; and *not antisemitic*. It also has a flag for content concerning Israel or Zionism. The primary label is $L_c = 1$ for the first category only. Sensitivity analyses count borderline items as antisemitic, and report scope variants under the IHRA and JDA definitions where they differ (International Holocaust Remembrance Alliance, 2016; Jerusalem Declaration on Antisemitism, 2021), side by side and without a preferred variant. A third, senior coder adjudicates disagreements; items are not dropped. Annotators complete a calibration round on items outside the audit until their agreement reaches a registered threshold. Agreement is reported as Cohen's kappa (Cohen, 1960), overall and by stratum. Coders must be fluent in the languages of the items. The protections for annotators are those of Section 9.3.

Use. For each puppet, the adjudicated labels give the *double-sampled exposure* $\widetilde{E}_u$: the right side of (6.3) computed from the puppet's own codes, with the unflagged stratum split into the strata above, each weighted by its share of the puppet's recorded positions. Each coded unflagged position thus stands for about 16 recorded positions. Given the puppet's record, $\widetilde{E}_u$ is unbiased for its true exposure $\bar{E}_u$ (Theorem 6.1(d)),

and it lies in $[0, 1]$. It assumes nothing about sensitivity or specificity, in either arm or in any persona, and it is the primary outcome. Aggregated over the puppets of a cell, the same codes give exposure-weighted estimates of $\mathrm{Se}_\pi(d)$, $\mathrm{Sp}_\pi(d)$, $\mathrm{PPV}_\pi(d)$ and $\mathrm{FOR}_\pi(d)$, and of the sensitivity ratio $\lambda_S(d)$. We report them by cell, with tests of equal error rates across arms within each persona and across personas, and we report naive, pooled Rogan–Gladen and double-sampled estimates side by side. For the pooled correction we also report $\mathrm{cv}_{\mathrm{Sp}}$ and a bootstrap over puppets and codes (Section 6.1). False negatives are rare among unflagged exposures, so under the inputs of Table 8, and before any gain from score bands, the double sample pins $\lambda_S(d)$ down only to within a factor of about 1.13 at 95% confidence on rung R4, and about 1.06 even on rung R1. The range registered for λ_S in the secondary analysis (Appendix D.6) is therefore an assumption that the audit can contradict but not confirm.

D.6 Estimands and analysis

Primary. For each confirmatory persona $d \in \mathcal{D}_A$, the amplification ratio $\mathrm{AR}(d)$ and the additive amplification $\mathrm{AA}(d)$ of Definition 3.1, comparing the algorithmic policy with the registered reference policy, with exposure measured in impressions (Appendix D.2). They are estimated from the double-sampled exposures $\widetilde{E}_u$, blocked by wave as in Theorem 5.1(d), with wave weights equal to the shares of pairs. Under drift, the estimands are the window-averaged $\overline{\mathrm{AR}}(d)$ and $\overline{\mathrm{AA}}(d)$ of Theorem 5.1(d) for the waves actually run, and every conclusion is indexed by the window (Section 5.4). Inference is by the confidence sequences of Appendix D.7. No population weighting Λ of personas is registered, so no population-weighted estimand is reported (Section 3.2). If the pilot's upper Clopper–Pearson bound for $\mu_{\mathrm{ref}}(d)$ lies below the registered floor (0.002 illustratively), the ratio is unstable for that persona, and $\mathrm{AA}(d)$ becomes its primary estimand, with the stopping rules of Section 8.2 applied to the sequence for $\mathrm{AA}(d)$ and the registered additive margin δ_{A} (0.005 illustratively) in place of $\log \delta$.

Secondary.

(i) *Confirmed-flagged analysis.* Algorithm 1 as stated, on the confirmed-flagged exposures Y_u, with the registered range $[\lambda_-, \lambda_+] = [0.95, 1.05]$ for $\lambda_S(d)$. We also report the largest symmetric range $[1/\lambda^\dagger, \lambda^\dagger]$ for which its conclusion would still hold. This analysis uses no unflagged codes but assumes the range. The primary analysis assumes nothing about λ_S and pays for it with the unflagged codes.

(ii) *Persona-contrast decomposition*, only if all four personas are audited (rung R1 of Section 8.3). For each persona d relative to the baseline d_0, we report the supply contrast $\mathrm{AA}(d_0)$, the demand contrast $\mu_{\mathrm{ref}}(d) - \mu_{\mathrm{ref}}(d_0)$ and the interaction $\mathrm{AA}(d) - \mathrm{AA}(d_0)$, with their Shapley shares (Section 3.5). The cell means are those of $\widetilde{E}_u$, so each is corrected with its own persona's and arm's error rates, as contrasts across personas require (Theorem 5.1(a)). The demand contrast compares two scripted personas. It is not the demand-shaping effect τ_{dem} of Theorem 3.3, which no fixed-demand audit identifies (Proposition 3.4).

(iii) *Markov exposure summaries*, fed as described below and reported as exploratory, model-based analyses.

(iv) *Diagnostics*: the interference reporting of Appendix D.4; the drift statistic X^2_{drift} of Section 5.4, the wave-level estimates and the sentinel series; attrition by arm; error rates by cell and their equality across arms and personas; and the exact betting sequence of Theorem 6.2(b) at the stopping time.

Feeding the Markov model. Section 4 models a session as a chain of items, one per step. The protocol measures exposure on slates of k recorded positions. Under the avoid-on-flag rule, the chain of *opened* items visits $\mathcal{A}$ only through false negatives, so its exposure functionals would describe the residual footprint, not exposure. We therefore apply Section 4 with the slate of step t in the role of the item. The state $X_t \in \{\mathsf{N}, \mathsf{B}, \mathsf{A}\}$ is the most severe true state among the k recorded positions of step t: A if any of them is antisemitic, B if none is but one is borderline, and N otherwise. Then $\mathbf{1}\{X_t \in \mathcal{A}\}$ records whether the slate of step t shows antisemitic content, and $e_T(P, \nu)$ of (4.1) is the expected share of steps whose slate does. The matrix $P_\pi(d)$ describes transitions between consecutive slates, so it combines what the policy offers with what the persona opened. Let $\bar{k}_{\mathcal{A}}$ be the mean number of antisemitic positions in a slate in state A. If $\bar{k}_{\mathcal{A}}$ does not depend on the step, which we check by comparing early and late steps, the impression exposure of the primary analysis is $\mu_\pi(d) = e_T\{P_\pi(d), \nu_\pi(d)\}\, \bar{k}_{\mathcal{A}}/k$, and its long-run analogue is $\varpi(\mathcal{A})\, \bar{k}_{\mathcal{A}}/k$. A ratio of chain functionals between the arms must then be multiplied by the ratio of the arms' $\bar{k}_{\mathcal{A}}$, which is reported. If the order tests of Anderson and Goodman (1957) reject a first-order chain on pilot data, the state is enlarged to the pair formed by the slate state and the state of the item opened at the previous step, as Section 4.1 allows. The slate states are observed through the classifier, with the borderline state defined by the registered score band. The observed transition counts are corrected as in Section 4.4, with a slate-level confusion matrix Γ computed from each cell's item-level error rates under the independence of errors across positions assumed there. A neutral slate is misread as A with probability about $k(1 - \mathrm{Sp})$, so the correction is large. The correction is applied to the two states $\{\mathsf{A}, \text{not } \mathsf{A}\}$; the three-state chain is reported only if the pilot shows that the borderline score band separates B from N, because otherwise the slate confusion matrix is nearly singular. In simulation, the uncorrected slate chain erred by up to 150% and could reverse the comparison of the arms, while the corrected two-state chain erred by at most 7% in the long run, with a standard deviation of the log ratio of the arms up to 2.3 times that of the session average (Section 7.5). The variance of every chain functional is obtained by a bootstrap over puppets, because the sessions of one puppet are correlated. The stationary exposure, the hitting times of A and the sensitivities of Proposition 4.1 and Theorem 4.2 are model-based and are reported as such. At the audited horizon, the session averages of the primary analysis take precedence, as Section 4.4 recommends. The opened-item chain is reported descriptively.

Exploratory. Exposure by slate position, and heterogeneity by entry point.

D.7 Sequential monitoring

The primary sequences. After each wave, Algorithm 1 is run for every confirmatory persona with the outcome $O_u = \widetilde{E}_u$ and the range $\{1\}$ for λ, because the ratio of the arms' mean double-sampled exposures is $\mathrm{AR}(d)$ itself (Theorem 6.1(d)). The level is $\alpha/|\mathcal{D}_A|$ per persona, so that the family of sequences has simultaneous coverage $1 - \alpha$ to the same approximation. The registered burn-in is $b = 1$ wave: the first confirmatory wave updates the estimates but can neither narrow the running interval nor stop the audit. The tuning constant is $a = -2\log\alpha + \log(1 - 2\log\alpha)$ at the per-persona level, and the planned horizon n^* is the n fixed at the gate. The same construction applied to the pair differences $\widetilde{E}_{a,i} - \widetilde{E}_{r,i}$ gives a confidence sequence for $\mathrm{AA}(d)$. It needs no delta method, because the differences are independent and identically distributed with mean $\mathrm{AA}(d)$, so Lemma B.7 applies with $R_n = 0$.

What the guarantee is. The coverage of these sequences is approximate, not exact. Theorem 6.2(a) is a large-sample statement, in which the planned horizon and the burn-in grow together, and it assumes no interference (Assumption A3), stable policies (Assumption A4(a)) and scripted demand (Assumption A6).

At the protocol's sizes, the probability that monitoring after every wave and stopping early produces a wrong conclusion is therefore close to its nominal level but is not guaranteed to stay below it. Simulation shows that it can exceed the nominal level when there are few pairs per wave (Section 6.2), and the double-sampled outcome is heavy-tailed, because each coded unflagged position stands for about 16 recorded positions. In the simulation of the registered design (waves of 20 pairs, $W_{\max} = 6$, $b = 1$), $\widetilde{E}_u$ had excess kurtosis of up to about 13 in the arms and up to 47 in a null case, yet the sequences for AR and AA covered in 98.4–99.4% of audits at $\alpha = 0.05$ and 99.1–99.8% at $\alpha = 0.025$, and the betting fallback, though valid, was nearly powerless at 120 pairs (Section 7.6). The resampling check of Section 8.2 draws pairs with replacement from the ten pilot puppets per cell, so it reproduces only the tails that the pilot happens to contain. It can reveal a gross failure of the approximation; it cannot certify coverage. If it fails, the exact betting sequence becomes the primary sequence, and its horizon is found by the same resampling.

Drift. Under drift, the pairs are independent but not identically distributed. The sequences then target the ratio of window-averaged means (Section 6.2), here $\overline{\mathrm{AR}}(d)$ with wave weights equal to the shares of pairs, which is also the target of the blocked estimator, and the guarantee holds only approximately. Blocking by pairing within waves keeps drift from being mistaken for amplification, but it does not remove drift from the estimand (Section 5.4). We report the wave-level estimates and X^2_{drift} with every conclusion. If the service announces a change to the audited surface during the window, the waves before and after it are reported separately as well as together. No conclusion is extended beyond the window.

Robustness. The registered robustness checks are: naive, pooled Rogan–Gladen and double-sampled estimates; the confirmed-flagged analysis and its $\lambda^\dagger$; the exact betting sequence; scope variants of the label; analyses by wave block; the exclusion of puppets with attrition; the saturation-extrapolated estimator; and the sensitivity of each conclusion to interference through ξ^*.

D.8 How the sample sizes are computed

Illustrative inputs. Table 8 evaluates the closed forms of Proposition 6.3 for the design of this protocol. The inputs are illustrative assumptions, and the pilot replaces them. Reference exposure is $\mu_{\mathrm{ref}}(d) = 0.02$ for every persona. The session-level coefficient of variation of true exposure is 1, and the intra-puppet correlation is $\gamma = 0.3$. A puppet runs $S = 4$ measurement sessions of $kT = 100$ recorded positions ($k = 5$, $T = 20$). The classifier has Se = 0.85 and Sp = 0.98 in both arms. Power is 0.8 at the registered alternative $\mathrm{AR} = \delta^2$, with $\delta = 1.25$. Each puppet has $m = 10$ flagged and $m' = 25$ unflagged codes, and the cost ratio is $c_p/c_s = 3$, a puppet's training costing about three sessions. The candidate $S = 4$ rounds up $S^* = 3.6$ of Proposition 6.3(b). It needs more puppets than $S = 10$ (the row "R1, $S = 10$ sessions" of Table 8), but fewer sessions in all, and sessions are the audit's footprint. For the double-sampled statistic, the coding variance of a puppet does not fall with S, which lowers S^*, while the coding cost of a puppet raises it. The pilot fixes S by the minimisation of Proposition 6.3(b) with both included.

How the sizes are computed. For the primary analysis, the double-sampled outcome has variance $V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}}$ per pair, with V^{ds} as in Proposition 6.3(a) and $V^{\mathrm{ds}}_{\mathrm{val}}$ of (6.6) evaluated at $m_{1\pi} = nm$ and $m_{0\pi} = nm'$, so that $nV^{\mathrm{ds}}_{\mathrm{val}}$ does not depend on n. The horizon is that of Proposition 6.3(c) with this variance, $n = (c_\alpha + z_{1-\beta})^2(V^{\mathrm{ds}} + nV^{\mathrm{ds}}_{\mathrm{val}})/\Delta^2$. Materiality rungs use $r_0 = \delta$ and detection rungs $r_0 = 1$, both with $r_1 = \delta^2$. The other columns of Table 8 are the fixed-sample size with known error rates (Proposition 6.3(a) with $V_{\mathrm{val}} = 0$), the validation floors $n_0^{\min}$, $n_{0,\mathrm{st}}^{\min}$ and $m_0^{\min}$ of Proposition 6.3(a), and the horizons of the confirmed-flagged analysis with $\lambda_S = 1$ and with the registered range for λ_S (Proposition 6.3(d)).

Feasibility in detail. Rung R1, materiality for four personas, needs about 4,900 trained accounts and 170,000 coded exposures, and stays in the thousands of accounts in every sensitivity row; rung R2 needs about 2,200 accounts. Detection is much cheaper: rung R3 needs about 550 trained puppets and rung R4 about 250. A conventional small-lab design, with 40 puppets per cell over eight waves of five pairs, $S = 10$ and 2,000 coded items in all, would have power of about 17% for the four-persona materiality test at a fixed sample with known error rates, 2.4% with sequential monitoring, and about 1% with the double-sampled outcome, and its validation sample would be a small fraction of every floor in Table 8. Three further facts shape the gate.

(1) The pooled floor $n_0^{\min}$ is shown for comparison only. The protocol needs error rates by arm and persona, and estimated from coded negatives these need about 21,400 per arm and persona on rung R1. That is why the protocol uses double sampling, whose coding grows with the number of puppets.

(2) With $\lambda_S \in [0.95, 1.05]$, the confirmed-flagged outcome needs more than twice the pairs of the double-sampled one on rung R1, and about a fifth more on rungs R3 and R4. The comparison is not uniform: when sessions vary less (coefficient of variation 0.5), the confirmed-flagged outcome needs fewer pairs on rung R3 (66 against 77), because double sampling then pays mainly for its unflagged codes. Double sampling is primary because it assumes nothing about the sensitivity ratio, not because it is always the more efficient design.

(3) A horizon sized for detection rarely permits rule S2. Concluding "no material difference" at AR = 1 with power 0.8 needs about 650–800 pairs per persona under these inputs, so on rungs R3 and R4 "difference detected; materiality undetermined" is the conclusion that the audit is powered to reach.

The pilot. The pilot is one wave run under the full protocol, with all four personas, ten trained puppets per cell, two sentinels per arm and $S = 4$. Its coding is heavier than in confirmatory waves: all flagged positions and 50 unflagged positions per puppet. It estimates, for each persona and arm, $q_\pi(d)$, and $\mathrm{PPV}_\pi(d)$ and $\mathrm{FOR}_\pi(d)$ by score band, and hence $\mu_\pi(d)$, $\mathrm{Se}_\pi(d)$ and $\mathrm{Sp}_\pi(d)$; and, pooled over cells, the session variance, γ, the costs of a puppet, a session and a code, the load per puppet, the false-negative share of openings and the decay of sentinel exposure through the washout. Its data are not part of the confirmatory analysis.

D.9 Data minimisation and retention

The audit records only what the estimands need: item identifiers, slate positions, steps, timestamps, the item content needed for classification (text, titles, transcripts or images), and classifier scores. It does not record comments, viewer lists, profile information of other users or any other data about real users beyond the items themselves. Identifiers of the accounts that posted the items are replaced by keyed hashes at collection, and the key is held separately from the data. Raw item content is retained only inside the secure environment of Section 9.4, and only for the validation sample and for reproducibility checks. It is deleted twelve months after publication, after which only labels, scores and hashed identifiers are kept. Puppet accounts are closed and their credentials destroyed at the end of the audit, or after the pilot if the gate stops the audit. Public outputs are aggregate exposure tables, code, the codebook (with paraphrased rather than verbatim examples) and the classifier version. Raw items are not released.

D.10 Timeline

Table 16 gives the timeline. Months are counted from registration. It assumes rung R4 or R3; a larger rung takes no longer, because waves grow rather than multiply, but needs proportionally more coding within each washout.

Table 16. Audit timeline (months from registration).

Months	Activity
−2 to 0	Ethics approval, including the capacity caps; legal review of the service's terms and of the access route; DSA data-access application where applicable; registration of this protocol
0–1	Classifier freeze; codebook finalised; annotator training and calibration; seed pools screened
1–2	Pilot wave and washout, with sentinels throughout; pilot coding; feasibility gate; amendment registering the rung and the design before the first confirmatory wave, or a feasibility report and no further accounts
2–6.5	Confirmatory waves, up to six three-week cycles; each wave's sampled positions coded within the following washout, so that the confidence sequences are updated before the next wave; stopping per Section 8.2
6.5–7.5	Adjudication completed; analysis as registered; robustness checks
7.5–8.5	Responsible disclosure to the provider, at least 30 days before release (Section 9.5)
8.5–9.5	Publication; release of code, codebook and aggregate tables; deletion schedule begins